

CAPSTONE PROJECT 2

Final Report

Gamified Mobile App for Student Procrastination Management

by

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Abstract

Academic procrastination remains a prevalent issue among university students, negatively affecting academic performance, motivation, and psychological well-being. Existing productivity applications primarily provide generic task management and reminder functions without adapting to individual procrastination behaviours. This project presents Nudgr, an intelligent gamified mobile application that integrates machine learning, Generative Artificial Intelligence (AI), and gamification to predict students' academic procrastination risk and provide personalised support for self-regulated learning. The project adopted a design and development methodology comprising requirement analysis, machine learning model development, mobile application development, system testing, and pilot evaluation. A secondary dataset based on the Procrastination Assessment Scale–Student (PASS) questionnaire was used to train and evaluate seven supervised machine learning algorithms, where Random Forest was selected as the best-performing classifier after comparison using accuracy, precision, recall, F1-score, and cross-validation. The selected model was integrated into a Flutter-based mobile application with Firebase backend services and Google's Gemini API to provide AI-assisted task decomposition, personalised reminders, behavioural nudges, progress analytics, and gamified productivity features. The Random Forest model achieved 76.27% validation accuracy, 82.35% precision, 77.78% recall, and an F1-score of 80.00%, while maintaining 80.00% recall on the independent test set for identifying students at high risk of procrastination. Pilot testing demonstrated positive user acceptance, with participants reporting that the application improved motivation, simplified task management through AI-assisted planning, and encouraged consistent engagement through gamification. The application also achieved an average System Usability Scale (SUS) score of 70.75, indicating good overall usability. Overall, this project demonstrates that integrating behavioural prediction, Generative AI, and gamification into a single mobile application can provide proactive and personalised support for academic procrastination management while contributing to self-regulated learning and supporting the objectives of Sustainable Development Goal 4 (Quality Education).

Keywords: Academic procrastination, Machine learning, Random Forest, Generative Artificial Intelligence, Gamification, Self-regulated learning, Flutter.

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1.0 Introduction

1.1 Background Study

Academic procrastination was defined as delaying academic tasks intentionally although the students were aware that there would be negative consequences (Aldhi et al., 2026). Due to self-regulatory problems, this procrastination issue in academics was widely recognized among university students and was consistently related to poorer academic performance, increased stress, and inefficient study habits. Furthermore, students also felt a sense of aversiveness which can differentiate postponing the tasks intentionally because of their situation with academic procrastination (S. H. Lee et al., 2025). S. H. Lee et al. (2025) noted that interventions could help manage procrastination as it is a behaviour that can change with time according to how the students interact with the tasks and situations.

There were plenty of studies that highlighted that external factors that made the students delay tasks were because they assumed that the task was unimportant, too much burden, the complexity of the work assigned, unclear instructions, and difficulty of the work given. Most students preferred engaging in activities that were more enjoyable compared to tasks that are rewarding in the long-term. There were also other factors such as time management, jobs, family obligations, and too many school projects and assignments. There was also proof that the culture of school affected the level of procrastination among the students (Kemppainen & Oinas-Kukkonen, 2026).

With the widespread use of smartphones, apps were developed to manage procrastination such as time management apps or to-do apps had become very convenient to control the issue of delaying task completion. Such apps usually focus on improving users' behaviours to achieve their goals by effectively using their time by implementing time management strategies. These strategies included planning, managing and some apps also assisted in dividing the tasks into subtasks and as well as performing an assessment to evaluate themselves (S. Lee et al., 2025).

S. Lee et al. (2025) also mentioned that although these apps were convenient, they did face challenges especially in resolving the issue of procrastination. Time management apps did not really focus on factors that caused procrastination such as motivation, self-esteem, and mood

regulation because these factors were not the sole reason for poor time management but instead it is related to psychological problems like poor satisfaction of life, and low mental health.

Hence, interventions to manage academic procrastinations were needed so that the students were motivated to complete their tasks and also encouraged continuous engagement. One of the most effective interventions is gamification as it can help improve the user engagement. A systematic literature review by Pereira et al. (2025) investigated how gamification could influence procrastination. The authors researched 24 accepted articles out of 781 screened papers across ACM, IEEE, Scopus and many other major databases. Among the 24 articles, they noted that many of those studies were about mobile apps and gamified task managers such as Habitica, Kahoot!, chatbots that are gamified as well, and web apps that are adaptive. Besides that, the articles also acknowledged that there was a trend where gamification elements such as points, badges, progress bars, challenges, and leaderboards helped to target factors that caused procrastination such as poor time management, fear of failure, and perfectionism.

Additionally, other interventions such as artificial intelligence (AI) were also useful to manage procrastination in academics. This study highlighted that there should be more AI tools that should be integrated into academic workflows to reduce procrastination and enhance academic productivity (Jebasingh et al., 2025). AI tools that were useful as mentioned in the study are AI-powered task management systems, AI-powered reminder systems, AI-powered scheduling assistants and chatbots, and including peer motivation mechanisms in AI-powered personalized reminders. Thus, it was very encouraged to optimize AI adoption to enhance academic productivity.

Moreover, machine learning (ML) techniques had gained attention for their ability to predict students' procrastination. Hence, they had great potential to be used in apps to predict procrastination behaviours. ML consistently learnt from new data which helped to improve their predictions. On top of that, ML could not only analyze huge datasets but could also identify trends that are complex and the relationships among the variables (Sackey et al., 2025).

Although there were many apps that are productive and help tackle procrastination among the students, they still lacked the ability to sustain the motivation of users due to the many traditional task management systems implemented. This was why gamification strategies along

with AI tools and ML should be implemented to provide personalization and guidance to prioritize tasks.

Therefore, this project proposed a gamified mobile application development which was designed to help students reduce academic procrastination and at the same time improve management of tasks. Gamification elements and intelligent task prioritization were introduced in this app to encourage consistent task completion and provide recommendations that were personalized. The application was developed to help the users develop better study habits, increase productivity, and manage academic responsibilities effectively.

1.2 Problem Statement

Despite the availability of productivity tools and digital learning, academic procrastination remains a common issue in the university. Solutions that are currently available are mainly geared towards generic task scheduling, reminders or notation features and assume that everyone needs the same kind of support. But there are several behavioural, emotional and psychological factors involved in academic procrastination and these are different for various individuals (Li et al., 2026). So, static productivity tools can be ineffective in offering personalized interventions to overcome the root causes of procrastination. Given students' poor time management and lack of focus on academic tasks, academic stress and performance reduction may persist if students are not able to manage their academic tasks using productivity applications.

Recent years have shown that machine learning can have significant promise in the field of predicting academic outcomes or detecting behavioural patterns from educational data. Many studies have been able to create predictive models that are able to identify students who are likely to be academic procrastinators or underachievers (Imhof et al., 2023). Many of these studies are however finished once the models have been tested for their performance and are not expanded to practical software applications. The power of machine learning to predict is consequently not often turned into real-time decision support systems that can give instant and individualised help to students. This leaves a divide between research and the application of educational technology.

Moreover, while gamification has gained traction in enhancing the engagement of learners in educational apps, most of the currently available gamified systems offer the same rewards, reminders and motivational messages irrespective of the human behaviour of each learner (Dennis et al., 2024). In the same way, recent developments in Generative Artificial Intelligence have brought intelligent task planning and personalised content creation, while the affordances of these tools in relation to behavioural prediction models in educational applications are still relatively limited (Chen & Dethlefs, 2025; Kirchner-Krath et al., 2025). Therefore, available approaches take prediction, task management and motivation as isolated elements instead of a unified adaptive intervention approach that can adaptively react to each student's procrastination risk.

Hence, an intelligent educational support system is needed which integrates machine learning prediction, Generative Artificial Intelligence and gamification in a single mobile application. This system should be able to capture students at risk of academic procrastination, provide customized task structure, provide adaptive reminders and motivational nudges, and reinforce behaviours for long-term SRL. These challenges are the main reason to develop the proposed Nudgr application.

1.3 Aim and Objectives of the Project

1.3.1 Aim of the Project

The goal of this project is to create and test an intelligent gamified mobile app called “Nudgr” that uses machine learning, Generative Artificial Intelligence (AI) and gamification approaches to foresee students' academic procrastination risk and offer tailored intervention to promote self-regulated learning and task management. The proposed system is designed to guide the university students in building sustainable study habits and to alleviate their negative academic and emotional impact due to procrastination through the use of behaviour prediction, adaptive task planning, and motivational support.

1.3.2 Objectives of the Project

The project aims are achieved through the following objectives:

1. To investigate procrastination behaviours and gamification strategies by reviewing existing studies on academic procrastination and motivational design to identify key behavioural factors and effective gamification approaches.
2. To develop a machine learning model that will predict students' risk of academic procrastination by selecting suitable machine learning model, training predictive models by using behavioural data, and evaluate their performance in identifying students at risk of procrastination.
3. To create and build a gamified mobile app that supports self-regulated learning (SRL) by combining the predictions of the machine learning, task planning assisted by Generative AI, personalised reminders, behavioural nudges and gamified features to foster sustained engagement and better academic task management among University students.

1.4 Project Scope and Limitations

The project aims to design, develop and evaluate the intelligent gamified mobile application Nudgr, which will help to manage academic procrastination among the university students using machine learning prediction, Generative Artificial Intelligence (AI) and gamification techniques. The application is designed for the use of university students who want to enhance their task management, cultivate self-regulated learning skills and get personalized behavioural assistance during their learning processes. The system presented here is meant to be an instructional productivity resource and should not be used for the diagnosis or treatment of psychological or medical problems associated with procrastination.

The machine learning aspect of the project is built with a secondary data from Rusdi et al. (2020), which is a questionnaire called Procrastination Assessment Scale–Student, consisting of answers from undergraduate students in Indonesian universities. The prediction model is based on a classification of users as Low Risk and High Risk procrastination users using

selected behavioural and psychological indicators. Several supervised machine learning algorithms are tested such as Logistic Regression, Support Vector Machine (SVM), Decision Tree, Random Forest, Gradient Boosting, k-Nearest Neighbours (k-NN) and Multilayer Perceptron (MLP) before selecting the best performing model for inclusion in the application.

The mobile app is built with the Flutter framework, and Firebase is used as the backend for user authentication, cloud databases, cloud storage, and notification. The application combines the selected Random Forest prediction model with Google's Gemini API to enable AI-guided task decomposition and automatic deadline generation. There are many more features, such as personalized reminders, behavioural nudges, task management, progress analytics, achievement badges, a virtual garden gamification system, and user profile management. It is a functional prototype for mobile phones that are both Android and iOS.

There are two main components of the evaluation of the proposed system. First, the standard metrics of classification are used to assess the machine learning model: accuracy, precision, recall, F1 score, confusion matrix and cross-validation. Second, the mobile application is pilot tested with usability testing students from the University using System Usability Scale (SUS) questionnaire and additional user experience questions to evaluate usability, user satisfaction, perceived usefulness, and areas for further improvements.

While this is a contribution, there are some limitations in the current research. The prediction model is built from a secondary data set from a single geographical region and may not be generalisable to other student populations. Moreover, the data is imbalanced, so the prediction model, in fact, performs better at predicting students with high risk of procrastination than those with low risk of procrastination. There were further limitations in this study due to the time constraints of the project, time needed for further balancing of the datasets and time for further optimising of the model. Thus, the mobile application includes a temporary threshold fallback mechanism for providing more feedback to the lower risk users, and future work should consider a larger and balanced set of data to enhance prediction accuracy at all risk levels of procrastination.

1.5 Research Significance

The proposed project has a significant contribution to the field of educational technology as it shows the integration of multiple intelligent computing technologies in a single mobile application that can assist in self-regulated learning. Nudgr's AI-driven productivity application is different to traditional task scheduling and reminder software, by integrating machine learning, Generative AI and gamification to offer personalised behavioural support, depending on each user's procrastination risk. This integration highlights the potential of predictive analytics to move its theoretical machine learning research into a tangible application as a decision-making tool for solving an actual educational challenge.

An educational aspect of the suggested system is that it can help students to adopt proper study habits by identifying student procrastination risk at an early stage and taking appropriate action. The application does not just react when deadlines are missed, but rather offers proactive support with personalised reminders, motivational nudge, AI-enabled task decomposition, and progress tracking. These characteristics help students to better manage their workload, avoid being overwhelmed, and build better SRL skills, which in turn could result in better academic outcomes and learning experiences.

This project also builds on the computing discipline by providing a hands-on experience of an intelligent mobile application that adopts multiple software technologies in an integrated architecture. The system is designed using Flutter for cross-platform mobile development, Firebase as the backend services, a Random Forest machine learning model for behavioural prediction, Google's Gemini API for AI-based task planning, and gamification mechanisms to boost user engagement. The project is thus an example of how artificial intelligence, cloud computing, mobile application development and human-computer interaction principles can be combined to achieve the solution to a complex problem in the real world through a user-centred software solution.

The project is also socially relevant in terms of the benefits it offers for the promotion of healthier learning behaviours and for students' long-term learning growth. Procrastination has a significant effect on the stress level, motivation and academic performance among students in higher education, making it a crucial issue in higher education. The proposed system aims to enhance the educational experience and promote lifelong learning by encouraging students

to take charge of their academic responsibilities more effectively. The evaluation results further support this contribution, as participants reported that the application improved their motivation to complete academic tasks, made task management easier through AI-assisted planning, and encouraged consistent engagement with their studies using the gamified virtual garden. These outcomes demonstrate how intelligent digital technologies can promote self-regulated learning and student engagement, which are key objectives of SDG 4: Quality Education as it focuses on inclusive learning in technology-rich learning environments that foster self-regulated learning, lifelong learning and enhanced learning outcomes using intelligent digital technologies.

2.0 Literature Review

2.1 Introduction

This chapter provided a comprehensive overview of literature related to academic procrastination, and mitigation techniques such as machine learning techniques, and artificial intelligence (AI)-based task planning. Key concepts and theories were examined by referring to the existing studies, methodologies, and technologies, and its limitations and research gaps relevant to the proposed study as the purpose of this review. By synthesizing these topics, it provided a foundation to understand the factors contributing to academic procrastination and how it can be addressed by using machine learning and gamification techniques. By highlighting the studies, this chapter established the theoretical basis and conceptual framework to help guide the system's development.

2.2 Academic Procrastination

2.2.1 Definition and Concepts

Torrens et al. (2026) defines academic procrastination as an action of delaying something although the students are aware that there will be negative consequences. However, Hong et al. (2021) defines procrastination as self-regulatory failure. Research showed that procrastination is often related to poor control of emotions, motivation and habits other than just time management. Thus, procrastination is considered to play a significant role in students' academic performance and overall well-being.

A study by Ahmed et al. (2023) investigated the prevalence of academic procrastination among university students and found high rates of behaviour of procrastination across various student tasks. These tasks included doing revision for exams and completing assignments. The common pattern among these students was the tendency of delaying tasks despite understanding it will affect their academic performance. This study highlights that procrastination is widespread and how it appears in different academic fields, proving that it is a serious problem among students.

Another significant study by Sirois (2023) proposed a model of procrastination to prove that risk of procrastination can increase due to stressful contexts such as academic pressure, personal stressor or the constant shift to digital learning. This perspective views procrastination as a path for students to regulate their mood. This is where they avoid completing tasks to escape negative emotions rather than improve their time management.

Furthermore, Fentaw et al. (2022) examined how procrastination was shaped by emotional avoidance and behavioural pattern by analyzing multiple studies. It was found that the students often delay completing their assignments when they feel overwhelmed, overburdened, or unsure. The concept of delaying is often used by them as a temporary escape rather than planning a strategic decision.

Together, these studies demonstrated that academic procrastination involves emotional avoidance, recurrent behavioural cycles, and vulnerability in stressful situations. This understanding helped to explain how procrastination develops and persists among students. It also highlighted the importance of creating intervention so that it can identify behavioural patterns, assist with emotional regulation, and adjust to students' actual learning environment.

Table 1 shows a summary of the academic procrastination definitions discussed in this section.

Author(s)	Main Definition	Key Findings
Torrens et al. (2026)	Academic procrastination is an intentional postponement of academic work that is known to have negative consequences.	Highlights procrastination as a conscious delay that has detrimental effects on academic performance.
Hong et al. (2021)	Procrastination is considered not merely as a failure of time management skills but as a failure in self-regulation.	Emotional regulation, motivation, and behavioural control are factors that contribute to procrastination.

Ahmed et al. (2023)	Studied the prevalence of academic procrastination among students at the university.	Procrastination was seen about assignments and preparation for examinations.
Sirois (2023)	Suggests that procrastination is used as a way to regulate emotions in stressful situations.	Students put off doing things because of negative emotions, not poor time management.
Fentaw et al. (2022)	Analysed the emotional avoidance and behavioural approach to procrastination.	Repeated procrastination behaviour is caused by feelings of overwhelm and uncertainty.

Table 1: Summary of Academic Procrastination Definitions and Concepts

2.2.2 Causes and Contributing Factors

Procrastination among university students is influenced by personal characteristics, psychological processes and contextual factors. Research continuously shows that students' tendency to put off academic tasks is greatly influenced by maladaptive perfectionism, self efficacy, self-regulated learning, and resilience. Designing solutions that are both successful and personalized requires an understanding of these aspects. Although there are multiple factors that are being examined by each study, this section focuses on the most influential factors that were recognized in each study. However, the full range of the examined factors is summarized in Table 2.

A study reveals that academic procrastination among Chinese nursing undergraduate students is strongly correlated with maladaptive perfectionism using a cross-sectional study among 587 undergraduate nursing in two undergraduate universities in China, and this effect is somewhat mitigated by students who have higher self-efficacy (Huang et al., 2023). However, self efficacy was a partial mediator, which indicates that students with greater confidence in their capabilities were better equipped to deal with perfectionism. The relationship is further

moderated by resilience as with greater resilience weakens the perfectionism-procrastination link. These results highlight how academic procrastination may be more likely in people with maladaptive perfectionism, poor resilience, and low self-efficacy. This demonstrates how crucial it is to support both emotional and cognitive resources while dealing with procrastination.

Another study investigated how self-regulated learning predicts procrastination and academic outcomes which was conducted on 728 students in Eastern Spain (García-Ros et al., 2023). The study used structural equation modelling (SEM) approach where the participants filled questionnaires on self-regulated learning, procrastination, academic stress, well-being, and achievement. Procrastination was proved to be acting as the mediator when two competing structural models were tested. Procrastination had direct negative effects on achievement and well-being and a positive effect on stress. It also partially had an effect on the impact of self-regulated learning on these outcomes. The study emphasizes that poor metacognitive strategies and low academic self-efficacy, which are the weaknesses in self-regulated learning, function as key factors that lead the students to engage in procrastination.

Jing et al. (2025) investigated how negative perfectionism contributes to academic procrastination among Chinese university students and whether self-efficacy acts as a mediating psychological mechanism in this association. A quantitative cross-sectional survey was designed to collect data from 460 students from Southwest China at four universities. Validated tools like Aitken Procrastination Inventory, an academic self-efficacy scale and the Negative Perfectionism Questionnaires were used for this study. It was predicted that higher the negative perfectionism the worse the academic self-efficacy is. This regression-based mediation analysis was done using Hayes' PROCESS Macro. The direct impact of perfectionism on procrastination became non-significant when self-efficacy was considered. This suggests full mediation whereas the indirect effect was statistically significant. Overall, the study demonstrates that the increase in academic procrastination is led by negative perfectionism due to low self-efficacy in academics.

Collectively, these studies show that academic procrastination is not just a result of poor time management. It is a behaviour that is complex which is influenced by personal, emotional and cognitive factors. Maladaptive perfectionism and self-criticism increase the tendency to postpone tasks related to academics by students. Low self-efficacy, weak self-regulation, and

limited resilience also make procrastination even worse. However, having better self-regulated learning, higher resilience and an increase in self-efficacy in academics can help lessen these negative effects.

Table 2 summarizes the findings clearly by outlining the key factors that affect procrastination as mentioned by the studies reviewed in this section.

Author(s)	Participants	Main Factors Investigated	Key Findings
Huang et al. (2023)	587 nursing undergraduates (China)	Maladaptive perfectionism, self-efficacy, resilience	Maladaptive perfectionism is positively linked to procrastination, with self-efficacy as a partial mediator, and resilience is a modifier.
García-Ros et al. (2023)	728 students (Spain)	Self-regulated learning, academic self-efficacy	Lack of self-regulation is linked to higher levels of procrastination, higher stress, and lower academic performance.
Jing et al. (2025)	460 university students (China)	Negative perfectionism, academic self-efficacy	The role of academic self-efficacy is fully mediated between negative perfectionism and procrastination.

Table 2: Summary of Factors Affecting Procrastination

2.2.3 Effects on Academic Performance and Well-being

Academic performance and psychological well-being of the students are often negatively influenced by academic procrastination. There are recent studies where it was indicated that academic tasks can lead to psychological distress other than poor academic achievement, and low effective learning. Hence, timely interventions are needed so that these consequences can be minimized by identifying the students at risk of procrastination.

Kumar et al. (2025) have investigated whether performance in education and the overall mental health of the students in India can be predicted by academic procrastination in their recent study. There were 150 undergraduates, postgraduates, and doctoral students who participated in this study. Employed standardized instruments such as Academic Procrastination Scale, Academic Performance Scale, and Mental Health Continuum-Short Form were utilized in this study as well. It was proven that academic performance, psychological well-being, emotional well-being, social well-being and overall mental health can be significantly and negatively associated with academic procrastination as revealed by the analysis of correlation and regression. Moreover, procrastination in academics also predicted across all measured dimensions that there were poorer academic achievement and lower levels of mental well-being. These findings suggest that both academic success and psychological well-being can adversely be affected by procrastination which makes the need for early interventions to promote student well-being even more important.

Additionally, a study by Vlachopanou et al. (2025) investigated the relationships among academic motivation, procrastination, adjustment to university, and grade point average (GPA). Tools that were used to conduct this study was Academic Motivation Scale, Procrastination Assessment Scale Questionnaire, and Student Adaptation to College Questionnaire. After analysing using K-Mean cluster analysis and decision tree methods, it was identified that there were three groups of student profiles which are highly motivated and well-adjusted students, moderately motivated and adjusted students, and procrastinated and poorly adjusted students. The third group which is the procrastinated group obtained the lowest GPAs meaning they demonstrated low intrinsic motivation, and their experience in adapting to university life is poor. The study highlighted that increase in academic stress, disengagement, and poor psychological adjustment leads to the students having a high level of procrastination. These factors influence the overall well-being of the students. Thus proving that reducing

procrastination can help promote healthier psychological adjustment and provide better university experience.

Recent evidence by Geronimo et al. (2025) suggested that academic procrastination affects both responsibilities in education and poor emotional health. The study investigated the relationships between academic procrastination, mental well-being, and psychological distress especially in a hybrid learning environment in Metro Manila. The study involved 100 among university students. It was found that students with higher levels of academic procrastination are affected by poor mental health and higher psychological distress. More importantly, mental health was negatively affected by procrastination when it is related to studying for exams and completing weekly reading assignments. This indicates that to improve the students' psychological well-being and their overall academic life, actions must be taken to reduce procrastination.

Overall, the studies that were reviewed in this section implies that academic procrastination is often affecting the students' education performance and their well-being. It is clear that students who procrastinate frequently experience greater stress and psychological distress. These findings justify strongly developing early intervention systems so that students with procrastination risk can be identified and able to encourage positive behavioural change.

Table 3 below compares the academic and psychological effects discussed in this section.

Author(s)	Academic Effects	Psychological Effects	Overall Findings
Kumar et al. (2025)	Lower academic performance	Poor mental health, emotional well-being, social well-being	Academic procrastination has a negative correlation with academic performance and a negative correlation with well-being.

Vlachopanou et al. (2025)	Lower GPA	Poor university adjustment, increased stress	Students' academic adjustment and intrinsic motivation is poorer for students with higher procrastination.
Geronimo et al. (2025)	Poor study performance	Higher psychological distress and poorer mental well-being	Laziness has a detrimental influence on studies and mental health.

Table 3: Comparison of academic and psychological effects

2.3 Machine Learning for Academic Procrastination Prediction

2.3.1 Introduction to Machine Learning

Machine learning (ML) is a subgroup of artificial intelligence (AI) that enables computer systems to learn patterns from data and make predictions or decisions without needing explicit programming (Kufel et al., 2023). ML focuses on automating cognitive processes like decision making and problem-solving. It allows machines to learn from experience and improve over time.

ML comprises various techniques such as supervised learning, where the models are trained using labelled data so that it can predict outcomes or categorize new observations. Another technique is unsupervised learning where it identifies the patterns or grouping in unlabelled data. Besides these two techniques, reinforcement learning allows models to learn the best course of action through trial-and-error interactions with an environment. These methods have been used across fields like engineering, finance, medicine and many more. This is where they demonstrate their capacity to identify trends, support decision-making and forecast results from complex datasets.

ML has great potential to understand and influence human behaviour. It has been utilized extensively in healthcare to diagnose diseases, predict patient outcomes and optimize treatment plans. In education, ML can analyse students' learning habits, identify behaviours such as procrastination, and offer insights to help prompt interventions in educational settings. In a study by Baylin and Lomibao (2024), K-means clustering and filter-based feature selection were used to uncover patterns in students' procrastination tendencies and then categorized them into four different behavioural categories. The results showed that ML techniques can effectively identify procrastination levels and highlight key factors. This includes perfectionism, time management, and anxiety for maths based on this study. This emphasized that ML has the potential to recognize procrastination-related behaviours and produce useful insights for educators.

Overall, machine learning (ML) offers a strong framework to extract meaningful information from large and complex datasets. At the same time, it also provides chances to understand, predict, and influence behaviours in various contexts.

2.3.2 Statistical and Structural Modelling Approaches

Research on academic procrastination have utilized numerous mathematical and statistical frameworks to measure the behaviour of students. These include models like multi-dimensional Hawkes process and Structural Equation Modeling (SEM). The main advantage of using SEM is its effectiveness in identifying the psychological factors such as anxiety or low self-regulation. These factors often correlate with task delay. Meanwhile, Hawkes process effectively models the timing and self-exciting nature of procrastination events over time. These methods provide a strong basis to understand how procrastination triggers work in a controlled environment.

The study by Jamil et al. (2021) investigated the structural relationship between procrastination and several psychological factors that are relevant to college students. This was done using a cross-sectional design with questionnaires that are self-report and using structural equation modelling (SEM) to test a causal model of procrastination. The sample included 390 students who reported on their procrastination, difficulties in emotional regulation, depression, anxiety,

perfectionism, worry, rumination and experiential avoidance. The results of SEM showed that perfectionism was affected by emotional dysregulation. This factor was not only associated with anxiety and depression but also worry, rumination, and experiential avoidance. All these factors contributed to procrastination. The results showed that worry is the strong predictor since the model fit indicates a good fit with RMSEA of 0.038 and CFI value of 0.985. The study concluded that factors such as perfectionism, negative emotions, emotion dysregulation and cognitive avoidance processes are causally related to procrastination. This suggests that addressing these factors can prove to be effective in reducing procrastination and provide a starting point for future research adopting a transdiagnostic framework.

Another study by Yao et al. (2020) used a multi-dimensional Hawkes process to investigate student procrastination in Massive Open Online Courses (MOOCs). This process models the timing of interactions between students and different learning materials such as assignments, video lectures, and discussion forums. This approach captures external factors like assignment deadlines but also internal responses where the activity sequences are triggered by previous interaction. This allows for a better understanding of patterns of procrastination. Yao et al. (2020) identified two distinct behavioural clusters which are a delay group and a non-delay group. Delay group is characterized by high excitation and bursty activity with long pauses and the non-delay group is marked by lower excitation and evenly distributed activities. The findings showed that the Hawkes model parameters represent the delays more accurately and correlate more strongly with student grades compared to traditional measures. This shows that the timing and interdependence of activities significantly impact the performance of academics. The study gathers that multidimensional Hawkes processes provide a strong framework for modelling procrastination in online learning platforms. Furthermore, this approach offers valuable insights for predicting procrastination and suggesting potential solutions.

Noviana Khoirunnisa et al. (2024) carried out a quantitative study where the authors examined the relationships between future time perspective, academic self-efficacy, academic self control, and academic procrastination. The authors used self-report instruments and structural equation modelling (SEM) on a sample of 517 university students to test mediation effects. The study found that both future time perspective and self-efficacy on academics positively associate with academic self-control. Whereas self-control on academics significantly mediated the effects of the two predictors on academic procrastination. This indicates that students who perceive their future goals more clearly and have stronger beliefs in their

capabilities in academics tend to have greater self-control. This reduces procrastination behaviours. The SEM model showed that it has acceptable fit indices where the values of NFI, CFI, NNFI are all more than 0.90. As a result, it supports the proposed relationships among the variables. The study concluded that one of the effective interventions to target is to enhance self-control in academics as it can also help reduce procrastination by leveraging students' self efficacy and future time orientation.

In summary, these studies provide a comprehensive overview of how SEM and multi dimensional Hawkes process works to identify psychological drivers and temporal patterns of academic procrastination. However, these models often lack the computational flexibility and real-time scalability that is required for an application that is dynamic and gamified. Machine learning (ML) surpasses these traditional approaches by providing prediction that is accurate by using its ability to process high-dimensional, non-linear data. Consequently, using a ML based approach is much better as it allows for a more personalized and adaptive intervention that can respond to a student's behaviour as it happens.

2.3.3 Machine Learning for Behavioural Analysis and Change

Machine learning has increasingly been used to recognize, evaluate, and predict student behaviour especially in educational settings. ML algorithms can find trends that are hidden in student engagement, learning habits and tendencies like procrastination by leveraging large and complicated datasets. These insights make it possible to create personalized interventions to support change in behaviour. These can help educators to understand how students engage with classroom settings. Machine learning is especially great for modelling complex human behaviours as it can handle versatile and nonlinear data, unlike traditional statistical methods. Lately, patterns in academic performance, engagement and motivational behaviours have been successfully identified using a variety of methods from supervised learning such as Random Forest, Support Vector Machines (SVM), and Naive Bayes to unsupervised learning such as clustering.

A study by Mohd Talib et al. (2023) predicted student success comprehensively by integrating both academic and non-academic factors such as learning program outcomes, self-efficacy,

learning style and demographic information. A combination of supervised and unsupervised machine learning techniques was used to classify students into groups of low-, average-, and high- performance. Support Vector Machines (SVM) was used for classification after the K-Means Clustering was so that it will be possible to identify the behavioural patterns of students over the course of a semester. This revealed that changes in self-efficacy and learning style had a significant impact on overall performance, especially during the final weeks. The predictive model was able to identify subtle variations in student behaviour by clustering students according to multiple factors instead of solely relying on final grades which led to increase in classification accuracy. This study proved that machine learning can identify trends in student behaviours which provides educators with useful information to implement early interventions and support techniques and strategies into practice that improve academic results and the overall development of their students.

Similarly to the previous study, a research by Shen and Yuan (2021) explored how data collected from a digital campus can be leveraged to assess and control college student behaviour using machine learning. Their research emphasized how digital solutions can collect comprehensive data about students' daily activities, consumption, and learning. The researchers hoped to gain a thorough understanding of student behaviour that goes beyond conventional management techniques. These techniques often rely on administrators' personal experience and fail to provide tailored guidance. Adaptive K-means clustering algorithm was utilized to accomplish this. This enhances the standard K-means technique by automatically adjusting the number of clusters and using the elbow rule to identify outliers. This method allowed for better classification of student behaviours in terms of consumption, lifestyle, and learning patterns more accurately and effectively. Findings from this study revealed four learning behaviour groups, three lifestyle patterns and five types of consumption behaviours. These insights can help school administrators to improve overall student management, optimize learning methodologies, and provide focused support. The research highlights the potential of machine learning to predict academic success and guide proactive, tailored interventions that promote better study and lifestyle practices.

A study by Yuan et al. (2024) investigated the integration of behavioural analysis with machine learning to evaluate online learning performance more effectively. Their research emphasized traditional predictive models that overlook the variation of students' behaviours. These variations are crucial for understanding engagement and identifying potential areas where

support is needed. The researchers were able to classify students into different behavioural patterns such as driven learners, and students with poor autonomy. This classification was done by using clustering algorithms on online learning data. Random forest models and LightGBM were used to forecast learning outcomes. Crucially, the study also analysed how various characteristics of behaviour contributed to these predictions. This enabled the teachers to identify which behaviours had the most influence on the achievement of the student. This approach accentuates how machine learning may be used not just to predict results but also to identify significant trends in student behaviour. These provide useful information to the educators to customize intervention and successfully assist the learning of the students.

In summary, the studies reviewed mentions that machine learning approaches to predict behaviour in educational settings mainly use supervised, unsupervised, and ensemble learning techniques. Supervised learning algorithms are commonly used to predict academic outcomes and identify at-risk students by analysing labelled behavioural and performance data. Supervised learning algorithms include SVM, Random Forest, Naïve Bayes, and Gradient Boosting models such as LightBGM. Furthermore, unsupervised learning methods are often employed to reveal behavioural patterns that are hidden and divide students into different profiles of engagement or learning. Hybrid methods are also recognized in several studies, where they combine clustering with classification. It resulted in improved prediction accuracy by capturing behavioural diversity before applying predictive models. Together, these algorithms are not only able to predict the outcome but also provide a foundation for personalized and timely interventions in educational systems.

Table 4 below summarizes the key studies that were reviewed in this section where supervised, unsupervised and hybrid approaches were studied for behavioural analysis and prediction in educational settings.

Author (Year)	Educational Context	ML Approach	Algorithms Used	Behavioural Factors Analysed	Key Findings
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Mohd Talib et al. (2023)	Higher education	Hybrid (Supervised + Unsupervised)	K-Means, SVM	Academic performance, self-efficacy, learning style, demographics	Clustering students before classification improved accuracy, changes seen in self-efficacy and learning style significantly influenced performance
Shen and Yuan (2021)	Digital campus behavioural management	Unsupervised Learning	Adaptive K-Means Clustering	Learning behaviour, lifestyle patterns, consumption behaviour	Identified distinct behavioural clusters, enabled tailored student management beyond traditional administrative approaches
Yuan et al. (2024)	Online learning environments	Unsupervised Learning and Ensemble Supervised Techniques	Clustering, Random Forest, LightBGM	Engagement patterns, autonomy, behavioural variation	Behavioural profiling improved outcome prediction and revealed key behavioural contributors to learning success

Table 4: Comparison of Machine Learning Approaches

2.3.4 Predictive Modelling of Procrastination

The previous section 2.3.3 demonstrated how machine learning can be used to identify significant patterns in student behaviour such as student engagement levels, study habits and activity trends. Subsequently, the next step is to translate these insights into actionable predictions. Predictive modelling provides a means of identifying at-risk students before problems become worse. This is because student procrastination is often influenced by minute changes in motivation, learning behaviour, and engagement in tasks. Supervised machine learning models can predict an individual's procrastination risk more accurately by building on the behavioural patterns that were identified by clustering and other techniques that are analytical. Hence, this section 2.3.3 focuses on research that specifically uses predictive algorithms to predict the risk of procrastination.

The study by Öztaş and Akçapınar (2025) is extremely pertinent as it investigates the possibility of accurately predicting students' academic procrastination tendencies based on their online learning behaviours. The researchers worked with 51 students from university over the course of 14 weeks to gather two types of data. The first type is a self-report questionnaire based on nine identifiable assignment-submission behaviour patterns. The second is more than 70,000 behavioural logs from Moodle activities including SCORM modules, forums, quizzes, and assignments. Features were extracted pertaining to the frequency and duration of students' interactions with course materials. Logistic Regression produced the best results with 90% accuracy in differentiating between students procrastinating and those who don't among the seven machine learning algorithms that were tested using 10-fold cross-validation. The most significant predictors were early-week engagement features indicating that students who interact regularly and early with the course materials are significantly less inclined to procrastinate. Overall, the study reveals that behavioural data can accurately predict procrastination.

Another study by Sartana et al. (2025) predicted procrastination behavior and academic performance using Moodle log data of 377 students. The study extracted features related to how frequently and regularly students participated in course activities. Indicators among the features were early engagement, delayed task initiation, and total interaction intensity. Cluster

groups of three were identified by applying K-means clustering using the elbow method which are labelled as (1) low procrastination / high performance, (2) high procrastination / poor performance, and (3) medium procrastination / average performance. This was found after standardizing the data using Z-score scaling and eliminating outliers. Seven machine learning regressors were used to build the predictive models and evaluated them using the 10- and 15-fold cross validation. The machine learning regressors includes Decision Tree, Random Forest (RF), Gradient Boosting (GB), Multilayer Perceptron (MP), Support Vector Machine (SVM), K-Nearest Neighbours (K-NN), and LightGBM (LGBM). The findings showed that Random Forest performed the best with an R^2 of 0.812, a mean absolute error (MAE) of 6.248, and a root of mean square error (RMSE) of 8.456. It also revealed a strong negative correlation of 0.7777 between delayed tasks and academic outcomes. The findings emphasized that when a student delayed completing their task, lower scores were achieved. This highlights the harmful effect of procrastination. Overall, the study showed that the risk of procrastination may be correctly predicted using behavioural engagement patterns, with ensemble-based models yielding the best performance.

Abidi et al. (2020) conducted a study to investigate how machine learning can predict procrastination by analysing students' activity patterns within an online learning management system. Rather than relying on single-event variables, the researchers identified behavioural characteristics such as when students began their assignments, how often they access course materials, and how much they delay starting required activities. The characteristics reflects the students' consistency, timeliness and frequency of engaging with course works. Ensemble machine learning models such as Logistic Regression, Decision Tree, Gradient Boosting, and Forest were used to identify and predict procrastinating students. Kolmogorov-Smirnov (KS) statistics was used as a selection criteria for classes and Average Squared Error (ASE) was used for validating data to find the best model. Gradient Boosting (GB) model that was autotuned outperformed all the other models with a KS value of 91.77% and an ASE of 2.79. The default GB model remained highly effective with a KS value of 90.74% and an ASE value of 3.66. These results show that GB techniques effectively identify students at risk of procrastinating based on their online learning behaviours, especially when optimized.

Since online learning has expanded significantly during the COVID-19 pandemic, the study by Memon et al. (2020) investigated how procrastination can be pinpointed and predicted in Virtual Learning Environments (VLEs). The researchers used the Open University Learning

Analytics Dataset (OULAD) that contains 32593-student profiles enrolled for courses in STEM in VLE. This dataset is used to extract the key behavioural and demographic features such as number of prior attempts made by students for the same module, final results, day of the assessment submitted, total number of credits studies and the students' assessment score to build a comprehensive dataset for procrastination. At first, clustering was intended to be used to find patterns in student behaviour. However, visualizations revealed that unsupervised approaches produced linearly separable groups and were unable to capture the complex, non linear correlations present in tendencies of procrastination. Annotation by experts produced labels that more accurately reflected actual procrastinating tendencies considering a combination of students' involvement, performance, and submission habits. Two supervised classifiers, which are Logistic Regression and Artificial Neural Network - Multilayer Perceptron (ANN-MLP), were trained and optimized using the expert-annotated dataset. The ANN-MLP using Adam optimizer with a batch of 100 obtained 83.5% training accuracy and 82.9% testing accuracy whereas Logistic Regression with a similar performance obtained 82% for training accuracy and 81.9% for testing accuracy. These results proved that the most effective method to predict procrastination is expert-guided annotation in conjunction with a well-tuned supervised model.

Table 5 below summarizes recent studies mentioned in this subsection on predictive modelling for procrastination. It highlights the features used, machine learning models applied, evaluation metrics, and the approaches that perform the best.

Authors (Year)	Features Focused	Models Used	Performance	Best Results
Öztaş and Akçapın ar (2025)	Digital traces from LMS such as course-activity logs, assignment submission patterns, timing, and frequency of access	Support Vector Machines, Decision Tree, Logistic Regression, Naive Bayes, Artificial Neural Networks, Random	Comparing Models: classification accuracy, F-score, precision,	Logistic Regression predicted with best accuracy of 0.90.

		Forest, and K-Nearest Neighbour	and recall, Test Model Performance: 10-fold cross validation method	
Abidi et al. (2020)	Interaction data from an ITS (e-learning platform) by capturing log data usage, time spent, intercation features	Logistic Regression, Tree Ensembles (Decision Trees, Gradient Boosting, Forest Model)	Kolmogrov-Smirnov (KS) statistic, Average Squared Error (ASE)	Gradient Boosting is the best model with KS and ASE value of 91.77% and 2.79% respectively.
Sartana et al. (2025)	Moodle activity logs (features were derived from timestamps of task access, submission, and engagement duration)	Decision Tree regressor (DT), Random Forest Regressor (RF), Gradient Boosting Regressor (GB), Multilayer Perception Regressor (MP), Support Vector Regressor (SV, K-Nearest Neighbours Regressor (K-NN). LightGBM Regressor (LGBM)	R^2 , MAE, RMSE	Random Forest (RF) achieved the highest accuracy, with an R^2 score of 0.812, MAE of 6.248, and RMSE of 8.456
Memon et al. (2020)	Features focused was student_Id, Highest_Education, studied_credits,	Logistic Regression (LR) and Artificial Neural Networks (ANN)	Accuracy on training and testing	ANN training accuracy has the best

	Number_of_previous_attempts,final_result, disability, date_submitted, and score			model with the value of 83.3%
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Table 5: Summary of Predictive Modelling for Procrastination

2.3.5 Limitations and Challenges

One of the limitations seen in the study from the previous section is that since the sample size is small with only 51 students, it restricts the extent to which the results can be applied to larger or more diverse student populations (Öztaş & Akçapınar, 2025). Even though it is common and typical for a single class to have a small sample size and is widespread in educational research. Another limitation by this study is the data was gathered during the COVID-19 pandemic. This was when students faced numerous troubles adapting to new or distant learning environments. It is challenging to generalize these findings to normal, post-pandemic conditions. This is because factors such as limited access to educational materials, increased stress, and declining motivation may have influenced their procrastinating habits. Overall, future studies should benefit from using larger and more diverse samples to improve the generalizability and the relevance of the results across various educational settings.

There were several limitations seen in the study by Sartana et al. (2025) although it presents a structured and data-driven approach. Firstly, the dataset comes from a single private university in Indonesia, hence it is limited in its applicability to other institutions, learning cultures, or structure of courses. Since the study is relying on Moodle activity logs, only the observable online behaviours were recorded. Thus, unable to take into consideration factors including psychology, motivation and other external factors that affect procrastination. Assuming that all students regularly engage with Moodle is another limitation. This is because they might be working offline, reviewing content elsewhere, or even complete their tasks without regularly using the platform, which would reduce the accuracy of the behavioural indicators. Furthermore, using heavily engineered time-based features runs the risk of oversimplifying procrastination leading to presenting procrastination as a series of timestamps rather than a

complicated behaviour influenced by environment. Since there was an absence of external validation or testing on other datasets, overfitting is a risk even though the prediction model performs well. Finally, since how the prediction can be incorporated into educational interventions were not investigated, there is a gap between algorithm output and real-world application.

The study by Abidi et al. (2020) acknowledges several limitations could restrict the applicability of its conclusion. It still depended on a limited set of behavioural variables although many other relevant indicators such as delaying of tasks, levels of interaction, or time spent engaging with learning materials could have been included to build a more comprehensive model. Although autotuned algorithms introduce constraints since choosing hyperparameters automatically does not ensure the finding of the truly optimal model, even though it offers a starting point for further refinement. The study could have improved prediction accuracy if mental and health problems of the students were taken into consideration. Future work would probably increase the model's robustness and provide a more comprehensive knowledge of procrastination patterns by including a wider range of behavioural, contextual, and wellbeing-related characteristics.

One of the other limitations seen by the study by Memon et al. (2020) relied solely on a single secondary dataset (OULAD). This dataset may not fully represent student behaviour across various institutions, cultures, or online learning platforms. Detailed guidelines by the expert driven label are still subjective and based on assumptions that might not hold true for all students. Hence, it might introduce bias into the classification of procrastination. The clustering method and visual comparisons also highlight that many of the patterns from the data are unclear or overlapping. Thus, making it challenging to tell procrastinators from non procrastinators with confidence.

Table 6 below presents the focus of the research, methodologies, and limitations that were identified when discussing the studies in this section.

Author (Year)	Research Focus	Key Findings & Methodology	Identified Limitations
Öztaş and Akçapınar (2025)	Prediction of academic procrastination tendencies	Used behavioural trajectories in online learning traces to classify students	Small sample size (n = 51) Data collected during COVID-19 may lack external validity due to abnormal learning patterns
Sartana et al. (2025)	LMS-based prediction using machine learning	Random Forest and L-Means clustering applied to Moodle logs	Single institution context Relies only on observable logs Lacks translation of algorithmic results into active interventions
Abidi et al. (2020)	Behavioural data-driven performance prediction	Applied autotuned algorithms to behavioural variables and procrastination levels	Limited variable set Excluded task delay patterns and psychological factors that influence behaviour
Memon et al. (2020)	Learning analytics-based behaviour analysis	Utilized secondary datasets to analyse student interaction and procrastination.	Relies solely on a single secondary dataset (OULAD) which restricts generalizability across different systems of educations

Table 6: Summary of Research Focus, Methodology and Limitations

2.4 Gamification and Its Role in Behaviour Change

2.4.1 Concept of Gamification

Gamification has become a popular strategy to improve results of learning, enhance motivation and increase participation in various fields, especially in education. Gamification, which derives from principles of game design, is the process of introducing game-like elements such as points, badges, levels, and challenges. These elements are incorporated into a non-gaming environment to promote engagement and persistence. Gamification as a method has become more common to address issues like low motivation, procrastination, and inconsistent engagement in the learning environment. This is due to the growth of digital technology and online platforms.

Hebebcı and Alan (2021) defines gamification as the application of game design features in a non-gaming environment. This was first proposed by Nick Pelling in 2002. In 2008, the idea gained more attention, and it became even more popular after the 2010 Design, Innovate, Communicate, Entertain conference. Applications like digital badges, and points systems sparked interest in gamification marketing, trade and advertising. Gamification has gradually expanded into educational environments. For example, platforms like ClassDojo and Khan Academy use virtual characters, badges, and progress tracking to improve student involvement and track their performance. This demonstrates how gamification uses game mechanics to improve engagement and motivation in both business and educational contexts.

However, Staller and Koerner (2021) argue that gamification lacks a widely agreed definition and they emphasize the differences and similarities between play and games. They note that educational gamification involves both experiential factors like joy, engagement, and motivation as well as structural elements like rules and mechanics. The authors encouraged educators to take a reflective approach that considers context-specific factors, the visibility of gamified aspects, and students' acceptance in their "non-definition" of gamification. From this perspective, gamification is a flexible and adaptable tool that can be utilized to enhance engagement and persistence in learning. Its effectiveness not only depends on game mechanics but also on how the students view and interact with the gamified environment.

Gamification in eLearning systems should follow a clear framework based on instructional design principles (Yamani, 2021). The study identifies key components of gamification such as points, rewards, feedback loops, and progression mechanics. It discusses how their organized use might improve learner motivation and participation. Gamification can create a sense of achievement, encourage active participation, and provide ongoing rewards by integrating these components into course design. The framework also emphasizes the importance of user-centered, game-centered, and technology-centered methods to ensure that gamification aligns with the learners' aims and goals. This structured approach can improve information retention, promote skill development, and create an environment that minimizes negative habits like procrastination.

In summary, gamification combines game-based tactics with behavioural and educational goals to increase motivation, engagement and learning effectiveness. It also offers a flexible framework to influence the learners' behaviour by using psychological concepts, structured game mechanics, and historical roots. According to the research, gamification has strong potential to address academic issues like procrastination. Gamified interventions encourage timely engagement and consistent study habits by offering rewards, progress tracking, and enabling interactive learning.

2.4.2 Gamification for Procrastination

Recently, gamification has gained more attention because it has become a potential strategy to control and overcome procrastination. Gamification seeks to increase initiation of tasks and sustained effort by leveraging features such as points, rewards, and many more game elements. This method is appealing as it may have the ability to transform effortful or difficult tasks into experiences that are more organized and fulfilling. Researchers have explored how different types of game elements can strengthen commitment to tasks that are scheduled, promote regular productivity behaviour, and assist self-regulation as digital platforms continue to develop. Therefore, this section will investigate some of the studies that examines how gamification has been created and used to lower procrastination in academic settings.

Pereira et al. (2025) carried out a thorough literature analysis to investigate how gamification has been applied to procrastination and the true efficacy of these tactics. Their research synthesizes results from other numerous studies where they demonstrate that although gamification is often used to increase motivation, elements like points, badges, leaderboards, and progress bars make the results inconsistent. These game mechanics might help make activities more engaging and manageable particularly in academic settings. However, their efficacy mostly depends on how the elements are created and whether they activate intrinsic motivation instead of only depending on rewards from external sources. This study also emphasizes that many interventions have strong focus on extrinsic motivators, which is great for improvements for a short-term but not in the long-term as it may lose its effectiveness. Crucially, the authors stress that to create gamification systems that result in significant and long-lasting behavioural change, personalization, thoughtful design, directly addressing the underlying cause of procrastination are essential. Underlying cause of procrastination refers to the low motivation, poor emotional regulation and time management.

Another study by Amit et al. (2021) presents a useful, gamified framework intended to decrease procrastination by incorporating several motivational techniques into a mobile app called ProScore. The system created a structured yet engaging environment that encouraged starting tasks and maintaining focus by combining a modified Pomodoro timer with game elements like points, leaderboards, and live activity streams. By dividing work into manageable intervals, the Pomodoro approach creates a sense of urgency, while the gamification layer reinforces productive behaviour by giving points to the users for finishing sessions and penalizing them 26 when they stop doing the activities in the middle. The leaderboard increases peer influences and uses social comparison to combat procrastination by motivating the users to improve their ranks and match the productivity levels of their peers. The system also includes personalized task-duration estimates, which helps users determine realistic time requirements based on past data. This aims to reduce planning uncertainty, which is a common cause of procrastination. The framework from this study shows that combining time-management strategies with gamification, incentives, and social accountability can lead to an engaging, user-friendly intervention that helps individuals overcome procrastination and maintain consistent productivity.

A recent research by Sun (2024) studied the app called Forest: Focus for Productivity to explore how gamification affects academic procrastination. The study involved surveys and interviews

with five active users of the app. It investigates how gamification elements including rewards, punishments, personalization and collaboration affect the behaviour of procrastinating. The study found that the gamification significantly improves the user experience and supports the app's primary intervention strategies rather than directly enhancing motivation. Although the cooperation and punishment helped the users stay focused and restrain impulsivity, rewards and personalization improved user satisfaction and the sense of achievement. Participants noted that the core intervention tactics play a key role in the app's effectiveness. This is because gamification elevates the significance and engagement of the experience. In summary, gamification effectively boosts user engagement and offer meaningful support in managing academic procrastination. This is feasible by offering rewards, collaboration, customization, and other interactive elements. Game elements mainly enhance the effectiveness of intervention measures rather than directly addressing procrastination. However, its impact varies based on motivations and characteristics of each individual user. Despite some limitations, the studies mentioned in this section showed the potential of gamified application to create more engaging and dedicated learning experiences. This suggests that careful integration of gamification can be a useful tool for addressing procrastination.

Table 7 below reviews the gamification approaches in relation to procrastination and productivity. It details the specific game elements implemented and their effects on engagement and motivation.

Author (Year)	Context	Gamification Elements Used	Key Findings
Pereira et al. (2025)	Systematic review of gamification in procrastination /	Common elements aggregated across many studies: Points, Badges, Leaderboards, Progress Bars, Challenges, Avatars, Narrative, Friend Interaction / Social, Group Competition, Mini-games,	Points/Badges/Leaderboards/Progress Bars ("BPL + Progress") are most used. Social, narrative, and collaborative elements are much less frequent. suggests

	productivity contexts	Levels/Phases, Visual representations, Resource Management, Chatbots	over-reliance on extrinsic motivators.
Amit et al. (2021)	Online learning environment (assessing which gamification components improve engagement)	Standard gamification components: likely including progress tracking, points/badges, maybe feedback (details as typical gamification toolkit.)	Showed that certain gamification components correlated with higher engagement in online learning. Supporting that well-chosen mechanics can enhance user interaction.
Sun (2024)	Case study of productivity app targeting academic procrastination	“Forest”-style gamification: reward/penalty for focus sessions (gamified timer), virtual “planting” metaphor, visual forest, progress tracking, collaboration/ co-planting, unlockable content / customization, achievement / reward, social/ group accountability	Gamification helps with focus and engagement, but effectiveness depends on user motivation. Gamification alone not a guaranteed fix.

Table 7: Summary of Gamification Approaches

2.4.3 Limitations of the Gamified Approaches

Even though Pereira et al. (2025) provide valuable information about the connection between gamification and procrastination, the study also notes several limitations that should be

acknowledged. Since the study is a systematic literature review, the findings solely depend on the quality, consistency, and thoroughness of the included studies. Many of these studies may vary in design, sample size, context and measurement of procrastination. This makes it challenging to draw firm conclusions about which gamification techniques are successful or under what condition they work the best. The review also points out that a significant number of existing studies focus on short-term intervention. This means that long-term behavioural change and the sustainability of gamification effects are still unknown.

Next, a limitation of the ProScore study is that the framework may limit its effectiveness since individual and environmental factors are not fully accounted for (Amit et al., 2021). Although the Pomodoro technique, peer pressure, gamification, and incentives are all integrated in the application, the responses by the users may vary greatly based on their levels of motivation, personality attributes, and past time management experience. Furthermore, the reliance on user-defined task durations and arithmetic mean calculation to estimate time may limit the accuracy of task planning, especially for complicated or highlight variable tasks. Since historical data is required to enhance time estimates, the framework also faces issues for new users, which could reduce initial usefulness.

As for the study Sun (2024), there are also a few limitations that need to be acknowledged. Firstly, the study only consists of a very small sample size of five participants making it challenging to draw generalization conclusions about the broader user population. Purposive and convenience sampling reduces representativeness because participants with very high engagement or extreme usage habits may skew the results. Additionally, the study mostly relied on self-reported data from surveys and interviews, which can introduce biases like individuals giving socially acceptable answers or errors in memory recall. As the study only focuses on the Forest app, the conclusion might not accurately reflect how gamification works in other productivity apps where there are different user bases and game designs. Finally, even though the study effectively investigated user experiences and opinions, the sequencing of survey and interview analyses may have been enhanced to find and explore interesting patterns in a more systematic manner among the participants. These limitations imply that the results provide preliminary insights rather than offering conclusive proof of the efficacy of gamification in lowering academic procrastination.

Table 8 below summarizes the key limitations that was identified by the studies reviewed in this section on examining gamified approaches to address academic procrastination.

Author (Year)	Context	Key Limitations Identified
Pereira et al. (2025)	Systematic literature review on gamification and academic procrastination	Studies included varied widely in research design, sample size, educational context, and methods used to measure procrastination. Most studies focus on short-term outcomes on interventions.
Amit et al. (2021)	Gamified productivity and time management application	Experience such as motivation, personality traits, and prior time management not considered as everyone are different. Task duration estimates rely on user input and simple averaging methods, thus reducing accuracy for complex or variable tasks. The study uses historical data which limits effectiveness for new users.
Sun (2024)	Qualitative study of the Forest productivity app	Uses a small sample size of 5 participants. Purposive and convenience sampling limits representativeness. Data relies heavily on self-reported surveys and interviews that can lead to introducing bias. The focus on a single app restricts generalisability to other gamified systems.

Table 8: Key Limitations Identified in Prior Studies on Gamification-Based Procrastination and Productivity Interventions

2.5 AI-Based Hierarchical Task Decomposition and Planning

The decomposition of complex tasks into manageable subtasks is one of the long-standing problems in artificial intelligence (AI). Notably, this problem is within planning and decision-support systems. Decomposing tasks hierarchically enables the system to transform high-level, abstract objectives into sequences of executable actions that are structured. This enhances interpretability and feasibility in complex environments. Recent research has advanced in this area by combining traditional hierarchical planning techniques with optimization methods and adaptive AI reasoning. This highlights how the relevance of such approaches for application that need intelligent task breakdowns and adaptive support.

A study by Elkawkagy et al. (2025) proposed using Hierarchical Task Network (HTN) and combining it with Ant Colony Optimization (ACO) to create an integrated framework during the refinement phase. Candidate task decompositions are evaluated and refined using swarm-intelligence principles using the ACO-HTN approach. This allows the planners to navigate large solution spaces more effectively. Both quality of the solutions and computational efficiency provide improvements in their empirical results compared to HTN planners. This study emphasizes the value of rule-based hierarchical decomposition enhanced with adaptive optimization mechanisms, especially for complex tasks. Naive decomposition may lead to inefficient or suboptimal plans.

The next study by Kienle et al. (2025) introduces LODGE which is a framework that jointly performs hierarchical task planning and domain model learning. This framework is supported by grounded execution and simulation. LODGE learns hierarchical abstractions that connect low-level actions to higher-level tasks instead of relying solely on status, hand-crafted planning rules. Meanwhile, LODGE also validates their feasibility through execution feedback. If there are long horizon planning problems, then this hierarchical structure is very effective whereas flat planning approaches often fail. This study demonstrates that combining structured task decomposition with consistency checks across planning levels significantly improves planning success rates. Thus, reinforcing the importance of hierarchical representations for complex task management systems.

ADAPT framework which is an as-needed decomposition framework was proposed by Prasad et al. (2024) which dynamically breaks down tasks based on execution difficulty. ADAPT

recursively decomposes subtasks only when execution fails thus allowing the system to adjust decomposition depth according to the complexity of the task and the capability of the system. This approach differs from fixed hierarchical planners, which decompose tasks upfront. This adaptive decomposition strategy has proven to outperform traditional planning and executing tasks across multiple benchmark environments. The findings from this study suggest that effective task breakdown does not need exhaustive upfront planning. Instead, it needs context-aware decomposition that responds to execution constraints.

In summary, the studies mentioned demonstrate that hierarchical task decomposition is a strong method for managing complex tasks in intelligent systems. Approaches like optimized rule-based planning, learned hierarchical domain models or adaptive as-needed decomposition strategies, highlight the importance of structured task breakdown in supporting execution, decision-making, and user guidance. These insights gained from these studies provide a strong theoretical foundation to integrate AI-driven task decomposition mechanisms into student-focused app. This is where the large academic tasks must be broken down into smaller, actionable subtasks to reduce cognitive overload and procrastination risk.

The table 9 below presents representative AI-based hierarchical task decomposition approaches based on the studies explored in this section and their relevance to managing tasks that are complex.

Author (Year)	Proposed Approach	Key Contributions
Elkawagy et al. (2025)	Integration of Hierarchical Task Network planning with Ant Colony Optimization	The framework refines task decomposition using swarm intelligence principles as it improves solution quality and computational efficiency compared to traditional HTN planners.
Kienle et al. (2025)	LODGE framework combining hierarchical	The framework learns hierarchical abstractions from execution and simulation rather than

	task planning and domain model learning	relying solely on manually defined rules as it improves planning success for long horizon tasks.
Prasad et al. (2024)	ADAPT framework with as-needed task decomposition	Tasks are decomposed dynamically only when execution fails. This allows decomposition depth to adjust based on the complexity of the task and capability of the system. This approach outperforms fixed decomposition strategies.

Table 9: AI-Based Hierarchical Task Decomposition Approaches

3.0 Methodology

This chapter discusses the methodology that was adopted for the development of the mobile application called Nudgr. There are six main phases to develop this application which includes requirement analysis and planning, development machine learning model, development of application, integration of system, system testing, and pilot testing with usability evaluation. These stages make sure that the application built is technically working and is accomplishing the users' requirements. This application features prediction of procrastination risk, AI planning of tasks, gamification, reminders and nudges notification, and progress analytics page. These are the intentions behind these features: to aid in the management of the procrastination shown by the students in academics.

3.1 Overall Research Methodology

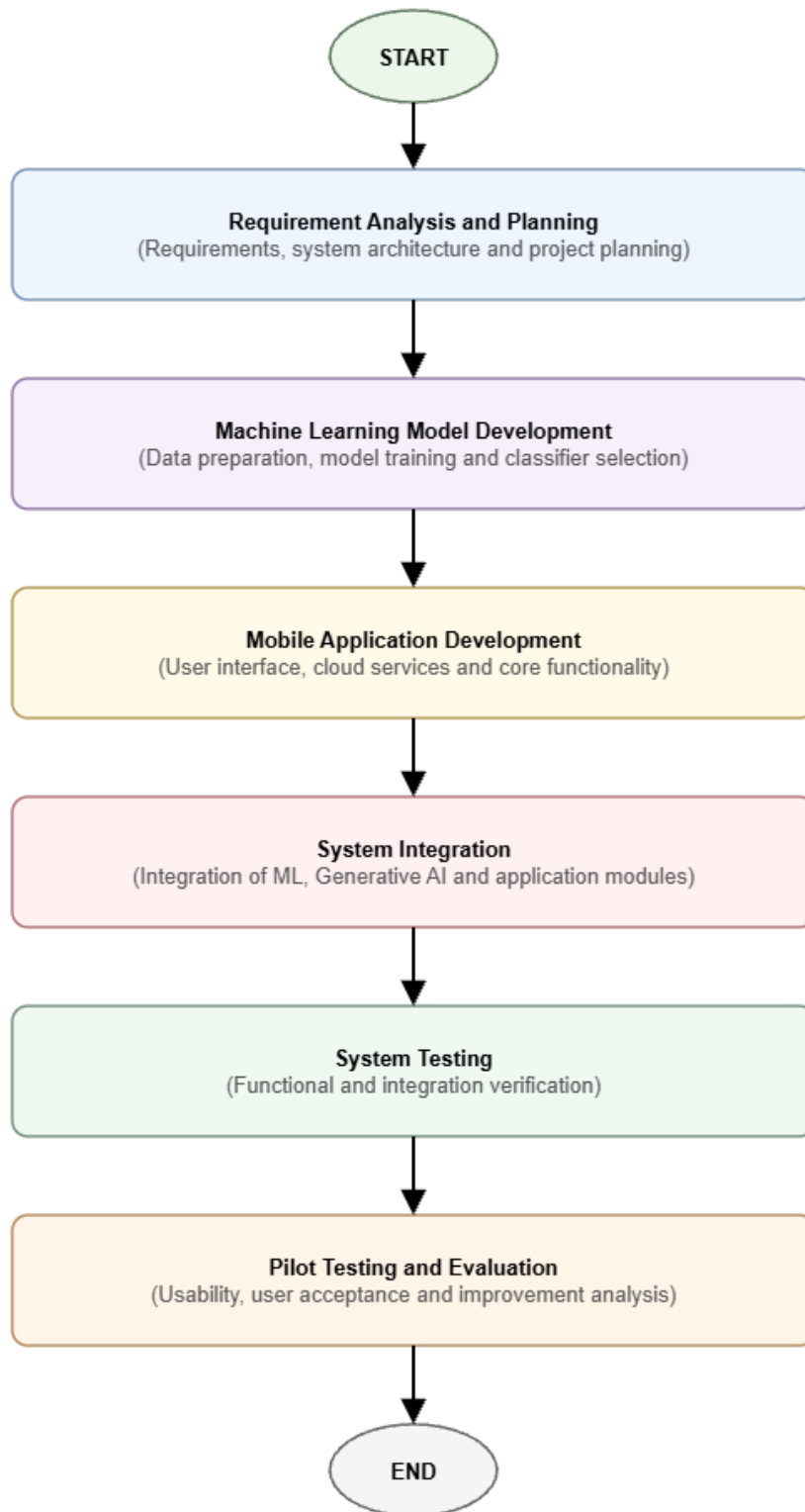


Figure 1: Overall Research Methodology Flowchart

This study adopted a structured software engineering and machine learning methodology to design, develop, integrate, and evaluate Nudgr, an intelligent mobile application for managing academic procrastination among university students. The methodology combined supervised machine learning, mobile application development, cloud computing, Generative Artificial Intelligence (Generative AI), and user-centred evaluation within a sequential development framework. This approach ensured that both the predictive intelligence and application functionalities were systematically designed, implemented, tested, and evaluated before deployment.

The overall methodology consisted of six sequential phases: Requirement Analysis and Planning, Machine Learning Model Development, Mobile Application Development, System Integration, System Testing, and Pilot Testing and Evaluation, as illustrated in Figure 1. Each phase built upon the outputs of the previous phase to ensure that the developed system met both the functional requirements of the application and the research objectives of this study.

Requirement Analysis and Planning formed the foundation of the project by identifying both the functional and non-functional requirements of the proposed application. Existing literature on academic procrastination, behavioural intervention strategies, machine learning applications in education, and gamification techniques was reviewed to determine the system requirements and establish the overall project objectives. During this phase, the system architecture, application modules, database structure, and overall development roadmap were defined to guide the subsequent implementation stages.

The second phase focused on the Machine Learning Model Development, where a supervised learning approach was adopted to construct a predictive model capable of identifying students who were at risk of academic procrastination. A secondary questionnaire dataset derived from Rusdi et al. (2020) was pre-processed through data cleaning, feature engineering, and feature selection before multiple classification algorithms were trained and evaluated. Several machine learning models were compared using recognised classification performance metrics, including accuracy, precision, recall, F1-score, and confusion matrices. Since the primary objective of the application was to identify students requiring early intervention, particular emphasis was placed on recall during model selection to minimise the likelihood of high-risk students being incorrectly classified as low risk. Based on the evaluation results, the Random Forest classifier

was selected as the final prediction model and exported for integration into the mobile application.

Following the development of the prediction model, the Mobile Application Development phase focused on implementing the proposed Nudgr application using Flutter as the cross-platform development framework and Firebase as the cloud-based backend service. Flutter was selected because it enables a single codebase to support both Android and iOS platforms, thereby improving maintainability and reducing development effort. Firebase Authentication, Cloud Firestore, Cloud Functions, and Firebase Cloud Messaging were utilised to manage user authentication, cloud data storage, backend processing, and push notifications respectively. During this phase, the user interface, task management module, procrastination assessment, analytics dashboard, notification system, and gamification components were implemented.

The fourth phase involved System Integration, where the independently developed software components were combined into a unified intelligent system. The trained Random Forest model was integrated into the application's backend to generate personalised procrastination risk predictions based on users' questionnaire responses. In addition, Google's Gemini API was incorporated to provide AI-assisted task decomposition by automatically generating subtasks and suggesting internal deadlines from user-defined academic tasks. Other application modules, including personalised reminder notifications, motivational nudges, experience point calculation, virtual garden progression, leaderboard functionality, and productivity analytics, were integrated to provide a comprehensive user experience.

After completing the system implementation, System Testing was conducted to verify the correctness and reliability of the developed application. Functional testing was performed to ensure that each module operated according to its intended requirements, while integration testing evaluated the interaction between the mobile application, Firebase services, the machine learning prediction model, and the Gemini API. Testing also verified data synchronisation, authentication, cloud communication, notification delivery, and AI-assisted task generation to ensure that the integrated system functioned correctly before user evaluation.

Finally, Pilot Testing and Evaluation were conducted to assess the usability and practicality of the completed application from the perspective of its intended users. University students were invited to use the application by completing representative tasks, including account

registration, procrastination assessment, task creation, AI-assisted task planning, task completion, and interaction with the gamification and analytics features. User feedback was collected using a structured evaluation questionnaire together with the System Usability Scale (SUS). Quantitative and qualitative analyses of the collected responses were subsequently performed to evaluate the application's usability, user satisfaction, perceived usefulness, and areas requiring further improvement.

Overall, the adopted methodology provided a comprehensive framework that combined machine learning, software engineering, cloud computing, and user-centred evaluation to develop an intelligent mobile application capable of delivering personalised interventions for managing academic procrastination.

3.2 Requirement Analysis and Planning

The first phase of this study involved requirement analysis and project planning to establish a clear foundation for the design and development of the proposed Nudgr mobile application. This phase aimed to identify the functional requirements, non-functional requirements, system architecture, development technologies, and overall project scope before implementation began. A structured planning process was adopted to ensure that the developed system addressed both the behavioral challenges associated with academic procrastination and the technical requirements of an intelligent mobile application.

The requirement analysis was conducted by reviewing existing literature on academic procrastination, machine learning applications in education, gamification techniques, behavioral change support systems, and intelligent productivity applications. In addition, several existing task management and productivity applications were analyzed to identify commonly implemented features, limitations, and opportunities for improvement. Findings from the literature review presented in Chapter 2 were subsequently translated into system requirements that guided the overall application design.

The proposed system was designed to achieve three primary objectives. First, the application should be capable of identifying students who are at risk of academic procrastination using a

machine learning prediction model. Second, the application should assist students in managing academic tasks through AI-assisted task planning and personalized interventions. Third, the application should encourage long-term engagement through gamification techniques that promote self-regulated learning behaviors.

Based on these objectives, both functional and non-functional requirements were identified. Functional requirements describe the core services that the application must provide, whereas non-functional requirements define the quality attributes and operational characteristics that the system must satisfy.

3.2.1 Functional Requirements

The functional requirements specify the features that enable users to interact with the application and receive personalized academic support. The proposed application was designed to provide the following core functionalities:

- User registration and secure authentication.
- User login and profile management.
- Completion of the initial procrastination assessment questionnaire.
- Machine learning-based prediction of procrastination risk.
- AI-assisted task decomposition using Google's Gemini API.
- Task and subtask management.
- Personalized reminder notifications and motivational nudges.
- Experience point (XP) calculation and streak tracking.
- Virtual garden gamification system.
- Productivity analytics dashboard.
- Leaderboard and ranking system.
- Cloud-based synchronization of user data through Firebase.

These functional requirements collectively support the primary objective of assisting students in organising academic tasks while delivering personalized behavioral interventions based on their predicted procrastination risk.

3.2.2 Non-Functional Requirements

In addition to application functionality, several non-functional requirements were considered throughout the system design to ensure that the developed application remained reliable, usable, secure, and scalable.

Usability

The application should provide an intuitive and user-friendly interface that enables students to perform common tasks with minimal learning effort. The interface was developed using Flutter's Material Design components to ensure consistency across different mobile platforms. Usability was subsequently evaluated through pilot testing using the System Usability Scale (SUS).

Performance

Prediction results, task management operations, and cloud synchronization should be completed within an acceptable response time to provide a smooth user experience. Efficient communication between the mobile application, Firebase backend, and integrated AI services was therefore considered throughout the implementation.

Security

User authentication and access control were implemented using Firebase Authentication. Sensitive user credentials were managed by Firebase rather than being stored locally within the application, reducing security risks associated with credential management.

Privacy

The application stores users' assessment responses, academic tasks, and productivity information solely for providing personalized support. Personal data collected by the application were limited to information required for authentication and application functionality.

Scalability

A cloud-based architecture was adopted to allow future expansion of the application without requiring substantial architectural modifications. Firebase Cloud Firestore supports real-time synchronization while enabling additional users and application features to be incorporated as the system grows.

Maintainability

Flutter was selected as the development framework because a single codebase can support both Android and iOS platforms. Furthermore, the modular architecture adopted throughout the implementation allows individual system components, such as the machine learning model or AI service, to be updated independently without significantly affecting other application modules.

3.2.3 Technology Selection

Several modern computing technologies were selected to satisfy the identified system requirements.

Flutter was selected as the mobile application development framework because it enables cross-platform deployment using a single codebase, thereby reducing development effort and improving maintainability. Firebase was adopted as the cloud backend to provide secure user authentication, real-time cloud database services, push notifications, and serverless backend support. Google's Gemini API was incorporated to generate AI-assisted subtasks and suggested internal deadlines through natural language processing capabilities.

For the intelligent prediction component, supervised machine learning techniques were adopted to classify students according to their academic procrastination risk. Multiple classification algorithms were evaluated during the model development phase before selecting the Random Forest classifier for deployment based on its overall predictive performance and suitability for the application.

The technologies selected for this project collectively provide an integrated platform capable of supporting personalized prediction, intelligent task planning, cloud-based data management, and behavioral intervention within a single mobile application.

3.2.4 System Architecture Planning

The overall system architecture was designed using a modular approach to improve maintainability, scalability, and ease of integration. The Flutter mobile application serves as the presentation layer responsible for user interaction and business logic, while Firebase provides backend services including authentication, cloud database management, notification delivery, and data synchronization.

The Random Forest prediction model operates as the intelligent decision-support component by classifying users according to their procrastination risk based on questionnaire responses. Meanwhile, Google's Gemini API provides AI-assisted task planning by generating subtasks and internal deadlines from user-defined academic tasks. Communication between these components enables the application to deliver personalized interventions while maintaining a clear separation between user interface, cloud services, artificial intelligence, and machine learning components.

The planned system architecture formed the foundation for the implementation phases described in the following sections.

3.3 Machine Learning Model Development

This phase focused on developing a supervised machine learning model capable of predicting the academic procrastination risk of university students. The objective was to construct a reliable classification model that could accurately distinguish between students with low and high procrastination risk before integrating the model into the Nudgr mobile application. A structured machine learning pipeline was adopted to ensure that the resulting model was

accurate, reliable, interpretable, and suitable for deployment within a real-world mobile application.

The overall machine learning pipeline consisted of six sequential stages, namely data preprocessing, exploratory data analysis, feature engineering, model development, model training and evaluation, feature importance analysis and model deployment, as illustrated in Figure 2. Each stage was designed to progressively improve the quality of the dataset, optimize the prediction performance of the model, and ensure that the final classifier could be effectively integrated into the application.

Unlike conventional machine learning studies that focus solely on achieving the highest prediction accuracy, this study emphasized the practical deployment of the prediction model within an intelligent mobile application. Consequently, both predictive performance and model interpretability were considered throughout the development process to ensure that the selected model could support personalized behavioral interventions while maintaining acceptable computational efficiency.

The final machine learning pipeline is shown in Figure 2 below.

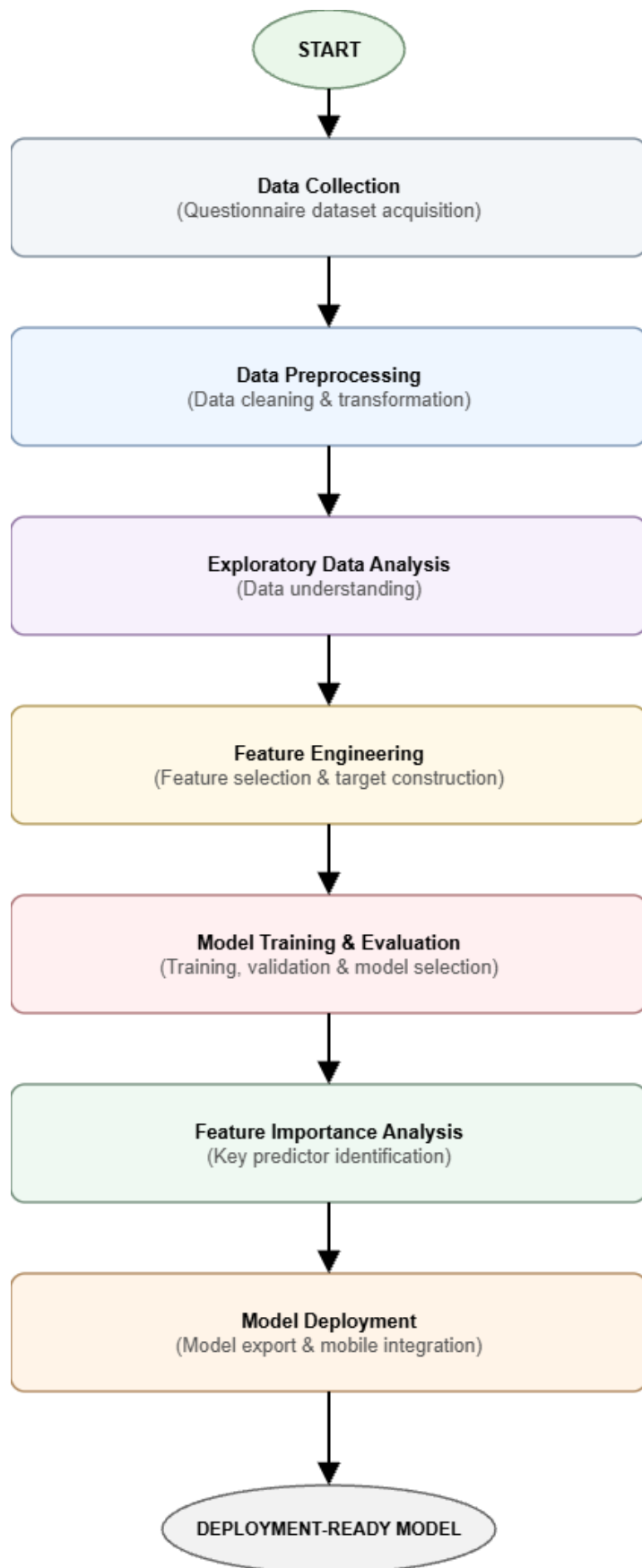


Figure 2: Machine Learning Pipeline

3.3.1 Data Collection

The data used in this research are from Rusdi et. al. (2020) who obtained the academic procrastination data of undergraduate students from several universities in Makassar, South Sulawesi, Indonesia. A structured questionnaire based on the Procrastination Assessment Scale-Student (PASS) was to collect information from 586 students where 310 of the students are males and the rest of 276 students are females. The five-point Likert scale ranging from strongly disagree to strongly agree was used to facilitate quantitative measurement of the procrastination-related behaviours and the reasons for them.

The dataset is selected because it directly collects the behaviours related to academic procrastination and mental factors that relate to academic procrastination. This makes it a good choice to model procrastination risk. This dataset is relevant in the context of the Nudgr mobile application because it is specifically about procrastination behaviour in an academic setting, rather than more generic datasets that are geared towards productivity or education.

This dataset consists of two sections. The first one is dedicated to academic procrastination, and the second one assesses two factors which are Fear of Failure and Task Aversiveness. The first one is Fear of Failure which is focused on anxiety about meeting the expected standards of self and others, fear of achievement and lack of self-confidence. The Academic Procrastination Behaviour section consists of 18 questionnaire items (A1 – A18) which assess students' level of academic procrastination behaviour. The second one is Task Aversiveness which emphasizes unpleasantness of tasks, risk taking and lack of assertiveness. This section contains 26 questionnaire items (R19 – R44). The two sections combined yielded both the outcomes of behaviour and the variables required for explaining and predicting the observed behaviour.

Additionally, the dataset is from a peer-reviewed and publicly available data article. The article mentions that the construct validity is reported, factor loadings, and high internal reliability where the Cronbach's alpha is over 0.90. This helps to ensure reliability and theory for the data. This dataset comprises factors that are generally acknowledged in higher education procrastination literature, and which could be adopted in the context of other universities

throughout the higher education sector despite having been obtained from Indonesian universities.

Although this data is from universities in Indonesia, the psychological constructs measured in this questionnaire are widely known in the academic procrastination literature, including the concept of procrastination behaviour, fear of failure, time management, and aversiveness of tasks.

This data set, therefore, can assist in the identification of procrastination by identifying common patterns and consequently make the system more responsive and smarter to the user needs. This data will be used to create customized nudges and feedback systems in the gamified mobile app. Hence, allowing the system to respond effectively when the tendency of procrastination is identified.

Table 10 outlines the structure of the dataset for this study which includes the demographics, target measures of procrastination and input features.

Section	Variable	Items	Description	Purpose in this Study
Demographics	SEX	N/A	Gender of the respondent	Descriptive information only
Demographics	UNIVERSITY	N/A	University attended by the respondent	Descriptive information only
Section A	Areas of Academic Procrastination	A1 – A18	Measures the frequency and severity of procrastination	A7 was selected as the target variable for machine

			across various academic activities	learning model development
Section B	Reasons for Procrastination	R19 – R44	Measures the psychological factors contributing to procrastination behaviour	Used as candidate input features during feature selection

Table 10: Structure of Dataset

A secondary dataset was selected instead of collecting primary data because it had already undergone psychometric validation and contained sufficient behavioural indicators for supervised machine learning. Using an existing validated dataset also ensured that the developed prediction model was based on established academic procrastination constructs while reducing the time required for large-scale data collection.

3.3.2 Data Pre-processing

The preprocessing of the raw academic procrastination dataset was the initial stage of the machine learning pipeline for this work. This was to ensure the data was complete, consistent, and appropriate for feature selection and model building. The original data set had 586 data points and 47 variables with the following components: Demographic variables, 18 items on areas of academic procrastination, and 26 items on the reasons for procrastination.

The first step to preprocessing was to clean the column names. Unnecessary text and extra spacing were deleted to give consistent names to the variables, A-1 through A-18 for the academic procrastination items and R-19 through R-44 for the reasons for procrastination. The

standardization of the column names minimized the risk of errors in selecting and referring to variables in the following parts of the machine learning pipeline.

This was followed by data checking for missing data. The records with incomplete data were marked for their removal when missing responses to any questionnaire could impact the building of the input variables and target variable. Also, removing incomplete records helped to keep consistency in the development of all questionnaire items with the model.

The variables A- and R- in the questionnaire then were transformed into numbers. All the answers were checked to ensure that they are within the acceptable range of 1 to 5 for the questionnaire used to get the answers were on a five-point Likert scale. Any record where a non-numerical value or a value was recorded outside the range was to be excluded. This validation process ensured that only valid responses to the ordinal items in the questionnaire were used for further analysis.

The demographic variable gender was also standardized for equalized demographic coding. Extra spaces were eliminated and all gender values were made caps. This process eliminated different formatting from being interpreted as different categories in the analysis.

In order to make the data more structurally regular, improve quality and uniformity, these pre-processing procedures were executed before feature selection was done. The cleaned data set was then used as the foundation for developing the secondary data set and analyzing various machine learning pipeline techniques.

Table 11 below summarizes the procedures done in the data pre-processing phase.

Pre-Processing Procedure	Method Applied	Purpose
Column heading cleaning	Removed unnecessary text and spaces from column names	Ensured consistent variable naming

Missing-Value	Examined the dataset and removed records that were incomplete when necessary	Prevented responses with missing value from affecting the model development
Numerical Conversion	A- and R- variables were converted to numerical values	Likert-Scale responses was prepared for analysis.
Ordinal-Range Validation	Values of questionnaire were checked so they are all between 1 and 5	Removed invalid or out-of-range responses
Gender standardization	Gender values were converted to uppercase and spaces were removed	Ensured consistent demographic encoding

Table 11: Data Pre-processing Procedures

Data preprocessing was performed to improve the overall quality and consistency of the dataset before model training. Removing inconsistencies, handling missing values, transforming categorical variables, and standardising numerical features reduced potential sources of bias and ensured that the machine learning algorithms learned from meaningful behavioural patterns rather than noisy or incomplete data.

3.3.3 Exploratory Data Analysis (EDA)

Exploratory data analysis (EDA) was done first to understand the data and to ensure it is appropriate for machine learning before the prediction model is built. The data provided 586 student responses obtained from the PASS questionnaire, with 47 variables including

demographic, academic procrastination behaviors, and psychological variables related to academic procrastination.

The exploratory analysis included a general examination of the data structure, variable types, descriptive statistics and missing data. The responses for the dependent variable (A-7) were also presented in graphical form to see the distribution of the procrastination level among the respondents. Demographic data like gender distribution was also explored to get the characteristics of the study population.

The results showed that there were no missing values present in the data set, and that there were 586 observations and 47 variables. The values of all the questionnaire items were recorded with the valid 5-point Likert scale type which showed that the obtained data set was complete and suitable for further preprocessing and model building.

Figure 3 presents the distribution of the target variable whereas Figure 4 presents the gender distribution.

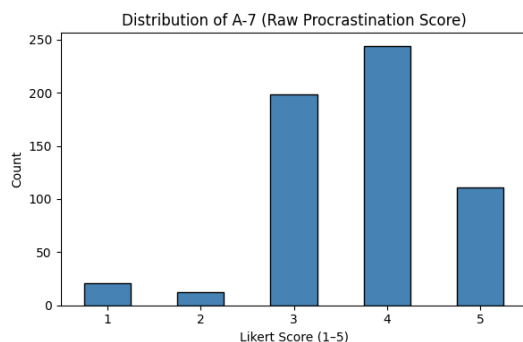


Figure 3: Distribution of A-7 (Target Variable)

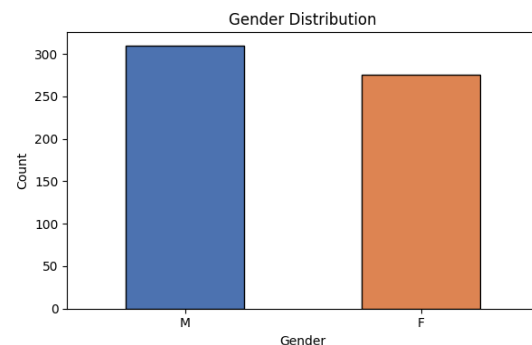


Figure 4: Gender Distribution Chart

3.3.4 Feature Selection and Target Construction

After the preprocessing stage, a second dataset was created for machine learning. Ten psychological factors, which were identified during the feature selection stage, were used as the predictor variables instead of using all the variables in the questionnaire. These variables have some behavioral and psychological attributes that are important in the context of academic procrastination.

The chosen predictor variables were:

- Overwhelm Score
- Peer Pressure Score
- Fear of Success Score
- Sensation Seeking
- Rebellion Score
- Time Management Score
- Social Anxiety Score
- Low Energy Score
- Perfectionism Score
- Social Distraction

The target variable was constructed from question number A-7 on the questionnaire which represents the tendency of students to procrastinate on reading assignments at the end of the week. For the Likert responses, the cutoff of 4 or 5 was defined as High Risk (1) and 1–3 as Low Risk (0) for the purposes of formulation of a binary classification problem.

Students who answer "Agree" or "Strongly Agree" in the response column confirm the reality of weekly reading tasks that are often missed, which are good indicators of a high level of procrastination risk, hence this threshold.

The secondary data was obtained, amounting to 586 observations, 10 predictor variables and a binary target variable, which served as the input for the models to be trained.

3.3.5 Model Training and Evaluation

To ensure the original class distribution, the prepared data set was partitioned into three independent subsamples with stratified sampling. The model training data comprised around 80% of the data, validation data 10% and final testing data 10%.

Table 12 below presents the distribution of the samples across the training, validation and test sets.

Dataset	Samples	Percentage
Training Set	468	79.9%
Validation Set	59	10.1%
Test Set	59	10.1%

Table 12: Distribution of Samples Across the Training, Validation, and Test Sets

The data set had a greater number of high-risk students than low-risk students, and therefore only the training set was used for applying the Synthetic Minority Oversampling Technique (SMOTE). By applying SMOTE to the training data alone, information leakage was avoided, and the learning algorithms were still provided with balanced class distributions when training.

The data was then balanced, and seven supervised machine learning algorithms were trained and tested:

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest
- Gradient Boosting
- k-Nearest Neighbours (k-NN)
- Multilayer Perceptron (MLP)

The models were tested with Accuracy, Precision, Recall and F1-score. During the model selection process, Recall was prioritized, as the main purpose of this study was to identify students who might be at high risk for procrastination.

The 5-fold Stratified Cross-Validation (5-fold SCV) approach was also used on the balanced training data set to assess the stability and generalization ability of each individual classifier.

Although accuracy provides an overall indication of classification performance, it was not considered the sole criterion for model selection. The primary objective of the application is to identify students requiring early intervention. Therefore, recall was prioritised to minimise false negative predictions. From an application perspective, incorrectly classifying a genuinely high-risk student as low risk would prevent timely behavioural intervention, whereas incorrectly classifying a low-risk student as high risk would primarily result in additional reminders and motivational support. Consequently, precision, recall, F1-score, confusion matrices, and cross-validation performance were collectively considered when selecting the final prediction model.

3.3.6 Model Development

During this research, several machine learning pipeline combinations were created, implemented and tested for their ability to predict procrastination. The limitations of the previous implementation guided the improvements made in each iteration for modifying the preprocessing procedures, model evaluation strategy and deployment readiness. This iterative process resulted in an end-to-end pipeline that could be trained to achieve acceptable predictive performance and was easily extensible, interpretable, and ready for deployment into the proposed Nudgr mobile application.

Throughout the development process, table 13 shows the iterations of machine learning pipeline.

Iteration	Objective	Major Improvements	Reasons for Further Refinement
Iteration 1 – Baseline Pipeline	Build a preliminary end-to-end forecasting system.	Performed Data Cleaning, Manual Feature Selection, Binary Target Construction, Train Validation Test Split, Feature Scaling, SMOTE, and comparison of seven supervised learning algorithms with the help of accuracy, precision, recall and the f1 score.	Developed a working prediction model, but less comprehensive evaluation, model interpretation, model validation.
Iteration 2 – Enhanced Model Evaluation	Improve model assessment.	Introduced ROC (Receiver Operating Characteristic) curves and ROC-AUC (Receiver Operating Characteristic - Area Under the Curve) analysis to assess the discrimination ability of classifiers.	Model selection was inconsistent, as ROC-AUC was not a good indicator of the primary goal of identifying students at high risk of procrastination.
Iteration 3 – Model Optimisation	Enhance prediction performance and deployment readiness.	Improved preprocessing techniques, optimized model parameters, created confusion matrices, added feature relevance, and deployed	The level of performance increased but the process of evaluation was not completely convincing of robustness and stability

		trained models for integration with applications.	of the models in repeated training.
Iteration 4 – Final Integrated Pipeline	Develop the final machine learning workflow.	Integrated all validated improvements into a single pipeline that includes exploratory data analysis, improved data preprocessing, target construction that follows the domain, SMOTE for the training set, cross-validation, feature importance analysis, extensive comparison of models and export of a model ready for deployment.	This iteration was selected as the final and is illustrated in the Section 3.3.2.

Table 13: Summary of Machine Learning Pipeline Iterations

3.3.7 Model Export and Deployment

After selecting the final Random Forest classifier, the trained model was serialized using the Joblib library together with the fitted feature scaler and predictor variable list. Exporting these components separately ensured that the same preprocessing pipeline used during model training could be consistently reproduced during application deployment, preventing discrepancies between training and inference.

Students fill in the starting procrastination assessment form when they are using the application. The predicted response is converted into the desired feature vector and passed on

to the trained Random Forest model. The prediction model then determines a student's risk level (Low Risk, Medium Risk or High Risk), and the application adapts reminder notifications, AI generated task planning, and gamification strategies to the student's predicted procrastination risk.

3.4 Mobile Application Development

Following the completion of the machine learning model, the next phase involved developing the Nudgr mobile application that integrates machine learning prediction, Generative Artificial Intelligence (Generative AI), cloud computing, personalized behavioral interventions, and gamification into a unified productivity platform. The application was designed to provide students with intelligent support for managing academic tasks while encouraging positive behavioral change through personalized interventions based on each user's predicted procrastination risk.

A cross-platform development approach was adopted to maximise accessibility while reducing development complexity. The application was implemented using Flutter, while Firebase provided the cloud backend infrastructure. Google's Gemini API was integrated to generate AI-assisted subtasks and suggested internal deadlines from user-defined academic tasks. The trained Random Forest prediction model developed in the previous phase was subsequently integrated to classify users according to their procrastination risk.

The overall application architecture follows a modular design in which individual components operate independently while communicating through well-defined interfaces. This architecture improves maintainability, scalability, and future extensibility by allowing individual modules, such as the machine learning model or AI service, to be updated without requiring substantial modifications to the remaining application components.

3.4.1 Technology Stack and Development Approach

Flutter was selected as the main mobile application development framework because it enables Android and iOS applications to be created from a single codebase. This reduced development effort while maintaining consistent application behavior and interface design across both platforms. Flutter's Material Design components were also used to create reusable interface elements for the task management screen, procrastination assessment, virtual garden, analytics dashboard, and other application features.

Dart was used to implement the application logic, navigation, data processing, and reusable widgets. The modular structure of Dart and Flutter allowed the application features to be developed independently before being connected into the complete Nudgr system.

Firebase was adopted as the cloud-based backend platform. Firebase Authentication handled account registration, login, password recovery, and user-session management. Cloud Firestore stored user profiles, assessment responses, prediction results, academic tasks, AI-generated subtasks, reminder information, experience points, levels, streaks, garden progress, and analytics data. Firestore's real-time synchronization enabled changes made in one module to be reflected automatically across the application.

Google's Gemini API was integrated to provide AI-assisted academic task planning. When users created a task, the relevant task information was submitted to Gemini to generate smaller subtasks and recommend internal deadlines. The generated output was returned in a structured format, validated by the application, and stored in Cloud Firestore after user approval.

The trained Random Forest classifier was integrated through a Flask-based prediction service. Assessment responses submitted through the mobile application were transformed into the same predictor format used during model training before being sent to the prediction service. The prediction result was then returned to the application and stored in Firestore for use by the personalized intervention components.

Joblib was used to export and load the trained Random Forest classifier, fitted scaler, and selected feature list. Saving these components ensured that the same model and preprocessing configuration used during training could be reproduced consistently during deployment.

Table 14 below summarizes the technology used to develop the application and its roles.

Technology	Role in the application
Flutter	Created and designed a cross platform mobile application and user interface
Dart	Developed application logic, navigation, data processing and reusable elements
Firebase Authentication	Handles user registration, login, and account authentication
Cloud Firestore	User profiles, tasks, subtasks, assessment results, progress, XP, levels and garden information are all stored.
Gemini API	Created AI generated subtasks and internal deadlines.
Random Forest model	Forecasted the user's procrastination risk level from the assessment answers
Joblib	Saved and loaded trained model, feature scaler and selected feature list

Table 14: Summary of Technology Used and its Role

3.4.2 System Architecture

The Nudgr application adopted a client–server architecture consisting of a Flutter mobile client, Firebase cloud services, a Flask-based machine learning prediction service, and Google’s Gemini API. The architecture was designed to separate the user interface, application logic, cloud data management, machine learning prediction, and Generative AI functionality into distinct components.

Users interact directly with the Flutter mobile application to register accounts, complete the procrastination assessment, manage academic tasks, generate AI-assisted subtasks, monitor progress, and interact with the virtual garden. The Flutter application functions as the main client and coordinates communication with the backend services.

Firebase Authentication verifies users during account registration and login. Each authenticated user receives a unique user identifier, or UID, which is used to associate the user with their corresponding records in Cloud Firestore. This ensures that assessment responses, tasks, subtasks, analytics, experience points, and garden progress remain linked to the correct user account.

Cloud Firestore operates as the central database of the application. It stores user profiles, assessment responses, prediction results, tasks, AI-generated subtasks, reminder schedules, experience points, levels, streaks, garden status, analytics, and notification preferences. Its real-time synchronization enables application modules to access the latest information without requiring manual data refreshes.

The machine learning prediction service operates separately from the mobile application. When a user completes the initial assessment, the responses are submitted to the Flask backend, where the trained Random Forest model processes the input and produces a procrastination-risk prediction. The result is returned to the Flutter application and stored in Cloud Firestore for subsequent personalization.

The application also communicates with Google's Gemini API during task creation. Task details, including the title, description, priority, and deadline, are submitted to the service, which returns a structured set of subtasks and internal deadlines. The generated content is validated before being displayed to the user and stored in Firestore.

The reminder and nudge components use task information, deadlines, completion status, and procrastination-risk levels to schedule notifications. The analytics dashboard retrieves completion data, experience points, streaks, and user activity from Firestore, while the virtual garden uses experience-point and task-completion records to determine the appropriate garden state.

This modular architecture improves maintainability and extensibility because individual components can be updated or replaced independently. For example, the Random Forest model or Gemini service could be replaced in future versions without requiring major changes to the Flutter user interface or Firestore database structure.

Figure 6 below shows the overall system architecture.

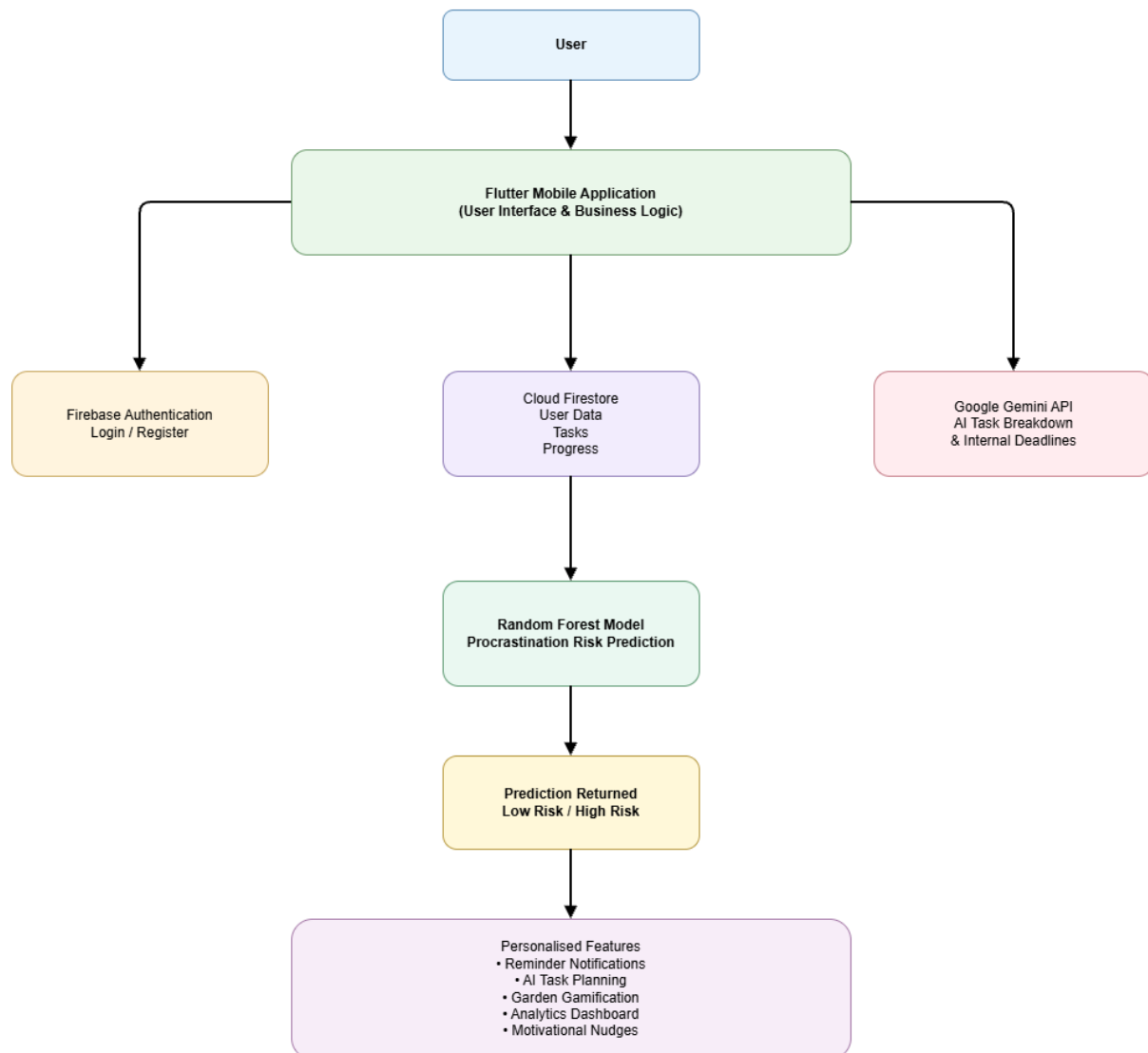


Figure 5: Overall System Architecture

3.4.3 User Authentication Module

Implementing the User Authentication module ensured that features and services provided by the Nudgr application are accessible to only registered users. This module is responsible for user registration, login, password recovery and user session management using Firebase Authentication. Personalized information like procrastination assessments, academic tasks, analytics and garden progress can be securely linked to individual user accounts when users have to authenticate before using the application.

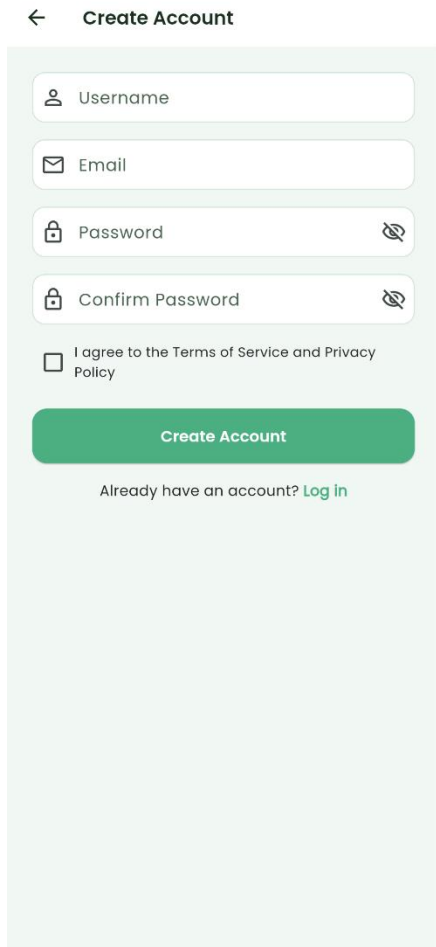
New users can register with their email and password. In the registration process, the application will validate the input and make sure all the required information is provided and that the email address is in a correct format. It also validates password strength to encourage users to make secure passwords. If the registration is successful, Firebase Authentication returns a unique identifier for the user, known as a "UID", that will be the key for Cloud Firestore's user-specific data.

Current users can log in to the application using the login interface and specify the email address and password that they have registered. The credentials are checked against Firebase Authentication before letting the application access. In case of authentication failure, due to the wrong credentials, the proper error message is shown, and user is informed about the failure so that the user can take corrective steps.

The user logs in and their session is remembered across the application. This means no need for users to re-enter their credentials and guarantees users' data stays securely attached to their accounts. Once the user logs out, the session is closed and access to personalized data is denied until the user logs in again.

The User Authentication module secures the accounts while allowing the services delivered throughout the Nudgr application to be personalized.

The figure 7 and 8 below present the login and register screens created in the application.



← Create Account

Username

Email

Password

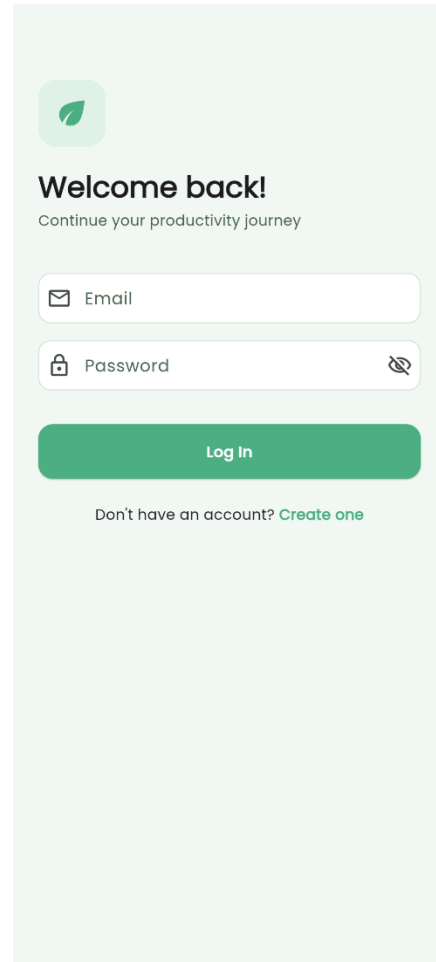
Confirm Password

☐ I agree to the Terms of Service and Privacy Policy

Create Account

Already have an account? [Log in](#)

Figure 6: Create Account Page



Welcome back!

Continue your productivity journey

Email

Password

Log In

Don't have an account? [Create one](#)

Figure 7: Login Page

3.4.4 Initial Procrastination Assessment Module

Once successful login, users must complete the Initial Procrastination Assessment before they can use the main features of the application. The assessment is used as input to personalize the behaviour of the application by calculating the risk level of procrastination for each user, as determined by the trained machine learning model.

The assessment is made up of 10 questions from the questionnaire, which are the psychological factors that were used in the feature selection step of the machine learning pipeline. These are the influences of overwhelm, peer pressure, fear of success, sensation seeking, rebellion, time

management, social anxiety, low energy, perfectionism, and social distraction. Answers given to each question are Always, Very Often, Often, Sometimes and Rarely.

After all responses are filled out, the selected responses are translated into the feature format needed by the Random Forest prediction model. These values are then sent to the prediction service, where the trained model predicts the user's procrastination risk. The prediction result is then "saved" in Cloud Firestore and added to the user profile.

This anticipated procrastination risk is then used by various parts of the application such as reminder scheduling, personal nudges, AI task planning, and progress monitoring. The application can offer customised assistance without having users fill in the same form again by conducting the assessment at only one point in the onboarding process.

For users with very low tendencies to procrastinate, an application-level decision mechanism was added to the machine learning prediction to increase the consistency of prediction. This process adds a second line of defence against the tendency of the prediction model to see some low-risk customers as high risk, which was discovered when the model was evaluated. The mechanism is further discussed in Chapter 5 for implementation and evaluation.

Overall, the Initial Procrastination Assessment provides for the application to tailor the next time the user will interact with the system, and for collecting the basic information needed for the intelligent parts of the system.

Figure 9 and 10 below presents the pages for procrastination assessment and the homepage of the application.

Procrastination Assessment

Question 1/10

I feel overwhelmed when I have multiple tasks to complete

Select your response:

☐ Rarely

☐ Sometimes

☒ Often

☐ Very Often

☐ Always

Next

Figure 8: Procrastination Assessment Page

Good morning, keekee

0 days

lv1 0 / 100 XP

Empty
Complete your first subtask to plant a seed.
0%
Tap to view full garden -->

Today's Nudges
Do not break the streak 🌟

Risk Level: Medium

Home Tasks Garden Ranks Analytics Profile

Figure 9: App's Homepage

3.4.5 Task Management Module

The Task Management module is the main module of the Nudgr application that enables this to happen, where users can organize and track their academic tasks. This module will allow the user to complete, edit, delete and create academic tasks and to keep a structured overview of their workload. Providing all task information in one place enables users to plan their studies and track their progress on a per-semester basis.

Users must enter multiple pieces of information to create a new task, such as the task title, task description, priority level, due date, and the estimated completion date. These data are checked before being saved to Cloud Firestore to prevent missing or incorrect data being uploaded to the database. Each is specifically linked to the authenticated user's account so that users can control their academic work.

Once a task is created, it appears on the main task management screen, allowing users to easily see the tasks they have left, completed and overdue. Tasks are displayed in chronological order of due dates to enable users to find those tasks that need to be done right now. Users can also edit information about tasks or cancel out tasks that are no longer needed. The information shown is always in sync with Cloud Firestore, and any changes made in the app are synced with Cloud Firestore live.

The module also has the ability to track task completion. Users can indicate when they have accomplished an academic task by completing a task using the application interface. Once a task is completed, it automatically updates the state of the task in Cloud Firestore, which in turn sends out several downstream processes, such as experience point (XP) allocation, progress through a virtual garden, analytics update, and calculation of streaks. This integration lets multiple application modules continue to stay in sync without having to bring in extra user input.

Overall, the Task Management module is the basis for the application's productivity functionality, which allows users to organize their academic workload, and supports the intelligent and gamified features that are introduced throughout the system.

Figure 11 to 14 presents the task page, task creation page, and the create task button.

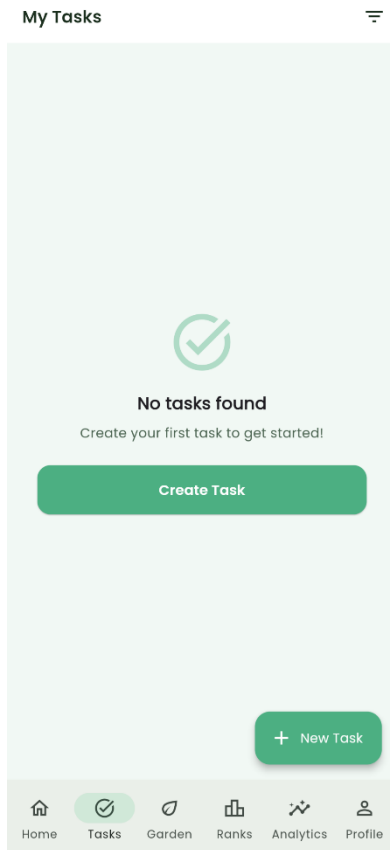


Figure 10: App's Task Page

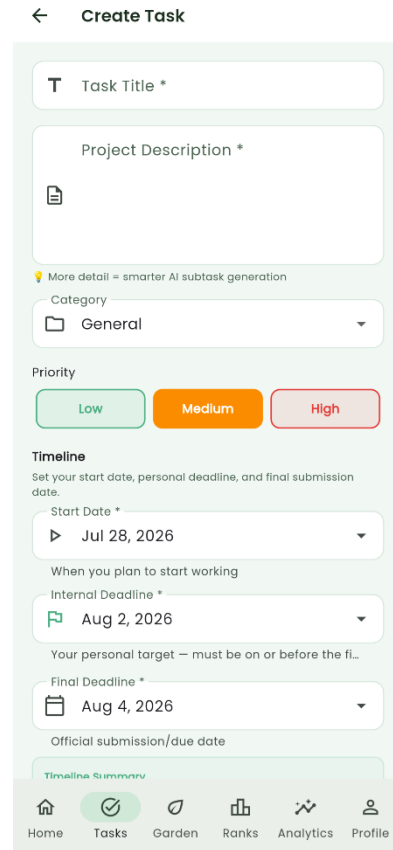


Figure 11: Create Task Page

Figure 12: Create Task Page with Information

Figure 13: Create Task Button

3.4.6 AI-Assisted Task Planning Module

The AI-Assisted Task Planning module was created to address one of the primary factors behind academic procrastination: the uncertainty about how to start a more complex or daunting task. This module automatically breaks down complex tasks into manageable subtasks and suggests internal deadlines for users to complete, eliminating the need for manual division, thanks to Google's Generative Artificial Intelligence (AI) model.

The application gathers task's title, description, due date, estimated completion date and priority level, then sends these details to the Gemini API via a secure backend service. The AI model is then given a well-designed prompt to explain to the model how to perform the task and return a structured response, including a set of subtasks and internal deadlines.

The subtasks generated are then sent back to the application in JavaScript Object Notation (JSON) format. This is then checked against the response to ensure that all the necessary fields are provided and correctly formatted before the subtasks are stored in Cloud Firestore. The application provides an error message if the AI response is not successfully parsed or is an incomplete response and the process can be repeated.

After validation, the subtasks created get automatically linked to the parent task and appear in the application. The generated content is always under the control of the user, and the user can edit, delete, reorder, and add more subtasks before starting work. This way, the AI is not just a replacement for the user's decision-making process, but also an assistant to help them make better decisions.

The AI-Assisted Task Planning module is designed to break down large academic tasks into smaller, more manageable activities, preventing students from feeling overwhelmed and motivating them to start working on their tasks sooner. This aligns with the broader goal of the Nudgr application, which aims to enable users to create more structured and attainable study plans.

Figure 15 to 18 presents the AI generation page, and the page after the subtasks are generated.

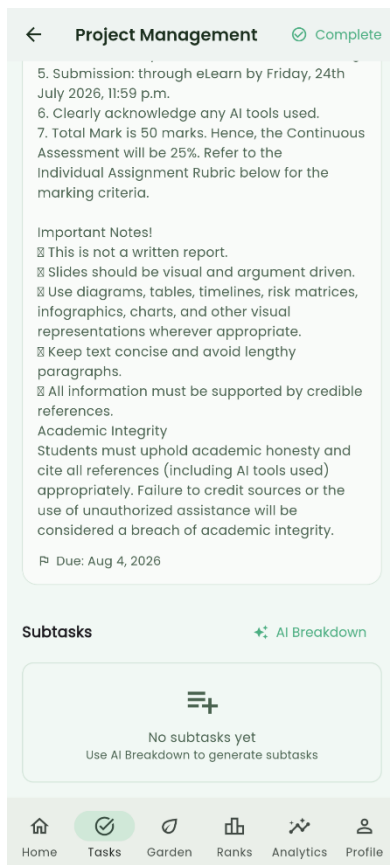


Figure 14: AI Breakdown Button for Subtasks Generation

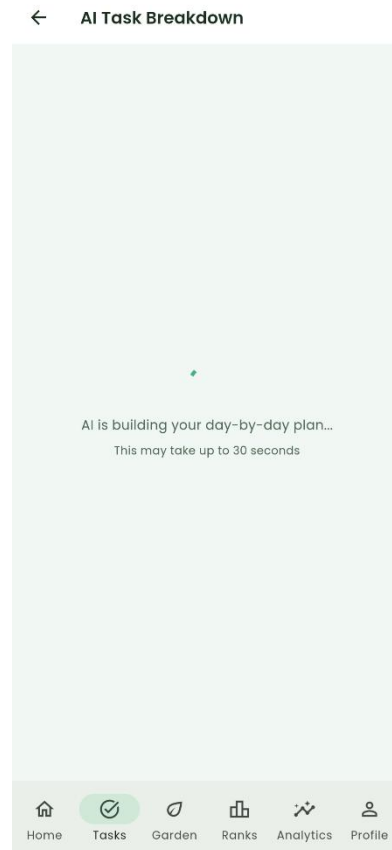


Figure 15: AI Generation Loading Page

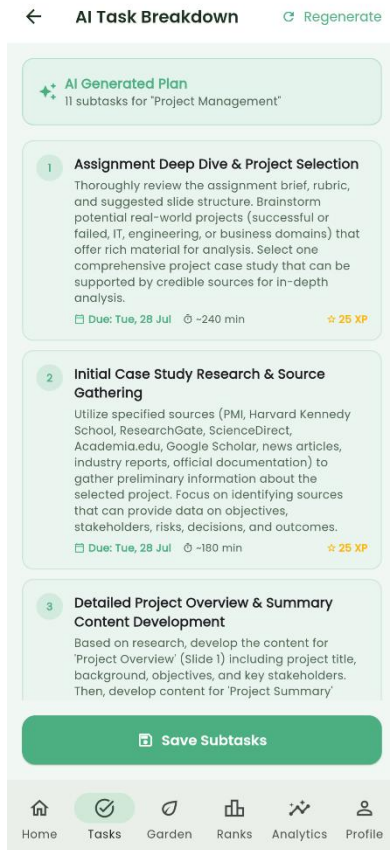


Figure 16: Subtasks Generated

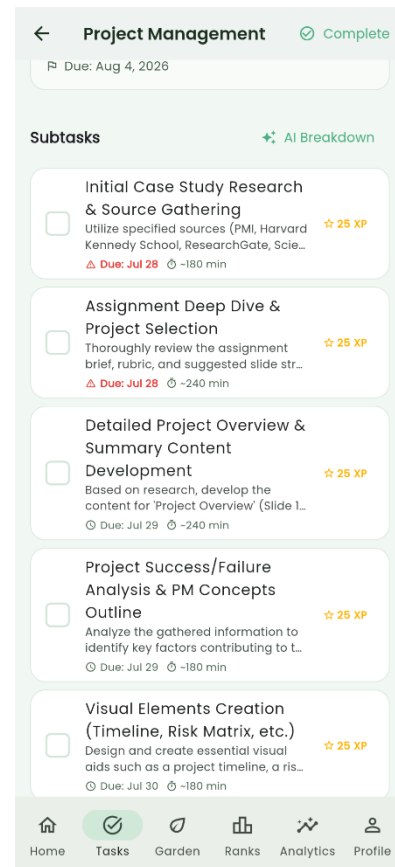


Figure 17: Saved Subtasks

These AI-generated subtasks pave the way for other intelligent functions in the application. After subtasks are created, reminder notifications, personalised nudges, gamification rewards, and progress tracking are based on the structure of the tasks created. For this reason, the AI-Assisted Task Planning module serves as the core element that links the task management function to the other intelligent services offered by the Nudgr app.

3.4.7 Reminder and Nudge Module

The module Reminder and Nudge was developed to facilitate user's involvement in academic activities and decrease procrastination by sending task-based reminder and motivational nudge. Both mechanisms are designed to encourage regular completion of the tasks but are used for

different purposes in the application. While reminder notifications give users details on what they are doing and how far they are in the application, nudges prompt them to change their behaviour depending on their predicted procrastination risk level.

3.4.7.1 Reminder Notifications

The reminder notification system was developed to keep users informed about their academic obligations during the task management process. Reminder information is stored with the task along with the reminder details in Firebase Cloud Firestore whenever a user creates a task. The application can schedule local notifications depending on the reminder settings and will deliver local notifications even when the application is in the background.

The reminder module is different from the traditional reminder systems which only remind users at a fixed time. It can send reminders by considering the status of the user's tasks and the progress of the application. Depending on the current situation, the user might get a notification to say that a task is coming up soon, overdue, or that it is complete, or that it is reminding them to keep working on unfinished subtasks. Moreover, the application offers notifications for the gamification system, for example, a user is notified when he or she has collected experience points (XP) or when the virtual garden is starting to wilt after being inactive for a long time. The app offers personalized reminders, making them more relevant to the user's context to help keep them focused on their studies and the gamification elements.

Figure 19 and 20 presents the notification indicating the status of the garden and real-time update when points are collected.

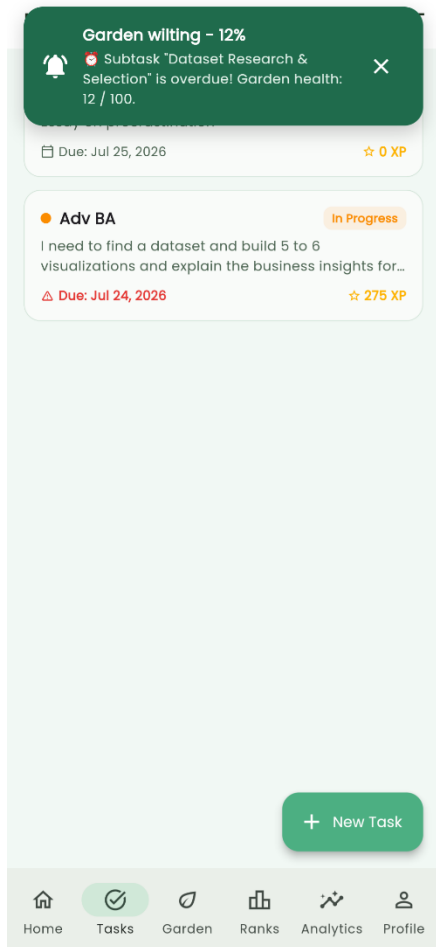


Figure 18: Garden Condition Notification

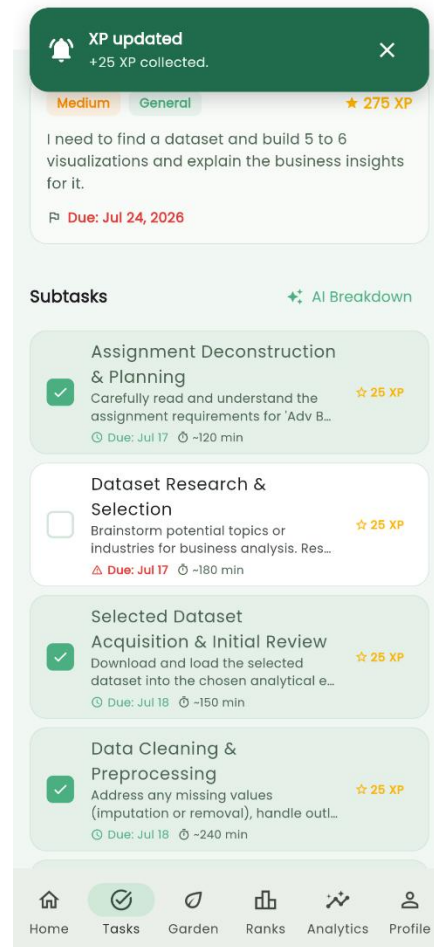


Figure 19: XP collected Notification

3.4.7.2 Nudges

Apart from reminding notifications, the app also includes nudges to help users build good study habits and curb procrastination. The nudges are tailored to the user based on his/her prediction of procrastination risk as calculated by the initial Random Forest model assessment.

The application doesn't produce a new message content dynamically but rather adjusts the frequency at which a motivational message is sent based on the expected risk level. Users are assigned a procrastination risk level one, two, or three nudges are given per day for a low-risk, medium-risk, or high-risk user, respectively, and spread over specific periods of time. This method gives users with higher risk of procrastination more frequent behavioural

encouragement while reducing the constant notifications to users with lower risk of procrastination.

The nudge messages are motivational statements, attempts to encourage productivity and elements to remind students of study. The nudge messages are motivational statements, attempts to encourage productivity and reminders of study with elements of the focus on positive academic behaviour. The nudges are shown at set time points during the day, so this means that the users do not have to actively engage with the application to receive the nudges.

In general, the Reminder and Nudge module serves as a complement to the machine learning prediction component and helps to translate the predicted procrastination risk into actual behavioural interventions. While the reminder notifications help in the management of on-going tasks, the user-specific nudges ensure long-term motivation according to the user's predicted level of procrastination.

Figure 21 presents the nudge notification.

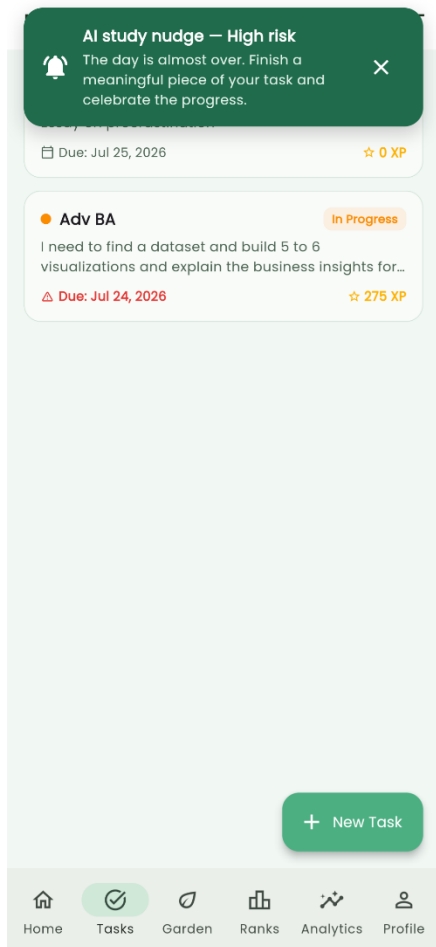


Figure 20: Nudge Notification

3.4.8 Gamification Module

The Gamification Module was designed to motivate users to be disciplined in their studies by adding game elements to the task management process. The app doesn't just present completed tasks in a checklist, instead, it makes users' academic progress into a virtual garden that constantly evolves as productivity goes up. The module has features such as experience points (XP), user levels, daily streaks, and a dynamic garden growth system, which give users immediate visual rewards and long-term motivation.

The virtual garden is the main sign of the student's academic work. Users receive experience points (XP) for completing their subtasks and tasks in the system, which are then used to

improve their level in the system and expand the garden. The more XP is gained, the more healthy the plants are, so that the garden can slowly grow over time. This visual progression enables users to observe the positive impact of maintaining consistent academic behaviour, making productivity more engaging than traditional task management applications.

XP are awarded automatically to the user whenever a task or subtask is successfully completed. The earned XP will be stored in the Firebase Cloud Firestore along with the user's XP levels, streak count and garden status. The more XP a user earns, the higher the level they move to, giving the users a further feeling of achievement and motivation to continue using the application.

The app also has a streak system that allows you to keep track of your study routine, encouraging you to keep up with your daily habits. The streak counter is incremented when users perform a minimum of one of the tasks within a day, thus motivating frequent activity rather than sporadic bursts. The longer the streaks, the more motivating it is for users to maintain their regular use of the app.

The virtual garden is not only a system of rewards for positive behaviour, but it also has consequences for procrastination, which is not found in many other gamification systems. If users do not finish their tasks before they were supposed to, then their experience points are deducted for the overdue tasks. The more XP is accumulated, the more the virtual garden becomes shabbier as the plants in the garden start to wither. The wilting garden is a visual representation that the user is no longer productive, and a reminder to them to get back to unfinished tasks. The state of the garden thus is a constant reflection of the user's academic behaviour where a productive user keeps a healthy and flourishing garden, while a prolonged procrastinator's garden will deteriorate.

The gamification system is also enhanced with notifications that alert the user when key events happen. These notifications will alert users when they have earned experience points for tasks completed, when the garden starts to wilt when a task has not been completed, and alert them to tasks that are coming up, have been completed, or are overdue. In addition to the tailored nudges mentioned in the previous section, these notifications persistently prompt users to stay active in the app.

In general, the Gamification Module acts as an interactive environment to motivate students to complete tasks, providing them with ongoing incentives for good behaviour and with immediate visual feedback for delaying tasks. The application fosters positive reinforcement, introducing a component of gentle behavioural intervention by directly tying task completion, experience points, and level progression, streak maintenance and garden growth together, leading to improved productivity over time.

Figure 22 to 24 presents the garden states, whereas Figure 25 presents the leaderboard page.

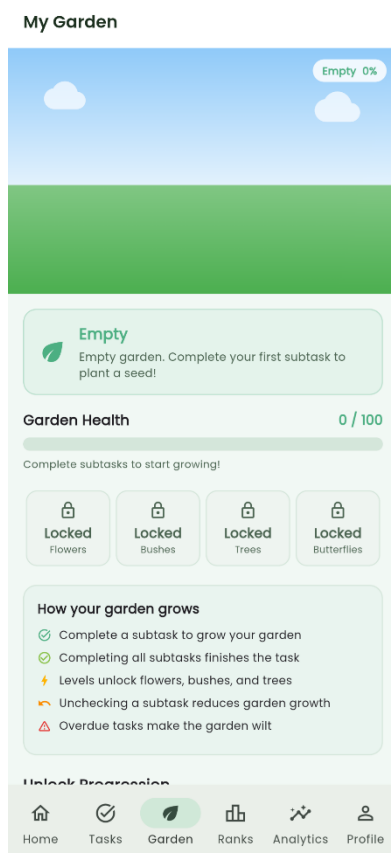


Figure 21: New User's Garden Page

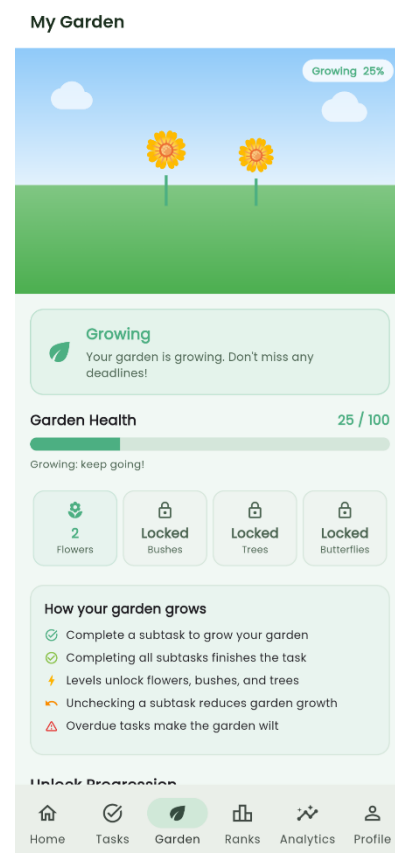


Figure 22: Garden Blooming

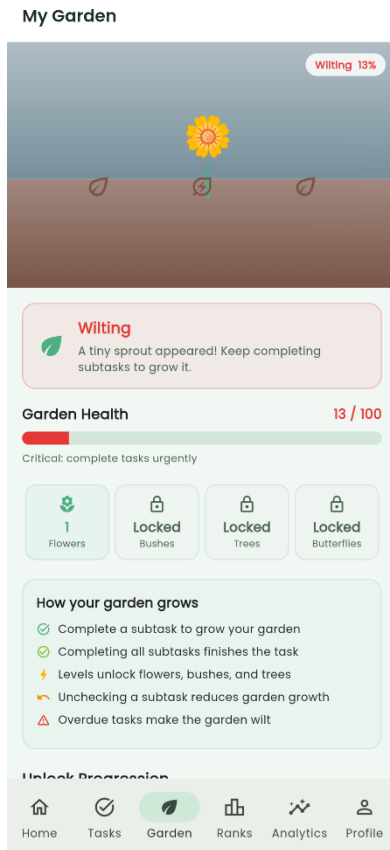


Figure 23: Garden Wilting

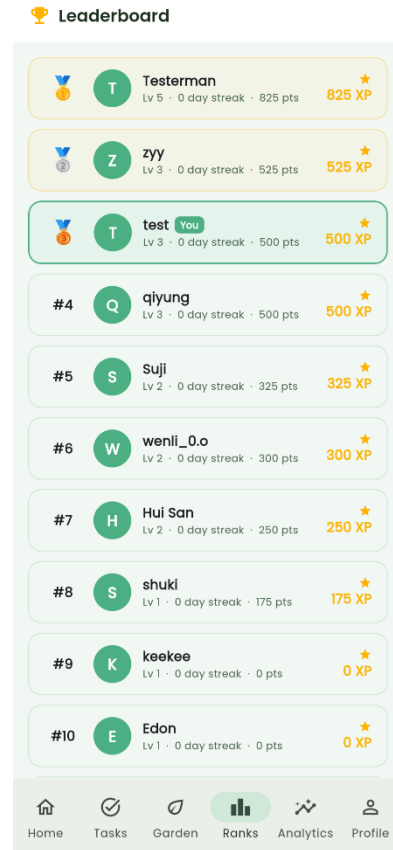


Figure 24: Leaderboard Page

3.4.9 Analytics Dashboard Module

The Analytics Dashboard module was developed to provide users with a visual overview of their academic productivity and application progress. The dashboard retrieves task, completion, streak, and experience-point information from Cloud Firestore and presents the information using summary cards and graphical visualizations.

The dashboard displays information including the number of tasks created, tasks completed, completion rate, tasks completed before their deadlines, current streak, accumulated experience points, and selected productivity trends. These indicators allow users to monitor their academic behavior without manually calculating their progress.

The dashboard is connected directly to the Task Management and Gamification modules. When a task or subtask is completed, the corresponding Firestore records are updated and the dashboard retrieves the latest values. This enables the dashboard to reflect changes in user activity with minimal delay.

The module supports self-regulated learning by encouraging users to review their task-management behavior, identify productivity patterns, and monitor progress over time. It also complements the virtual garden by providing numerical and graphical evidence of the same academic progress represented visually through garden development.

Figure 26 presents the analytics page of the application.

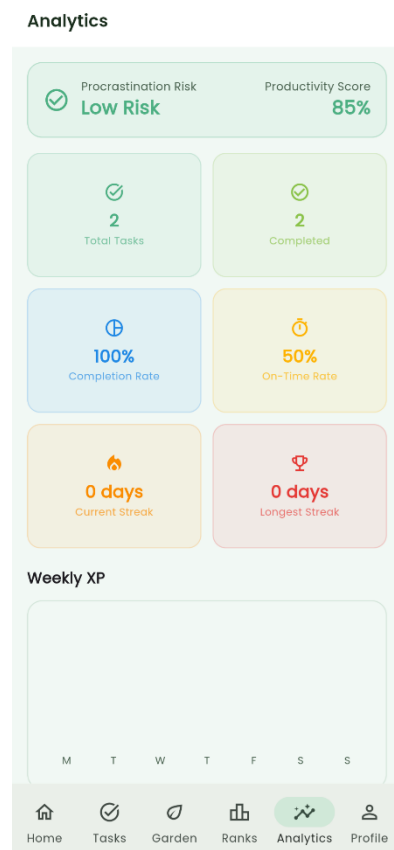


Figure 25: Analytics Dashboard Page

3.4.10 System Integration

During the System Integration phase, all the individual modules created during the project were integrated into a single mobile application. Each module has been developed independently but later connected to each other so that there is a smooth communication between frontend application, backend services, machine learning model, and artificial intelligence services. This integration allows users to enjoy a seamless process from account creation to predicting procrastination, organizing tasks and tracking progress.

All application components were integrated using the Flutter framework. Flutter was the main platform used to integrate all application components. Firebase Authentication was integrated with user login and registration system thereby enabling each user to access their own unique information on a secure manner. Once authenticated, the application pulls the users specific information from Firebase Cloud Firestore, such as assessment data, tasks, subtasks, garden status, experience points, streaks, analytics and notification preferences.

A trained Random Forest model obtained in the machine learning process was incorporated with Flask API in the back end. Once users finished the initial procrastination test, the answers provided on the questionnaire were sent to the prediction API where the model calculated the procrastination risk level of the user. The result of that prediction was then given back to the mobile app, and it was saved in Firebase for future reference. This prediction, in turn, had an impact on various application characteristics such as the personalized nudge frequency.

The application was also made to use the Google's Gemini API to help users split large academic tasks into smaller and more manageable subtasks. Securely sending the task title and description to the Gemini API generated a structured list of suggested subtasks and recommended internal deadlines that were sent to the user when the task was created. The users were shown these AI ideas and allowed to approve them before adding them to the database.

By using Firebase Cloud Firestore, all user interactions were synced to ensure data consistency across the application. As users finished tasks, changed deadlines, gained experience points, or made changes to their garden, the data was all stored in the cloud database. This allowed other modules like Analytics Dashboard, Reminder System and Gamification Module to get the latest information in real-time without having to manually sync that information.

The notification service was coupled with the task management and gamification modules, which helped to send notifications when required according to the status of the tasks and the progress of the application. Future to-do reminders were generated, as were Garden changes, earned experience points, completed tasks, and overdue reminders. Furthermore, the personalised nudge module was programmed to send motivational messages based on the users' estimated risk of procrastination.

The integration of these components enabled the application to become a unified intelligent ecosystem, not as separate modules. The machine learning model detected the risk of procrastination, the Gemini API helped in academic planning, Firebase handled user data and synchronisation, and the reminder and gamification systems constantly reminded users to adopt good study habits. The integrated architecture allowed all application components to interact, offering students help in managing their academic tasks all through.

3.5 System Testing

Following the integration of all application components, system testing was conducted to verify that the developed Nudgr mobile application operated according to its specified functional requirements. The objective of this phase was to ensure that the individual modules functioned correctly, that communication between integrated components remained reliable, and that the application was stable before it was evaluated by end users during pilot testing.

Two types of software testing were performed throughout this study: functional testing and integration testing. Functional testing was conducted to verify the correctness of individual application features, whereas integration testing focused on validating the interaction between different modules and external services such as Firebase, the Random Forest prediction service, and the Gemini API.

The detailed testing results obtained from these activities are presented and discussed in Chapter 4.

3.5.1 Functional Testing

Functional testing was performed to verify that every major feature of the Nudgr application operated according to the specified functional requirements established during the system analysis phase. The primary objective of this testing was to ensure that each application module produced the expected output when supplied with valid user inputs and handled user interactions correctly.

A black-box testing approach was adopted, whereby each function was evaluated based on its observable behavior without examining the internal source code. Individual application modules were tested independently by executing representative user actions and comparing the observed behavior against the expected functional outcomes.

The testing process covered all major application modules, including:

- User registration and authentication
- Initial procrastination assessment
- Machine learning prediction
- AI-assisted task planning
- Task management
- Reminder notifications
- Personalized nudges
- Gamification features
- Analytics dashboard

In addition to verifying the application's core functionality, testing was also performed on external service integrations. Communication between the Flutter application and Firebase services was verified to ensure correct user authentication, cloud data storage, and real-time data synchronization. The Random Forest prediction service was tested to confirm that assessment responses were successfully processed and prediction results were returned correctly. Similarly, the Gemini API was evaluated to ensure that AI-generated subtasks and suggested internal deadlines were successfully generated and stored within the application.

Each test case consisted of predefined input conditions, expected system behavior, and acceptance criteria. A test case was considered successful when the observed application behavior matched the expected outcome without generating functional errors or unexpected behavior.

The detailed functional testing results are presented and discussed in Chapter 4.

3.5.2 Integration Testing

Following the successful completion of functional testing, integration testing was conducted to verify that the independently developed modules operated correctly as a unified system. Unlike functional testing, which focuses on individual features, integration testing evaluates the communication and data exchange between interconnected application components and external services.

The integration testing process simulated realistic user workflows covering the complete application lifecycle. Representative scenarios included user registration and authentication, completion of the procrastination assessment, machine learning prediction, AI-assisted task generation, task management, gamification updates, reminder notifications, and analytics visualisation.

Particular attention was given to validating communication between the Flutter mobile application and the external backend services. These included Firebase Authentication for account management, Cloud Firestore for cloud data storage and synchronization, the Flask-based Random Forest prediction service for procrastination risk classification, Google's Gemini API for AI-assisted task planning, and the local notification service for reminder and motivational nudge delivery.

During each test scenario, data consistency, response accuracy, and real-time synchronization were verified to ensure that information generated by one module was correctly propagated to all dependent modules. This process ensured that task completion correctly updated experience

points, streaks, garden progression, analytics statistics, reminder schedules, and personalized behavioral interventions.

Each integration scenario was evaluated against predefined expected outcomes to determine whether the connected components exchanged information correctly without communication failures, data inconsistencies, or unexpected system behavior.

The detailed integration testing results are presented and discussed in Chapter 4.

3.6 Pilot Testing Methodology

Following the completion of the mobile application and system testing, a pilot testing study was conducted to evaluate the functionality, usability, and overall user experience of the proposed Nudgr mobile application. The pilot testing provided an opportunity to assess the application's performance in a realistic usage environment and identify potential usability issues prior to the final evaluation. Unlike the functional and integration testing performed by the developer, the pilot testing involved representative end users interacting with the application independently to provide feedback regarding its effectiveness and usability.

The pilot testing focused on evaluating the application's major components, including user registration, procrastination risk assessment, task management, AI-assisted task planning, reminder notifications, motivational nudges, gamification features, analytics dashboard, and overall user experience. Feedback obtained during the pilot testing was subsequently analyzed to identify the strengths and limitations of the developed application and to provide recommendations for future improvements.

3.6.1 Participants and Sampling

Pilot testing was conducted with ten voluntary university students, representing the intended target users of the Nudgr mobile application. Participants consisted of Diploma and Bachelor's Degree students aged between 18 and 23 years, reflecting the primary population for whom the

application was developed. Since the application aims to assist university students in managing academic procrastination, selecting participants from this population provided realistic evaluation scenarios.

A convenience sampling approach was adopted due to the limited duration available for pilot testing. Participants were recruited from the researcher's academic network and voluntarily agreed to participate after receiving information regarding the purpose of the study. Prior to participating, each individual received a Pilot Testing Information Sheet describing the objectives of the evaluation, the testing procedure, the expected activities, and the estimated completion time.

Before beginning the evaluation, all participants provided informed consent confirming that their participation was voluntary and that the collected responses would be used solely for academic research purposes. Personal information was treated confidentially, and participants were informed that they could discontinue their participation at any stage of the evaluation if they wished to do so.

The demographic information collected during pilot testing included participants' age, education level, previous experience using productivity applications, and self-reported frequency of academic procrastination. These demographic characteristics provide context for interpreting the pilot testing results presented in Chapter 4.

3.6.2 Pilot Testing Procedure

The pilot testing was designed to simulate a realistic user experience by requiring participants to interact with all major features of the Nudgr mobile application. Before beginning the evaluation, participants were provided with installation instructions and were guided through the overall testing procedure.

Each participant first registered a new account using the application's registration module before logging into the application using Firebase Authentication. After successful authentication, participants completed the Initial Procrastination Assessment consisting of ten

psychological and behavioral questions. The submitted responses were processed by the integrated Random Forest prediction model, and the predicted procrastination risk level was displayed within the application.

Following the onboarding process, participants explored the application's productivity features by creating one or more academic tasks. They were then instructed to use the AI-Assisted Task Planning module to generate subtasks and suggested internal deadlines using Google's Gemini API. Participants were allowed to review, edit, or delete the generated subtasks before saving them to the application.

Participants subsequently interacted with the Task Management module by completing subtasks and marking academic tasks as completed. During this process, they observed the allocation of experience points (XP), progression of the virtual garden, updates to productivity analytics, and changes to their achievement levels. Participants also evaluated the reminder notification system and personalized motivational nudges generated according to their predicted procrastination risk.

After completing all required activities, participants completed the evaluation questionnaire followed by the System Usability Scale (SUS) questionnaire. They were also encouraged to provide open-ended comments describing any usability issues, software defects, or suggestions for improving the application.

Table 15 below summarizes the content included in the questionnaire.

Step	Activity	Purpose
1	Register a new account	Evaluate user registration functionality
2	Log into the application	Evaluate authentication process

3	Complete procrastination assessment	Evaluate prediction workflow
4	View prediction result	Verify prediction display
5	Create an academic task	Evaluate task management
6	Generate AI-assisted subtasks	Evaluate Gemini API integration
7	Edit generated subtasks	Evaluate user control over AI output
8	Complete subtasks and tasks	Evaluate productivity workflow
9	Observe XP and virtual garden	Evaluate gamification features
10	View analytics dashboard	Evaluate productivity analytics
11	Receive reminder notifications	Evaluate notification functionality
12	Complete evaluation questionnaire	Collect feature evaluation
13	Complete SUS questionnaire	Measure overall usability

Table 15: Summary of Contents in Questionnaire

3.6.3 Evaluation Instruments

Two evaluation instruments were employed to assess the usability and effectiveness of the developed application.

The first instrument was a feature evaluation questionnaire specifically developed for this study. The questionnaire consisted of statements covering the application's major modules, including user registration and onboarding, task management, AI-assisted task planning, reminder notifications, gamification, analytics dashboard, and overall user experience. Participants indicated their level of agreement using a five-point Likert scale, where 1 represented Strongly Disagree and 5 represented Strongly Agree.

In addition to the structured questionnaire, participants were provided with an open-ended feedback section where they could describe software defects, usability problems, suggested improvements, or additional features that they believed would improve the application. This qualitative feedback provided deeper insight into user experiences beyond the numerical questionnaire responses.

To obtain a standardized measurement of overall usability, participants also completed the System Usability Scale (SUS) questionnaire. The SUS consists of ten standardized statements with alternating positive and negative wording that measure users' perceptions of system usability. Responses are converted into a usability score ranging from 0 to 100, allowing the usability of the developed application to be interpreted using internationally recognized SUS benchmarks.

3.6.4 Data Analysis

Both quantitative and qualitative analysis techniques were employed to analyze the data collected during pilot testing.

The responses obtained from the feature evaluation questionnaire were analyzed using descriptive statistics. For each questionnaire item, the frequency, percentage, and mean score were calculated to summarize participants' perceptions of the application's individual features. Mean scores were subsequently used to compare the overall performance of different application modules.

The responses collected through the System Usability Scale questionnaire were scored according to the standard SUS scoring procedure. Responses to positively worded statements were adjusted by subtracting one from the selected response, while responses to negatively worded statements were adjusted by subtracting the selected response from five. The adjusted values were summed and multiplied by 2.5 to produce a final SUS score ranging from 0 to 100. The resulting score was interpreted using established SUS guidelines to determine the overall usability of the developed application.

Qualitative responses obtained from the open-ended feedback section were analyzed using thematic analysis. Similar comments were grouped into common themes, including software defects, usability concerns, feature enhancement suggestions, and positive user experiences. These themes were used to identify practical improvements that could be incorporated into future versions of the application.

3.6.5 Ethical Considerations

Ethical considerations were incorporated throughout the pilot testing process to ensure that participants' rights, privacy, and confidentiality were protected. Participation in the evaluation was entirely voluntary, and participants were informed about the purpose of the study before commencing the testing activities.

Prior to accessing the application, each participant received a Pilot Testing Information Sheet describing the objectives of the research, the expected testing activities, the estimated duration, and the intended use of the collected information. Participants subsequently provided informed consent confirming that they understood the study and voluntarily agreed to participate.

The information collected during pilot testing was used solely for academic research purposes. Questionnaire responses were analyzed collectively, and participants were not identified individually within the dissertation. Personal information collected for application authentication remained confidential and was not disclosed to third parties.

Furthermore, participants were informed that the procrastination assessment integrated within the application was designed exclusively to support personalized productivity features and should not be interpreted as a medical, psychological, or clinical diagnostic instrument. The predicted procrastination risk generated by the Random Forest model represents an application-based behavioral estimate intended only to personalize application interventions.

3.7 Chapter Summary

This chapter presented the methodology adopted to develop and evaluate the proposed Nudgr mobile application. The methodology incorporated machine learning model development, cross-platform mobile application development, cloud-based backend integration, Generative Artificial Intelligence, gamification, system integration, and software testing. Functional testing and integration testing were conducted to verify the correct operation of individual modules and their interactions as a complete system. Finally, a pilot testing methodology involving representative university students, structured questionnaires, the System Usability Scale, and qualitative feedback was established to evaluate the usability and practical effectiveness of the completed application. The findings obtained from these evaluations are presented and discussed in Chapter 4.

4.0 Result & Discussion

4.1 Overview

This chapter presents and discusses the findings obtained from the development and evaluation of the proposed Nudgr mobile application. The results are organized according to the project's objectives, beginning with the performance evaluation of the machine learning models used to predict academic procrastination risk. The performance of the candidate classification algorithms is compared using multiple evaluation metrics to justify the selection of the final prediction model integrated into the application.

Following the machine learning evaluation, the results of the functional and integration testing are presented to verify that the developed application operated correctly and that all software modules communicated successfully with one another. These testing activities confirmed the correct implementation of the application's core functionalities, including user authentication, procrastination risk prediction, AI-assisted task planning, reminder notifications, gamification features, and cloud-based data synchronization.

The chapter then presents the findings from the pilot testing conducted with university students. The pilot testing results include participant demographics, feature evaluation using a five-point Likert scale, the System Usability Scale (SUS) evaluation, and qualitative feedback collected from participants. The quantitative results are analyzed using descriptive statistics, while the qualitative responses are discussed to identify the strengths of the application, usability issues encountered during testing, and potential areas for future improvement.

Overall, the findings presented in this chapter provide a comprehensive evaluation of both the technical performance and practical usability of the proposed system. The discussion demonstrates how the integration of machine learning, Generative Artificial Intelligence, cloud computing, personalized behavioral interventions, and gamification contributed to the development of an intelligent mobile application for supporting university students in managing academic procrastination.

4.2 Machine Learning Model Performance

4.2.1 Model Selection Results

The overall performance of the algorithms was assessed using the validation results, which indicated that the Random Forest classifier had the best performance. It achieved the highest value of Recall (0.7778), along with Accuracy (0.7627), Precision (0.8235) and F1 score (0.8000).

Table 16 below shows the summary of the results of the models.

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	0.7627	0.8235	0.7778	0.8000
SVM	0.6610	0.7667	0.6389	0.6970
Gradient Boosting	0.5593	0.6562	0.5833	0.6176
MLP (Neural Network)	0.5763	0.6774	0.5833	0.6269
Decision Tree	0.5763	0.7200	0.5000	0.5902

Logistic Regression	0.4915	0.6071	0.4722	0.5312
KNN	0.6102	0.8095	0.4722	0.5965

Table 16: Results of Models

Random Forest was chosen as the final model of prediction due to Recall being the main evaluation criterion in this study. This model was able to correctly classify the highest number of students with high procrastination risk and at the same time achieved competitive values of Precision and Accuracy.

This model was then tested on a test set not used for training. The final testing results showed an Accuracy of 57.63%, Precision of 60.87%, Recall of 80.00%, and an F1-score of 69.14%.

- Accuracy: 0.5763
- Precision: 0.6087
- Recall: 0.8000
- F1-Score: 0.6914

While the overall Accuracy score dropped when applied to test data, the model was able to maintain a high Recall score which allowed it to perform well at identifying students with high procrastination risk. The goal of the application is to deliver real-time interventions, not to achieve the highest overall accuracy of classification, therefore, a Recall focus was considered appropriate for this study.

4.2.2 Feature Importance Analysis

Feature importance analysis was done for the prediction model using Random Forest classifier to improve the interpretability of the prediction model. This analysis quantifies the relative importance of each predictor variable in the final prediction.

In addition to evaluating prediction performance, feature importance analysis was conducted to improve the interpretability of the selected Random Forest model. Understanding which behavioural factors contributed most strongly to the prediction provides greater transparency regarding the model's decision-making process and supports the design of personalised interventions within the mobile application.

The results showed a relationship between Perfectionism Score and Low Energy Score, Overwhelm Score, Social Anxiety Score and Time Management Score, with Perfectionism Score the most predictive.

Figure 26 below shows the chart for the Top 10 Feature Important Scores for Procrastination Prediction.

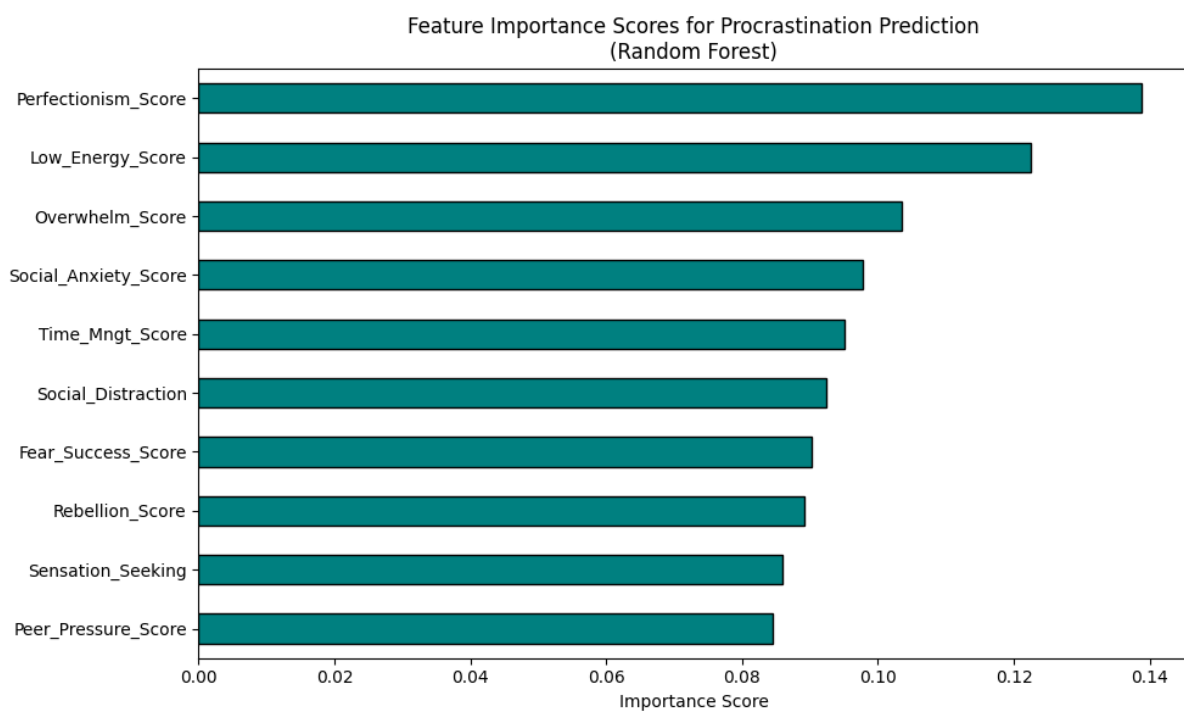


Figure 26: Top 10 Feature Importance Score

It is concluded that psychological factors related to perfectionism, fatigue, academic pressure, anxiety, and poor time management play a significant role in predicting students'

procrastination behavior. The factors identified also give useful insights into designing personalized interventions in the Nudgr mobile application.

4.3 System Testing Results

4.3.1 Functional Testing Results

The functional testing was performed to check that all application features worked as per the functional requirements. Each module was independently tested by performing its main functions and comparing the actual behaviour with the expected behaviour. This testing aimed to make sure that the users could execute all significant activities of the application without encountering functional errors.

All the essential modules built in the app, from user authentication, procrastination assessment, machine learning prediction, AI-generated tasks, task management, reminders, nudges, analytics dashboard, and the gamification system were covered in the functional testing process. In testing, each function was called multiple times with various input combinations to ensure that the application always returned the desired output.

Special care was taken in validating the external services integration. To make sure that the responses to the questionnaire have been properly sent out and the prediction results were returned to the application, the Random Forest prediction API was tested. Likewise, Google's Gemini API was tested to ensure that the AI-generated subtasks and suggested internal deadlines were properly generated and saved in Firebase.

Reminder notification system was also tested by setting up tasks with varying reminders and reminders settings. Notifications were checked for correctness with upcoming tasks, overdue tasks, completed tasks, experience point updates, and garden status changes. The nudge mechanism was validated by the simulation of users of varying procrastination risk level to ensure that the correct amount of motivational nudge was programmed each day.

In addition, the gamification module was tested by finishing and updating tasks to check that experience points were awarded and user levels increased, streaks were completed, and the

garden grew. Extra tests were carried out with overdue items to confirm that experience points were being deducted correctly and that the garden would end up in the wilting state as intended.

The functional testing results are shown in the Table 17 summarizes the tested functionality, expected outcome, observed outcome and testing status.

Test ID	Function Tested	Expected Result	Status
FT1	User Registration	User account created successfully	Pass
FT2	User Login	User successfully logged into application	Pass
FT3	Initial Assessment	Assessment responses successfully submitted	Pass
FT4	Random Forest Prediction	Procrastination risk prediction displayed correctly	Pass
FT5	AI Subtask Generation	Gemini successfully generated subtasks and deadlines	Pass
FT6	Create Task	Task saved successfully into Firebase	Pass
FT7	Complete Task	Task status updated and XP awarded	Pass
FT8	Overdue Task	XP deducted and garden updated accordingly	Pass
FT9	Reminder Notifications	Notifications triggered according to schedule	Pass
FT10	Nudges	Nudge frequency matches predicted risk level	Pass

FT11	Analytics Dashboard	Statistics updated correctly after task completion	Pass
FT12	Virtual Garden	Garden growth and wilting reflected user productivity	Pass

Table 17: Summary of Functionality Test

4.3.2 Integration Testing Results

While functional testing confirms that each function works as intended, integration testing tests the interaction between the various modules or external services when they are connected as a whole system. As the Nudgr application is developed using multiple technologies such as Flutter, Firebase Authentication, Firebase Cloud Firestore, Random Forest prediction model, Google's Gemini API, and the local notification service, integration testing was performed to make sure that data is exchanged properly between all above-mentioned technologies during the application flow.

The integration testing process was oriented towards testing the entire user journey instead of individual features. Firstly, for user authentication, login was tested to verify successful login and retrieve user information from Firebase Cloud Firestore. The first procrastination assessment was then carried out to verify that the questionnaire answers were properly sent to the procrastination prediction API of the algorithm (Random Forest) and that the corresponding level of procrastination risk was correctly returned and recorded in the database.

Evaluation of task management module to Gemini API interaction was also performed. The users successfully passed task information to the Gemini API when creating a new academic task, and the AI generated their own subtasks along with proposed internal deadlines. The generated subtasks were then uploaded to Firebase and shown in the app, which was used to verify successful communication between the frontend, AI service and the cloud database.

The interaction between the task management module and the gamification system was also verified by integration testing. The app accurately added or subtracted user XP, level, and virtual garden status when the user finished or modified a task. These updates were synced with Firebase Cloud Firestore and made instantaneously visible in the Analytics Dashboard, successfully showing the data synced across modules.

The notification service was also tested to verify that reminder notifications and individual nudges are correctly created when the task is ready and the anticipated procrastination risk. Task related notifications, such as upcoming deadlines, overdue tasks, completed tasks, experience point rewards, and garden wilting alerts were delivered successfully based on the configured notification schedules. Likewise, the number of nudges given to users differed according to the pro-challenge risk level, with the appropriate number of nudges being delivered in line with the user's pre-defined frequency of notifications.

On an overall basis, the integration testing showed that all the application modules worked cohesively as a system. A seamless interaction between the frontend interface, backend database, machine learning model, artificial intelligence service, notification system, and gamification module led to a smooth user experience across the application.

Test ID	Integrated Components	Test Scenario	Expected Outcome	Status
IT1	Flutter + Firebase Authentication	User logs into the application	User account authenticated and profile loaded successfully	Pass
IT2	Assessment + Random Forest API	Submit procrastination assessment	Risk prediction generated and stored in Firebase	Pass
IT3	Task Management + Gemini API	Create new academic task	AI-generated subtasks and deadlines returned successfully	Pass

IT4	Task Management + Firebase	Create, edit and delete tasks	Task information updated correctly in Firestore	Pass
IT5	Task Completion + Gamification	Complete a task	XP, level, streak and garden updated successfully	Pass
IT6	Task Status + Notification Module	Task becomes due or overdue	Appropriate notification delivered successfully	Pass
IT7	ML Prediction + Personalized Nudges	User assigned different procrastination risk levels	Nudge frequency adjusted correctly	Pass
IT8	Firebase + Analytics Dashboard	Complete multiple tasks	Dashboard statistics updated immediately	Pass
IT9	Firebase + Garden Module	XP increases or decreases	Garden growth or wilting displayed correctly	Pass

Table 18: Summary of Integration Test

4.4 Pilot Testing Results

4.4.1 Participant Demographics

The total number of university students that have been involved in the Nudgr mobile application pilot testing is ten. In order to provide realistic use scenarios, the participants were chosen from the intended target population, university students, who struggle with academic procrastination. The pilot testing questionnaire gathered demographic data to gain insights into the participants' characteristics and previous experience with productivity applications and academic procrastination.

Demographic data gathered covered participants' age, current educational level, prior experience with using productivity apps, and how often they procrastinated in school. These characteristics give context when looking at the usability evaluation and user feedback in the next sections.

Age Distribution

Figure 27 shows the age distribution of the participants who took part in the pilot testing. Most of the participants were aged 20-22 and the rest were aged 18-19. The age distribution shown is typical for university students on undergraduate studies, which are the main target users of the proposed application.

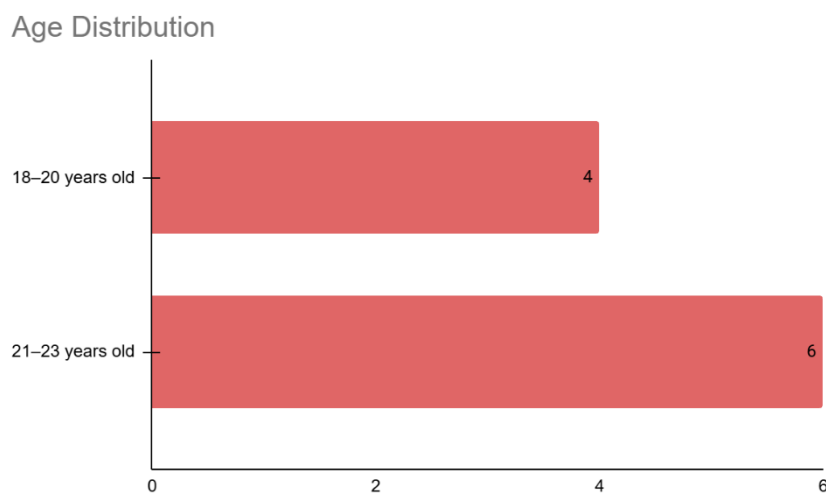


Figure 27: Age Distribution of Participants

Education Level

The participants were students of Diploma and Bachelor's Degree programmes. The majority of participants were studying a Bachelor's Degree course, with the remaining participants studying Diploma programmes. The involvement of students from various experience levels was beneficial for a wider evaluation of the app for students with varying academic experience.

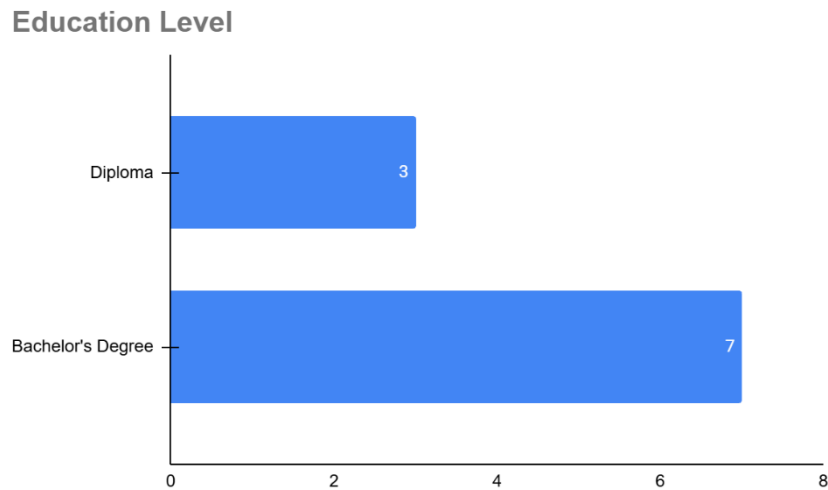


Figure 28: Education Level

Previous Experience Using Productivity Application

Prior to the study, participants were asked whether they had used productivity or task management applications before their participation. As shown in Figure 29, most of the participants stated that they had some experience with the use of productivity applications, but a minority reported they did not have any experience with the use of similar applications. An experienced and an inexperienced user were included in the evaluation to obtain feedback from users with varying levels of experience in using digital productivity tools.

The diagram below shows the previous experience that students have had with using productivity tools.

Previous Experience Using Productivity Application

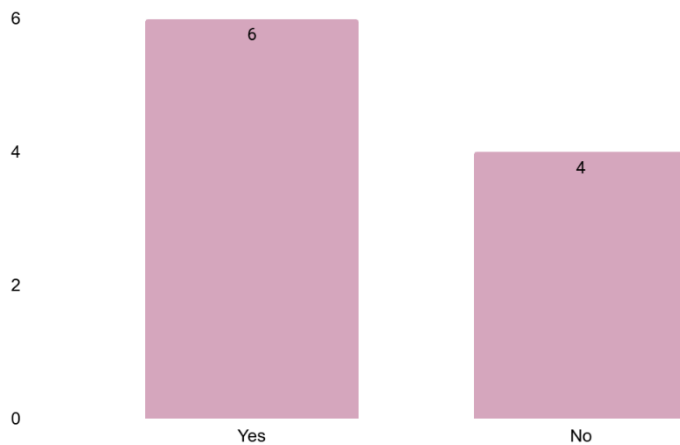


Figure 29: Previous Experience Using Productivity Applications

Frequency of Academic Procrastination

Next, the participants were asked how often they procrastinated academically. Answers are given in Figure 30. Most participants said that they sometimes or often procrastinated, which means that the users for the pilot testing were the ones who were supposed to use the app. A few took part indicating they rarely or very often procrastinated.

Frequency of Academic Procrastination Levels

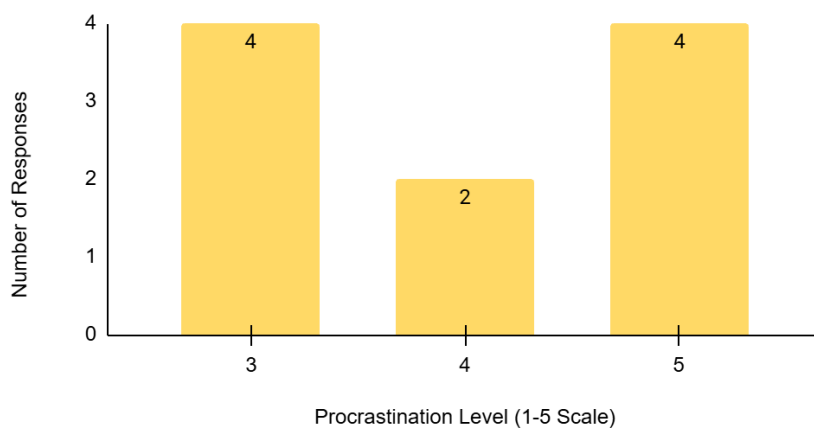


Figure 30: Frequency of Academic Procrastination

The demographic data gathered seem to have been composed by participants who were representative of the proposed target users of the Nudgr application. The subjects were mostly undergraduate students experienced with productivity applications and who frequently reported academic procrastination. Therefore, the feedback gained from the pilot testing can be used to assess the usability and effectiveness of the proposed application.

4.4.2 Feature Evaluation

The pilot testing questionnaire assessed the key features of the Nudgr mobile app using a 5-point Likert Scale from Strongly Disagree (1) to Strongly Agree (5). All core elements of the application were evaluated, such as onboarding, AI support for task management, personalised reminder notifications, gamification components, analytics dashboard, user experience and more. The aim of this evaluation was to find out the users' perception of each of the features and to find out the merits of the proposed application.

4.4.2.1 User Registration and Onboarding

The first section of the evaluation was on the application's onboarding process, which involved user registration, log-in, the initial procrastination assessment and showing the procrastination risk prediction. These components are the user's first contact with the application, and therefore an important part of establishing a positive initial experience.

The onboarding process was well-received by the participants in general. The majority of the participants agreed or strongly agreed that the registration process was straightforward, the login process worked well, the assessment of procrastination was easy to fill out and the results on prediction were presented clearly. These results indicated that participants were able to complete the onboarding without any major problems and knew the reason for conducting the initial assessment before accessing the rest of the features of the application.

Figure 31 to 34 below show the average participant ratings for each component of onboarding.

User Registration

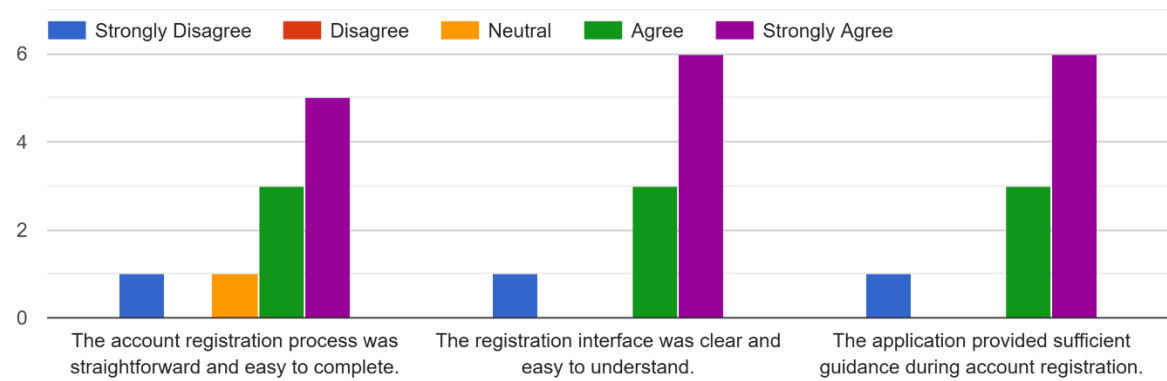


Figure 31: Rating for User Registration

User Login

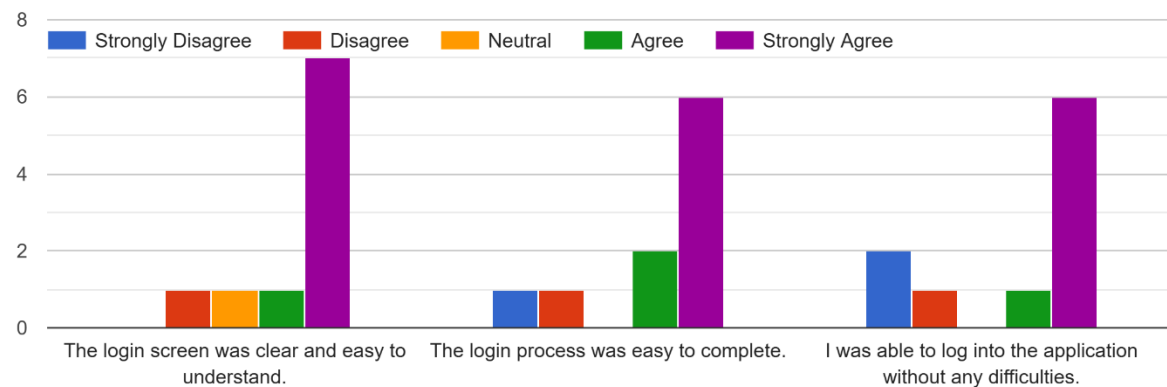


Figure 32: Rating for User Login

Onboarding Assessment

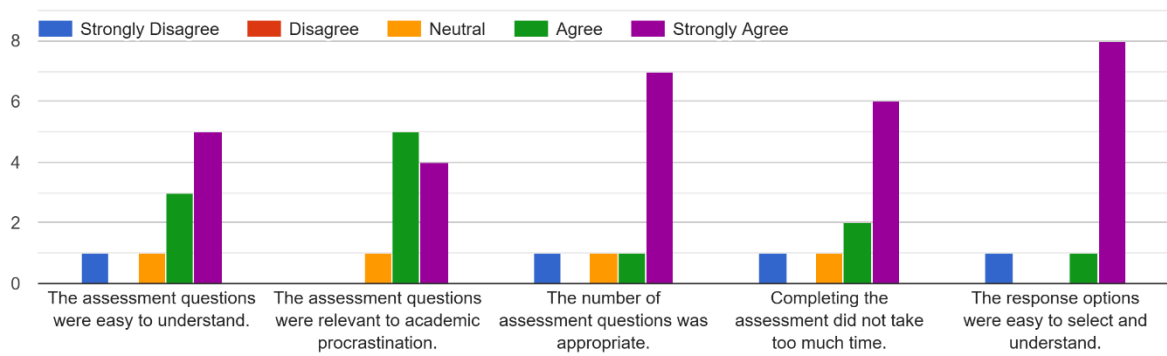


Figure 33: Rating for Onboarding Assessment

Overall, I was satisfied with the registration, login, and onboarding assessment experience.

10 responses

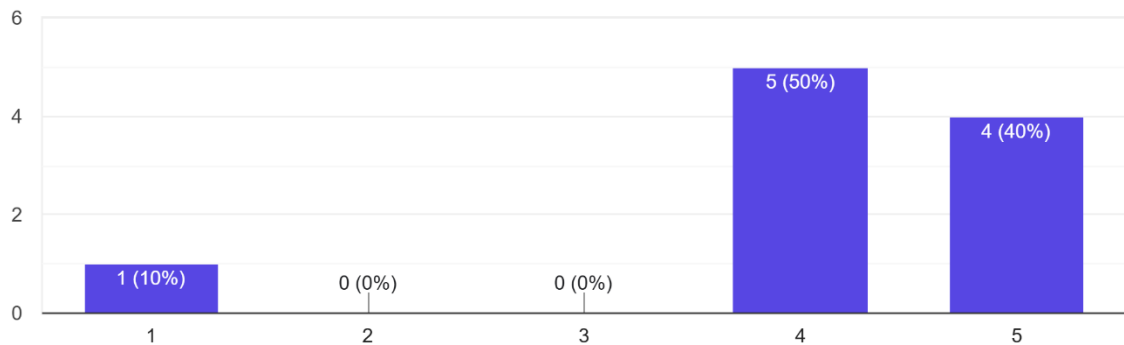


Figure 34: Overall Experience Rating for Onboarding Components

4.4.2.2 AI-Assisted Task Management

The AI-powered task management functionality was tested to see how well it might help participants manage their academic tasks. This is a combination of traditional task management and Google's Gemini API to automatically create meaningful subtasks and internal deadlines based on the user's task description. This feature aims to make even bigger academic tasks

more manageable and less daunting, by breaking them down into smaller, more manageable tasks.

The participants found the following aspects of the task management module to be easy to understand which includes creating new tasks, usefulness of the AI-generated subtasks, relevance of the internal deadlines suggested by AI, managing the AI-generated subtasks, and usefulness of the feature in organizing academic work. Additionally, the assessment took into account whether the AI-generated outputs were clear and useful for real-world implementations.

In general, participants were more positive about the AI-powered task management tool. The vast majority of participants agreed or strongly agreed that it was easy to create tasks and that the subtasks that were created were relevant to the original task descriptions. The suggested internal deadlines were also seen as helpful in allowing participants to allocate their work more effectively and not leave it to the last minute. The results of this study suggest that the incorporation of generative AI in the task management workflow is effective in reducing effort in task planning and in enhancing the users' performance in organizing tasks.

The feedback was overwhelmingly positive, but some participants recommended further enhancements to make the AI-generated task planning even more flexible. These recommendations were to give the user the ability to modify the subtasks created by the system, the ability to regenerate subtasks one by one, and the ability to set the suggested deadlines according to the user's own schedule.

Figure 35 to 38 below are the evaluation of AI-Assisted Task Management

Task Creation

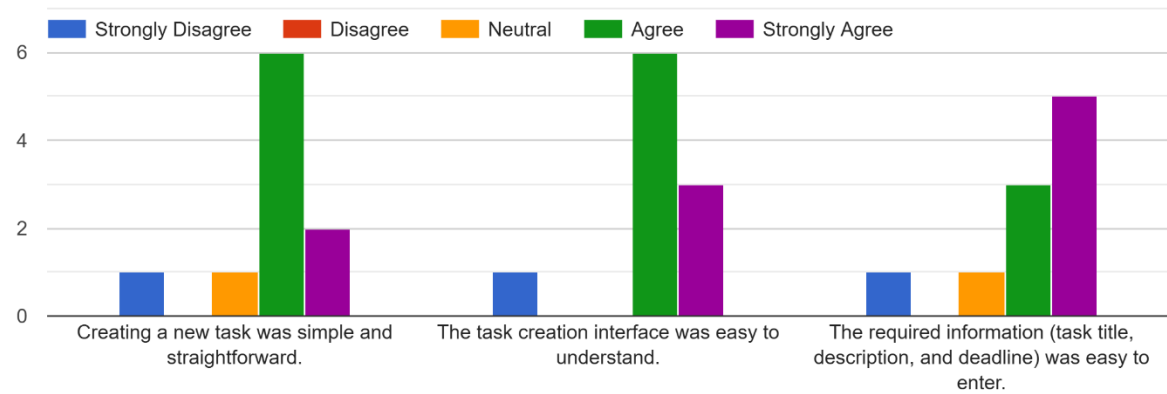


Figure 35: Ratings for Task Creation

AI-Generated Subtasks

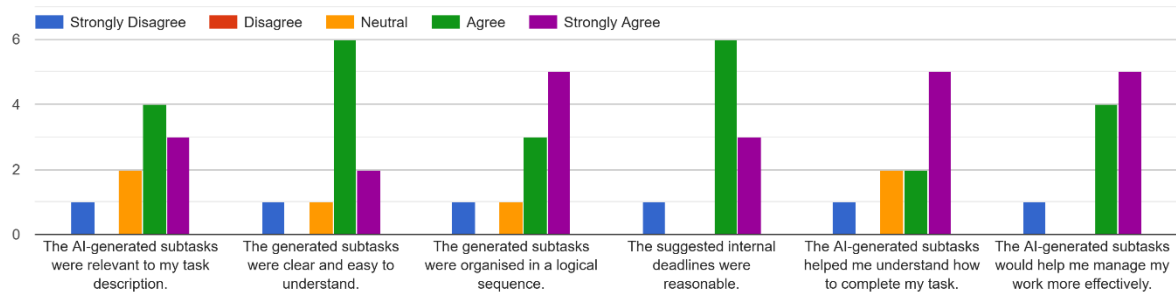


Figure 36: Ratings for AI-Generated Subtasks

Subtask Management

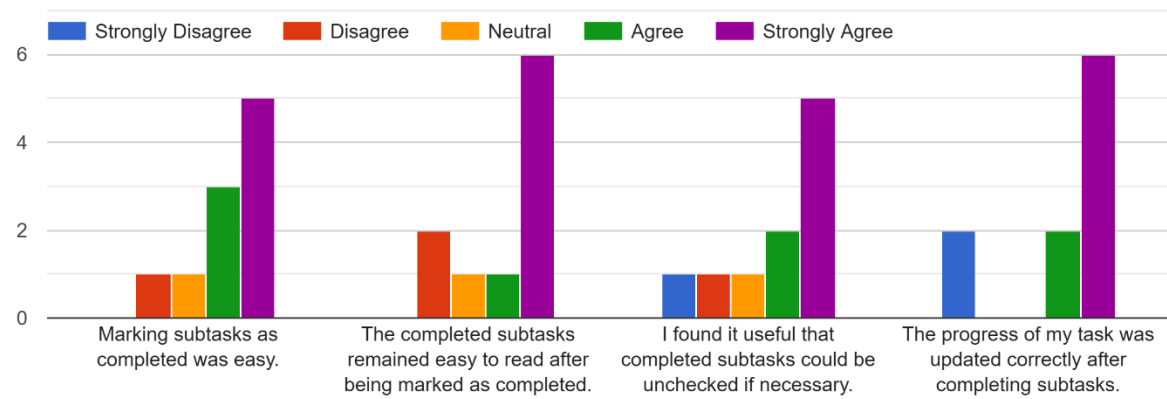


Figure 37: Ratings for Subtasks Management

Overall, I was satisfied with the task management and AI-assisted planning features.

10 responses

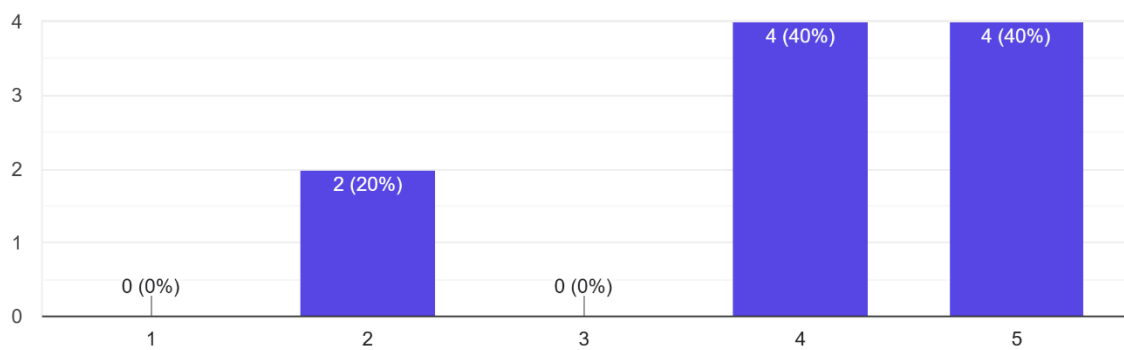


Figure 38: Overall Experience for AI-Task Management

4.4.2.3 Reminder Notifications and Nudges

The reminder notification and nudge feature was tested to check its effectiveness in motivating the participants to continue their educational activities and to decrease procrastination. The proposed application functionality is different from the traditional reminder systems which only inform the users of the approaching deadlines, it includes personalized reminders and motivational nudges depending on the user's expected procrastination risk level. The strategy is to offer timely interventions that promote timely task completion prior to overdue status.

Participants judged the reminder system to be effective at informing them of upcoming reminders, overdue tasks, completed tasks, experience point (XP) rewards, and changes in the virtual garden. Furthermore, participants evaluated the effectiveness of the personalised nudges in motivating them to stay productive during the day.

In general, the participants had good attitudes with regard to the notification system. Most participants indicated either an "agree" or "strongly agree" when asked whether the reminders were sent at appropriate times and whether they were effective in motivating the participant to remind him or herself of important academic tasks. The customised nudges were also felt to be useful to get tasks done and stay motivated. The results indicate that, by incorporating a personalised reminder notification system with the prediction of procrastination risk, users will be supported continuously during academic processes, not just when a set of tasks is due.

The notification system also helped to support gamification aspects of the application by providing information on experience points earned, garden growth, wilting of gardens, completion and overdue tasks. Participants had continuous feedback, which kept them on track with their progress and strengthened the association between completing the task and earning a virtual reward. This, in turn, helped to keep the user more engaged with the application as a whole and acted as a reminder system.

Participants tended to be positive about the reminder and nudge system, but some suggestions were made to make notifications more customisable, flexible in timing and a greater variety.

The average ratings for participants of the reminder notification and nudge feature are shown from Figure 39 to Figure 42.

Reminder Notifications

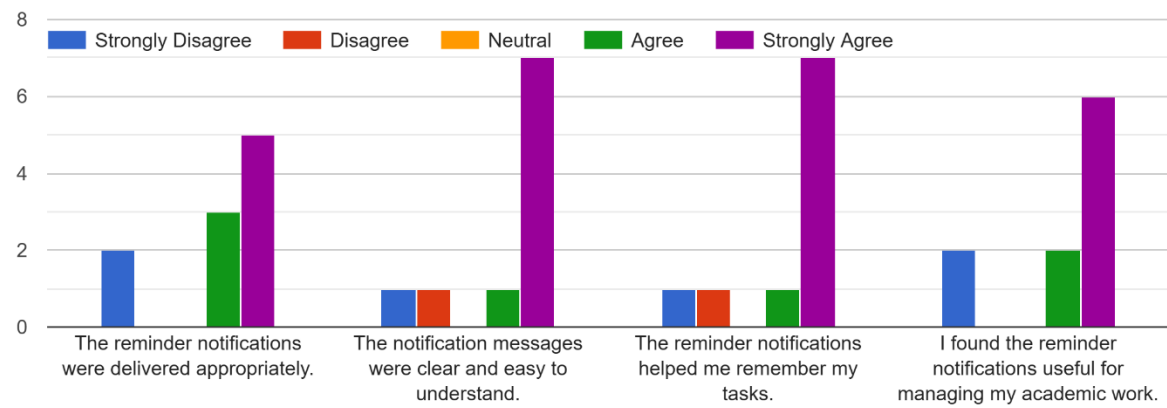


Figure 39: Ratings for Reminder Notification

Personalisation

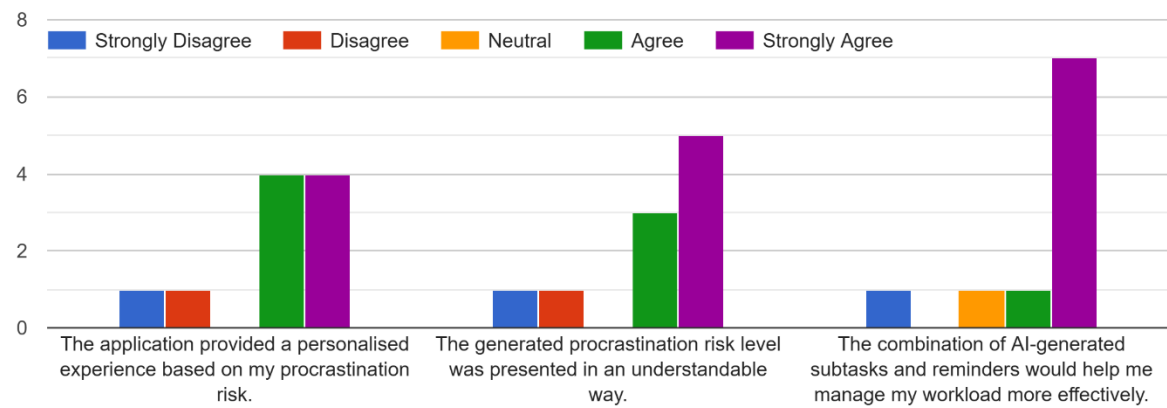


Figure 40: Ratings for Personalized Reminders

Motivation/Nudges

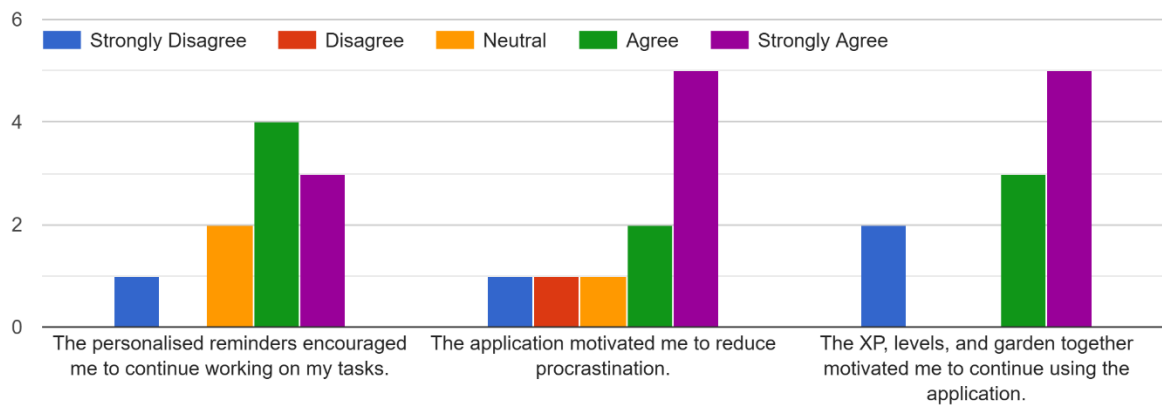


Figure 41: Ratings for Nudges

Overall, I was satisfied with the application's notification and personalisation features.

10 responses

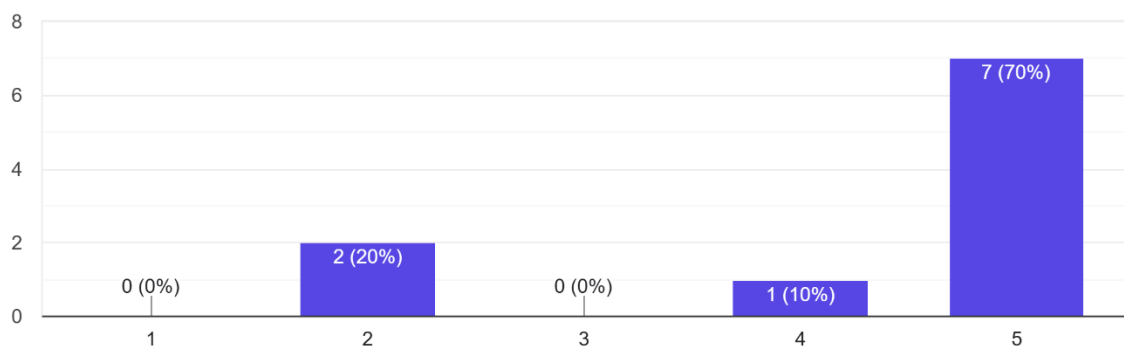


Figure 42: Overall Experience of the Notification and Nudges

4.4.2.4 Gamification Features

The gamification features were tested for their ability to motivate users and promote regular task completion. Nudgr adds gamification features to the typical task management app, such as levels, experience points (XP), streaks, and a virtual garden that shows the user's productivity. These features were intended to give instant feedback and reward, which will make task completion more interesting and rewarding.

Participants rated the usefulness of the experience point system, level advancement, virtual garden, and general motivation of gamification elements. The evaluation also considered whether these factors motivated the participants to complete their academic assignments more regularly and to maintain their engagement with the app in the long term.

In general, the participants had favourable attitudes about the gamification system. The majority agreed or strongly agreed that experience points/levels created extra achievement after completing tasks. The virtual garden also received positive feedback, as participants reported that watching the garden grow based on their experience points increased the fun and encouraged them to have productive study habits.

One of the main features of the virtual garden is that it is dynamic in nature and it is an indicator of the user's productivity. As users finish their tasks and subtasks, they collect experience points, which lead to levels and help the garden become healthier and more vibrant. However, if tasks are not completed on time, the user will lose experience points, and the garden will wither. This visual feedback mechanism helps users to see the effects of their procrastination and gives them feedback on doing things repeatedly. The application integrates the state of the virtual garden into the productivity of each user, thus motivating them to keep up with academic activities instead of postponing their work until deadlines.

Most participants also reported that they found this application to be more engaging than a typical task management tool due to the virtual garden, level-up, and XP. The results indicate a possible successful approach to user engagement by adopting gamification in the academic task management system, which makes it a continuous motivator to complete academic tasks.

The gamification features were largely appreciated by participants, but a few suggestions were made to improve the experience, such as adding more garden customisation options, adding more plant species, adding animations, more badges, and more achievement rewards.

Figure 43 to 47 below represents the evaluation of gamification features.

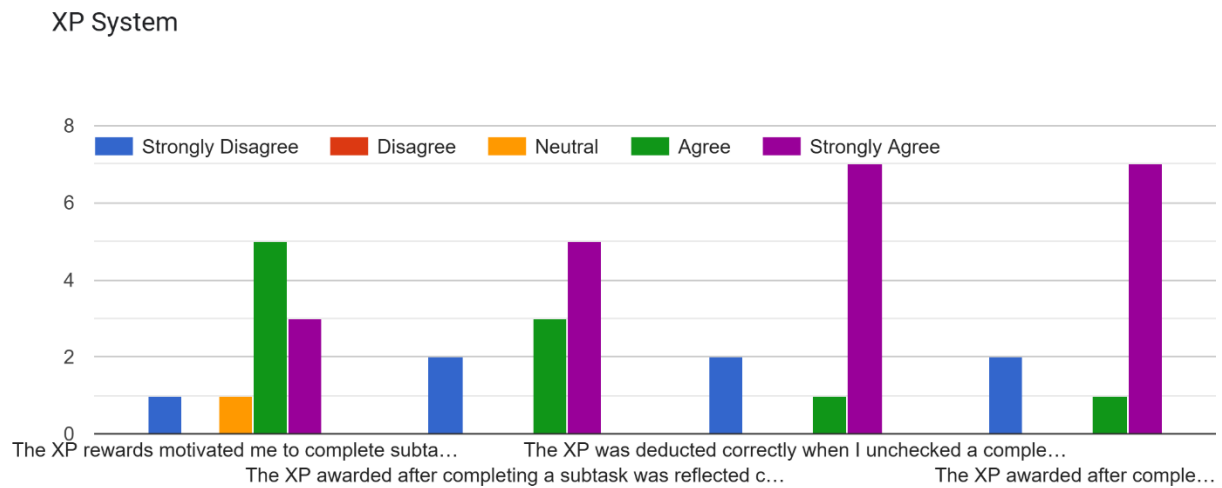


Figure 43: Ratings for XP System

Progress & Levels

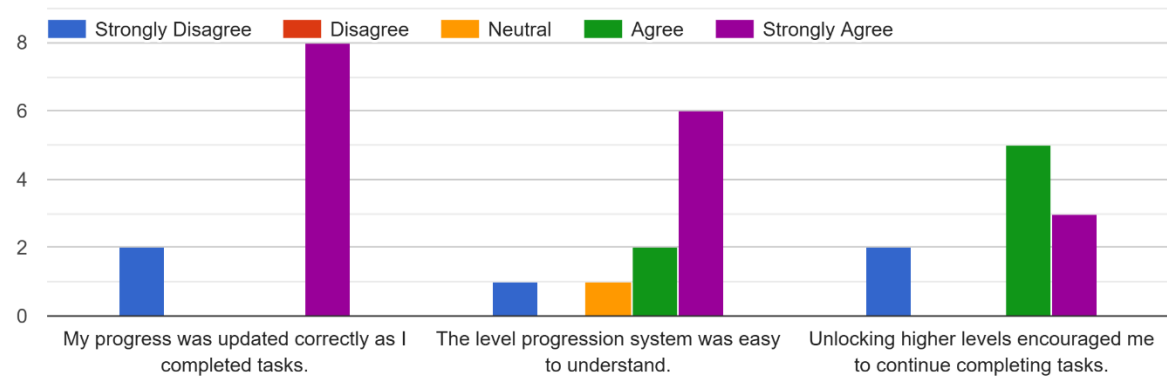


Figure 44: Ratings for Levels and Progress

Garden

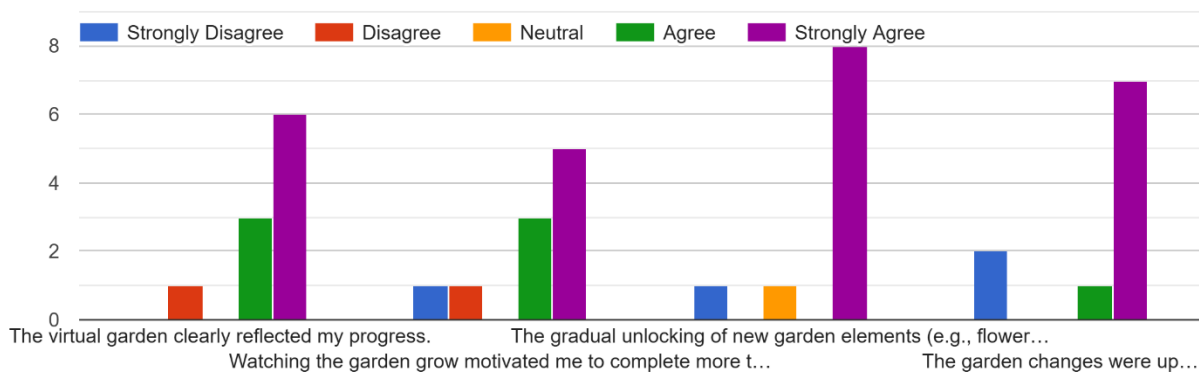


Figure 45: Ratings for the Garden

Recovery Mode

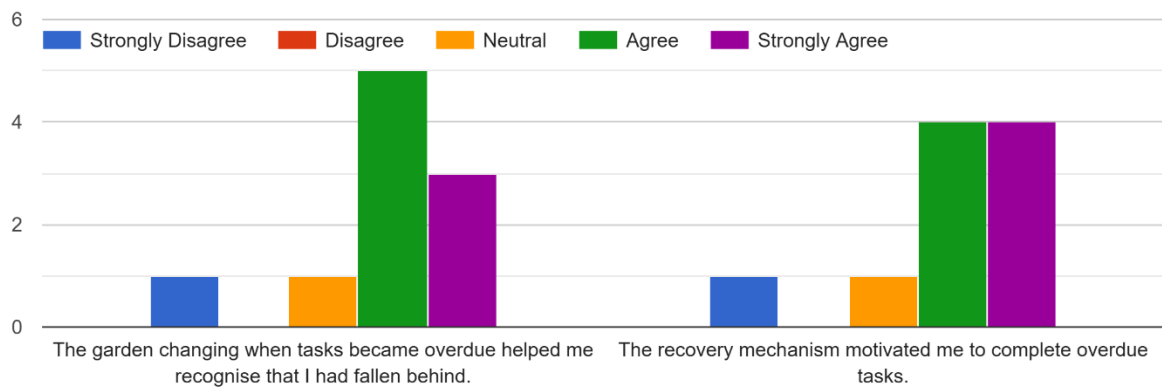


Figure 46: Ratings for Recovery Mode for the Garden

Overall, the gamification features made task management more enjoyable.

10 responses

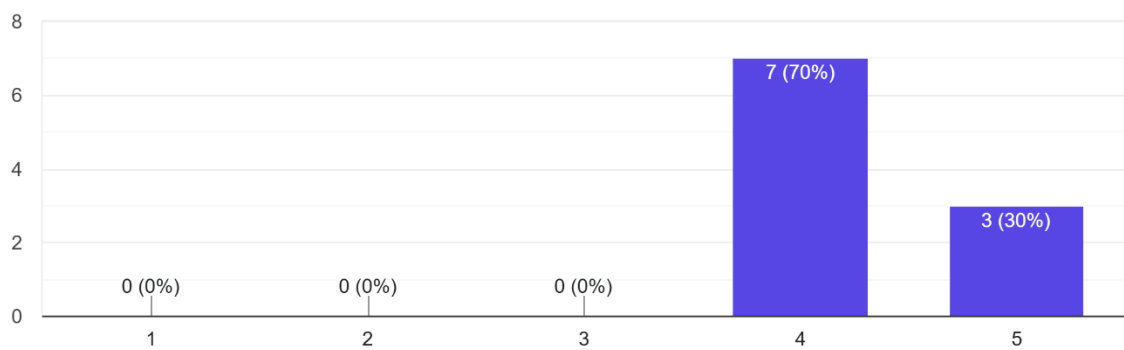


Figure 47: Overall Experience of the Gamified Features

4.4.2.5 Analytics Dashboard

The Analytics Dashboard was tested to see if it provided users with easily accessible data about their productivity and task completion progress. The dashboard was created to give users a visual overview of their academic performance, showing them information like the number of tasks they've made, completed tasks, task completion rate, active streak, experience points (XP) earned, and how tasks are distributed. The dashboard is designed to be visually appealing and easy to use, with data presented in a clear and organised format. The dashboard will be visually appealing and easy to navigate, with data organized in a clear and intuitive format.

Participants judged the dashboard on how easy it is to understand the information presented, the value of the statistics shown, and its ability to inspire them to continue with tasks. The visual representation of the productivity data was also evaluated to determine whether or not the data was easily understood by the participants in terms of their academic progress.

Overall, participants responded positively towards the Analytics Dashboard. The majority of people reported they agreed or strongly agreed that the dashboard made their productivity data easy to understand. The statistics shown were also found to be useful to track academic progress and to let participants see instantly how he or she was going at the tasks, how many XPs were earned and how he or she was doing in terms of productivity. Additionally, it was reported that the participants found the dashboard helpful in motivating them to complete activities and engage in regular studies.

These were complemented by the addition of the Analytics Dashboard to the Task Management and Gamification modules, making it even more useful. The dashboard was automatically updated in real time to show the latest productivity statistics as users progressed through the tasks and earned experience points. The immediate visual feedback allowed the participants to track instantly the results of their activities without having to track it manually, which meant they could always have an awareness of their academic performance.

The general opinion of the dashboard was positive, but some respondents offered further analytical capabilities that could be added, including more granular productivity trends, weekly and monthly performance summaries, and more extensive graphical visualisations.

The averages of the Analytics Dashboard's participant ratings are displayed in Figure 48.

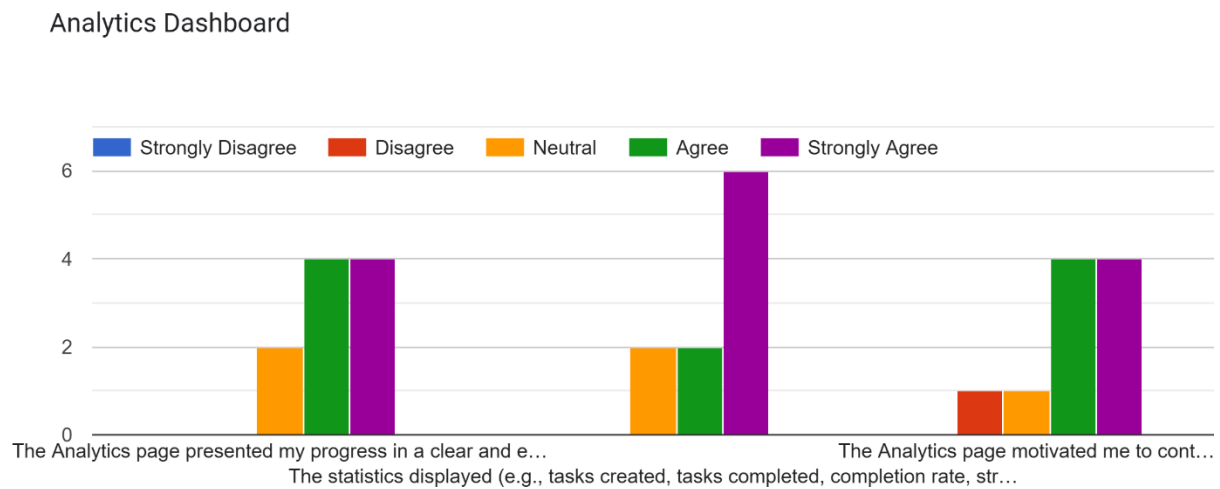


Figure 48: Ratings for Analytics Dashboard

4.4.2.6 Overall User Experience

The overall user experience was measured to assess the users' general impression about the Nudgr mobile application from the interaction with all the features. This evaluation differed from the previous sections, which were on individual parts of the application, as it measured the application as a whole system, covering aspects of ease of use, navigation, responsiveness, enjoyment, intention for continued use, willingness to recommend the application to others, and overall user satisfaction.

The participants rated a number of aspects of the overall user experience on a five-point Likert scale from Strongly Disagree to Strongly Agree. The assessment focused on the ease of learning the application, intuitive navigation between the various screens, smooth operation during use, ease of enjoyment of the application, and willingness to use and recommend the application in the future.

The general attitude of the participants towards the mobile application Nudgr was positive. The learning curve of the application was agreed or strongly agreed by most of the participants and the navigability of different screens was found to be intuitive and easy. The combination of task management, AI capabilities, personalised notifications, analytics and gamification also contributed to a seamless and engaging user experience, according to participants. The results indicate that the application integrated all the functionalities in a single interface without adding any unnecessary complexity.

The analysis also reveals that the participants were mostly satisfied with the use of the application and found that the application can be applied to assist them in performing academic tasks. Majority of participants said that they would use the application in their studies and would recommend it to other university students. The acceptance responses from the users indicate that the application could be used for students to help them cope with academic procrastination outside of the pilot testing context.

The overall feedback was positive, but some participants noted some minor usability problems and offered a few more features that could be added to make the application even better. The comments made were mostly about the improvements to be made to the interface, more flexibility, and performance enhancements rather than the system itself.

The average ratings for the overall user experience are shown in Figure 49.

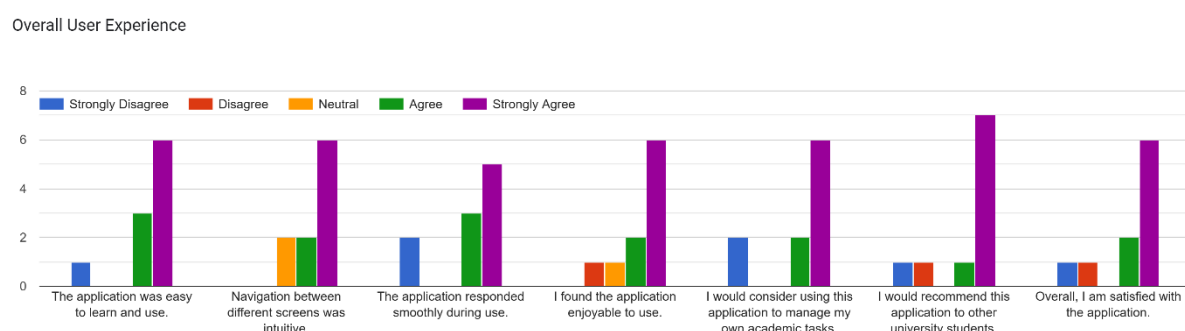


Figure 49: Overall User Experience of the Application

4.5 System Usability Scale (SUS)

To complement the feature evaluation questionnaire, the usability of the developed Nudgr mobile application was further assessed using the System Usability Scale (SUS). The SUS is a widely recognised standardised questionnaire consisting of ten statements designed to measure users' perceived usability of an interactive system. Unlike the feature-specific evaluation presented in the previous section, the SUS provides an overall measurement of the application's usability, allowing comparisons with established usability benchmarks.

The SUS questionnaire was completed by all ten participants after they had finished exploring the application's major features during the pilot testing session. Individual SUS scores were calculated using the standard scoring procedure described in Chapter 3.

Figure 50 below shows the chart for System Usability Scale.

System Usability Scale (SUS)

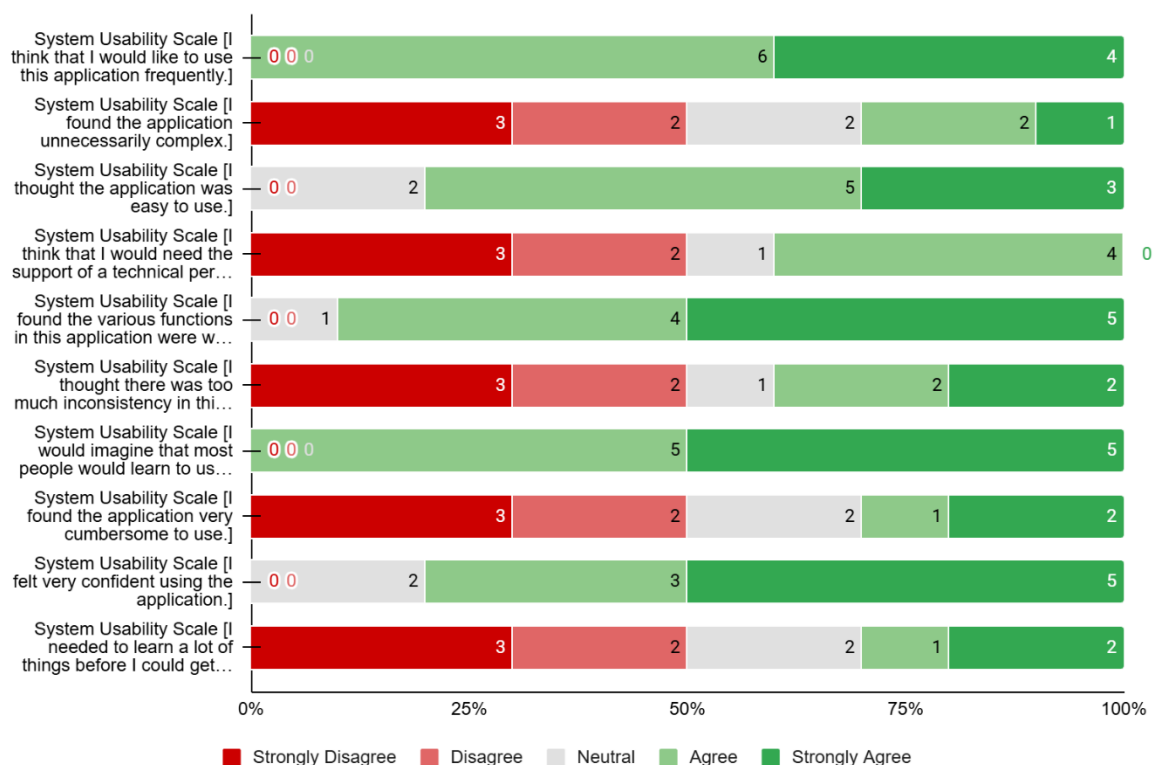


Figure 50: System Usability Scale (SUS) Summary Chart

4.5.1 SUS Score Results

The individual System Usability Scale (SUS) scores obtained from the ten participants are presented in Table X. The results show an average SUS score of 70.75, with individual scores ranging from 47.50 to 100.00. The standard deviation of 19.51 indicates some variation in participants' perceptions of the application's usability.

Table 19 summarizes the SUS results.

Measure	Value
Number of Participants	10
Average SUS Score	70.75
Standard Deviation	19.51
Highest SUS Score	100.00
Lowest SUS Score	47.50

Table 19: System Usability Scale (SUS) Results

The bar chart below in Figure 51 illustrates the individual SUS scores for each participant. Although several participants assigned high usability scores above 90, a few participants reported noticeably lower scores, resulting in a relatively wide distribution of responses.

SUS Score vs. Participant

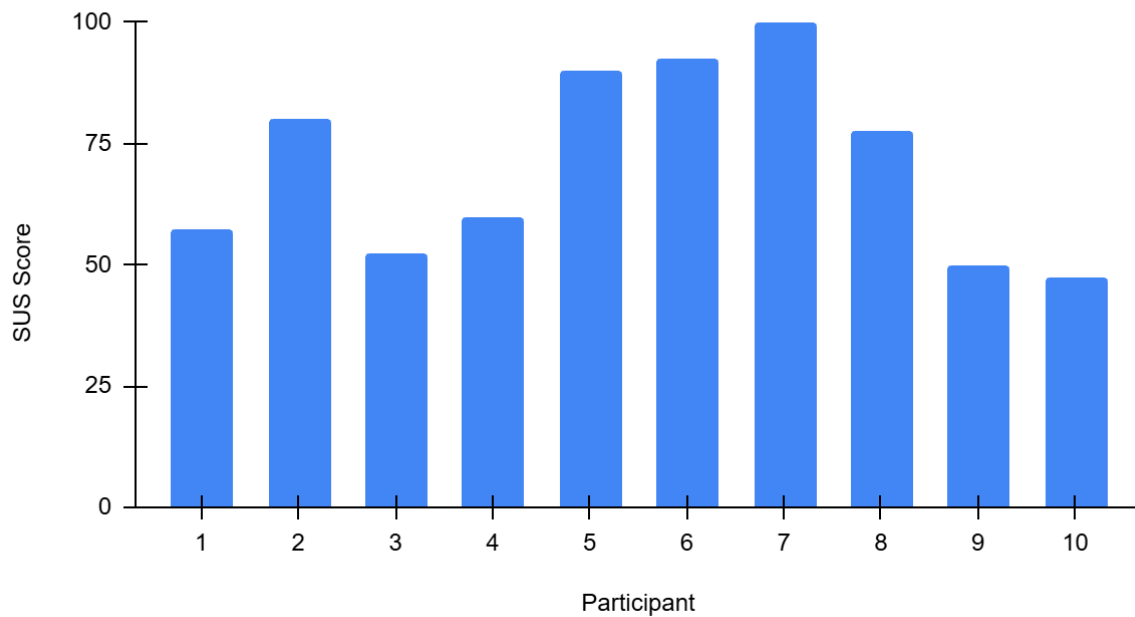


Figure 51: Individual SUS Scores

Overall, the average SUS score of 70.75 exceeds the commonly accepted benchmark score of 68, indicating that the developed Nudgr mobile application achieved good overall usability. The variation in participant scores suggests that while most participants found the application easy to use, some users experienced usability issues during the pilot testing, which are discussed in the following section.

4.5.2 SUS Discussion

The developed Nudgr mobile application achieved an average System Usability Scale (SUS) score of 70.75. Based on established SUS interpretation guidelines, this score falls within the Good usability category and is above the commonly accepted average benchmark score of 68. This indicates that participants generally perceived the application as usable, easy to learn, and suitable for supporting academic task management.

The responses to the individual SUS statements further support this finding. Most participants agreed or strongly agreed that they would like to use the application frequently, found it easy to use, believed that the application's functions were well integrated, and felt that most users would be able to learn the application quickly. Participants also generally expressed confidence while interacting with the application, suggesting that the user interface and navigation were relatively intuitive.

Although the overall usability evaluation was positive, the relatively large standard deviation of 19.51 indicates noticeable variation in participants' experiences. While several participants assigned very high usability scores, including one participant who awarded the maximum score of 100, others reported considerably lower scores. These differences can be attributed to several technical issues encountered during pilot testing rather than the overall application design.

Based on the qualitative feedback collected from participants, several usability issues were identified. These included occasional login failures caused by temporary Firebase quota limitations, duplicate account registration, duplicate task names, incorrect experience point updates after task completion, navigation issues within the virtual garden module, delays in AI-assisted task generation due to API processing limits, and notification backlogs when error messages were triggered repeatedly. Although these issues affected several participants' experiences, most were related to implementation constraints or external service limitations rather than fundamental usability problems.

Overall, the SUS evaluation demonstrates that the developed application achieved a satisfactory level of usability while identifying several practical improvements that can further enhance the user experience. Addressing the identified technical issues and improving the scalability of the backend infrastructure are expected to increase usability further in future versions of the application.

4.6 Qualitative Feedback

Besides the quantitative evaluation obtained through the feature evaluation questionnaire and the System Usability Scale (SUS), participants were invited to provide open-ended comments

regarding their overall experience using the Nudgr mobile application. These qualitative responses provided additional insights into the application's strengths, technical issues encountered during testing, and potential improvements for future development. The responses were reviewed and grouped into several common themes to facilitate discussion.

4.6.1 Positive Feedback

Overall, participants expressed positive opinions regarding the concept and functionality of the Nudgr mobile application. Many participants appreciated that the application combined multiple productivity features into a single platform instead of requiring separate applications for task management, reminders, and academic planning.

One of the most frequently praised features was the AI-assisted task planning module. Participants indicated that automatically generating subtasks made large academic assignments appear more manageable and reduced the difficulty of deciding how to begin their work. Several participants commented that the generated subtasks helped them organize their assignments more effectively and reduced feelings of being overwhelmed.

The gamification system also received favorable feedback. Participants reported that linking task completion with the growth of a virtual garden made the application more enjoyable to use and provided additional motivation to complete academic tasks. Rather than viewing task completion as a routine activity, users appreciated receiving immediate visual feedback through garden growth and experience point progression.

Participants also responded positively towards the analytics dashboard, stating that the productivity statistics helped them monitor their study habits and provided greater awareness of their academic progress. Similarly, the personalized reminder notifications were considered useful for helping participants remember upcoming deadlines and maintain regular engagement with their academic tasks.

Overall, the qualitative feedback indicates that participants considered the integration of machine learning, artificial intelligence, personalized reminders, analytics, and gamification to be one of the application's primary strengths.

4.6.2 Technical Issues Identified

Although participants generally reported a positive experience, several technical issues were identified during the pilot testing. Most of these issues were implementation-related and were subsequently investigated after the evaluation.

Several participants experienced authentication problems, including temporary login failures after registration and occasional difficulties accessing existing accounts. Investigation revealed that these issues were primarily caused by Firebase Authentication and Cloud Firestore quota limitations during periods of frequent testing rather than problems with the application's authentication logic. As multiple participants accessed the application within a short period, the allocated cloud resources occasionally reached their usage limits, temporarily preventing successful authentication and database synchronization.

Participants also identified several application-level issues. These included duplicate task names being permitted, experience points not always updating correctly after task completion, inconsistencies in completed task counters, navigation problems within the virtual garden module, and notification pop-up backlogs when error messages were triggered repeatedly. Although these issues did not prevent participants from completing the required testing activities, they highlighted opportunities for improving the robustness of the application before wider deployment.

Another commonly reported issue involved the AI-assisted task planning module. Some participants experienced "Failed to fetch" messages during the first or second attempts to generate AI subtasks, with successful generation occurring only after several attempts. Further investigation indicated that these delays were caused by Gemini API processing limitations and request quotas associated with the free-tier API rather than faults in the application's implementation. Participants also observed that detailed API error messages were displayed

directly within the application, unintentionally revealing information such as model names and quota limitations. Hiding these system-generated messages and replacing them with user-friendly notifications would improve both usability and security in future versions.

Overall, the majority of the reported issues were related to temporary service limitations or minor implementation bugs rather than fundamental architectural weaknesses. Most can be addressed through application refinement and deployment using production-level cloud infrastructure.

4.6.3 User Suggestions

Participants also provided several constructive suggestions for improving future versions of the application.

One frequently suggested improvement was expanding the AI-assisted task planning feature by generating more detailed and personalized subtasks based on different assignment types and individual study preferences. Participants believed that more context-aware AI recommendations would further improve the usefulness of the application when planning larger academic projects.

Several participants also recommended enhancing the analytics dashboard by providing more comprehensive productivity reports, including longer-term performance trends and additional visualizations of study behavior. More detailed analytics were considered useful for helping students better understand their productivity patterns over time.

Additional suggestions focused on improving the application's flexibility through greater customization of reminder notifications and motivational nudges. Participants expressed interest in controlling notification frequency, reminder timing, and other personal preferences according to their individual study habits.

Finally, several participants suggested expanding the gamification system by introducing additional rewards, achievements, badges, or unlockable content to further encourage long-term engagement. These recommendations indicate that participants viewed the existing

gamification features positively and believed that further expansion could increase user motivation.

Overall, the qualitative feedback demonstrates that participants were satisfied with the application's core functionality while identifying several practical enhancements that could improve usability, personalization, and long-term user engagement.

4.6.4 Discussion of Qualitative Feedback

The qualitative findings complement the quantitative results presented in the previous sections. Participants generally viewed the Nudgr mobile application as useful, engaging, and easy to use, particularly appreciating the integration of AI-assisted task planning, personalized reminders, productivity analytics, and gamification within a single platform. These findings are consistent with the positive feature evaluation scores and the average System Usability Scale (SUS) score of 70.75, indicating that the application achieved good overall usability.

The technical issues identified during pilot testing primarily reflected implementation and deployment constraints rather than conceptual weaknesses in the proposed system. Authentication failures, temporary database synchronization issues, and delays in AI-generated subtasks were largely attributable to the limitations of the free-tier Firebase and Gemini API services used during development. Meanwhile, several application-specific issues, such as duplicate task creation, experience point inconsistencies, and notification behavior, represent software refinements that can be addressed in future development.

Overall, the qualitative feedback demonstrates that the proposed application successfully achieved its intended functionality while also providing valuable insights for improving system robustness, scalability, and user experience before wider deployment.

4.7 Chapter Summary

This chapter presented the results obtained from the development and evaluation of the proposed Nudgr mobile application. The machine learning experiments demonstrated that the Random Forest classifier achieved the best overall performance among the evaluated models and was therefore selected for integration into the application. The feature importance analysis further identified several psychological and behavioral factors that contributed significantly to predicting students' procrastination risk.

Following the machine learning phase, the completed mobile application was evaluated through functional testing, integration testing, and pilot testing. The functional and integration testing results confirmed that all major application modules, including user authentication, procrastination risk prediction, AI-assisted task planning, task management, reminder notifications, gamification, and analytics, operated correctly and communicated successfully with the integrated backend services.

The pilot testing provided both quantitative and qualitative evidence regarding the usability and effectiveness of the application. The feature evaluation questionnaire indicated that participants responded positively towards the application's onboarding process, procrastination assessment, task management, AI-generated subtasks, reminder notifications, gamification features, analytics dashboard, and overall user experience. Furthermore, the application achieved an average System Usability Scale (SUS) score of 70.75, which exceeds the commonly accepted benchmark score of 68, indicating good overall usability.

The qualitative feedback further complemented these findings by highlighting the application's strengths while identifying several opportunities for improvement. Participants particularly appreciated the integration of AI-assisted task planning, personalized reminders, productivity analytics, and the virtual garden gamification system. At the same time, several implementation-related issues, including temporary cloud service limitations and minor software bugs, were identified during pilot testing. These findings provide valuable insights for refining the application and improving its robustness before wider deployment.

Overall, the results demonstrate that the objectives of this study were successfully achieved. The proposed Nudgr application successfully integrates machine learning, generative artificial

intelligence, cloud computing, personalized behavioral interventions, and gamification into a unified mobile platform that supports university students in managing academic procrastination. The limitations identified during the evaluation and the recommendations for future enhancement are discussed in the following chapter.

5.0 Conclusion and Future Work

5.1 Conclusion

5.1.1 Achievement of Objective 1

The first objective of this project was to investigate academic procrastination behaviors and suitable gamification strategies by reviewing existing literature on academic procrastination and motivational design. This objective was successfully achieved through a comprehensive review of recent studies that examined the psychological, behavioral, and cognitive factors contributing to academic procrastination, together with existing gamification approaches used to encourage behavioral change in educational applications.

The literature review identified several important factors associated with academic procrastination, including perfectionism, poor time management, low self-efficacy, self-regulated learning, fear of failure, resilience, and task aversiveness. These findings provided a theoretical foundation for selecting the behavioral features used in the machine learning model and justified the inclusion of an initial procrastination assessment within the mobile application. The review also highlighted the negative effects of academic procrastination on students' academic performance, motivation, psychological well-being, and learning outcomes, reinforcing the importance of developing an early intervention system capable of providing personalized support.

In addition, the review of gamification literature identified several motivational elements that have been shown to improve user engagement and encourage sustained behavioral change. Commonly adopted techniques included experience points, achievement systems, progress tracking, visual feedback, leaderboards, rewards, and personalized reminders. These findings directly influenced the design of the Nudgr application by guiding the implementation of the virtual garden, experience point system, productivity analytics, personalized nudges, reminder notifications, and progress visualisation.

Overall, the literature review established the theoretical and practical foundation for the entire project. It guided the selection of machine learning features, informed the design of the application's behavioral interventions and gamification mechanisms, and ensured that the

proposed solution was grounded in existing research rather than being developed solely from intuition. Therefore, the first project objective was successfully achieved.

5.1.2 Achievement of Objective 2

The second objective of this project was to develop a machine learning model capable of predicting university students' risk of academic procrastination by selecting a suitable machine learning algorithm, training predictive models using behavioral data, and evaluating their performance in identifying students at risk of procrastination. This objective was successfully achieved through the development and comparison of multiple supervised machine learning models using responses obtained from the PASS questionnaire dataset.

Seven classification algorithms, namely Logistic Regression, Support Vector Machine (SVM), Decision Tree, Random Forest, Gradient Boosting, K-Nearest Neighbours (k-NN), and Multi-Layer Perceptron (MLP), were evaluated using multiple performance metrics. Since the primary purpose of the proposed application is to identify students who may require early intervention, recall was prioritised during model selection to minimise the likelihood of failing to identify students with high procrastination risk. Based on the evaluation results presented in Chapter 4, the Random Forest classifier achieved the most suitable overall performance and was therefore selected as the final prediction model for integration into the mobile application.

Feature importance analysis further demonstrated that several behavioral and psychological factors, including perfectionism, low energy, feelings of being overwhelmed, social anxiety, and time management, contributed significantly to predicting academic procrastination. These findings not only improved the interpretability of the prediction model but also guided the selection of assessment questions implemented within the Nudgr application.

Although the evaluation identified limitations related to dataset imbalance, particularly the occurrence of false positive predictions among students with lower procrastination tendencies, the developed model provided an effective foundation for personalized behavioral interventions within the proposed application. The implementation of a temporary application-

level fallback mechanism further improved the practical behavior of the prediction system during deployment while highlighting opportunities for future improvement.

Overall, the machine learning component successfully fulfilled the second project objective by developing, evaluating, and integrating a predictive model capable of supporting personalized procrastination interventions within the Nudgr mobile application.

5.1.3 Achievement of Objective 3

The third objective of this project was to create and develop a gamified mobile application that supports self-regulated learning by integrating machine learning prediction, Generative Artificial Intelligence (Generative AI), personalized reminders, behavioral nudges, and gamification features to encourage sustained engagement and improve academic task management among university students. This objective was successfully achieved through the development of the Nudgr mobile application using Flutter as the cross-platform development framework, Firebase as the cloud backend infrastructure, Google's Gemini API for AI-assisted task planning, and the trained Random Forest model for procrastination risk prediction.

The completed application successfully integrates multiple intelligent technologies into a single platform designed to support students throughout their academic task management process. After completing the initial procrastination assessment, users receive a personalized procrastination risk prediction generated by the integrated machine learning model. This prediction is subsequently used to personalize several application features, including the frequency of motivational nudges and behavioral interventions. In addition, the AI-assisted task planning module automatically decomposes large academic tasks into smaller, manageable subtasks with suggested internal deadlines, reducing the cognitive effort required to begin complex assignments.

Beyond intelligent planning and prediction, the application incorporates several behavioral intervention mechanisms to encourage consistent study habits. These include personalized reminder notifications, motivational nudges, productivity analytics, and a gamification system consisting of experience points, user levels, daily streaks, leaderboards, and a virtual garden

that visually reflects users' academic progress. Rather than functioning as independent features, these components operate together to provide continuous encouragement, reinforce positive study behaviors, and increase user engagement throughout the academic semester.

The completed application was subsequently evaluated through functional testing, integration testing, and pilot testing involving university students. The evaluation confirmed that all major modules operated successfully and that the integrated system functioned as intended. Furthermore, the pilot testing demonstrated positive user acceptance across the application's core features, while the System Usability Scale (SUS) evaluation achieved an average score of 70.75, indicating good overall usability and suggesting that the application was generally easy to learn and use.

Overall, the successful development, implementation, and evaluation of the Nudgr mobile application demonstrate that the third project objective was fully achieved. By integrating machine learning, Generative Artificial Intelligence, cloud computing, personalized behavioral interventions, and gamification into a unified mobile platform, the proposed application provides an intelligent and engaging approach to supporting university students in managing academic procrastination and developing more effective self-regulated learning behaviors.

5.1.4 Overall Conclusion

This project set out to develop an intelligent mobile application capable of assisting university students in managing academic procrastination through the integration of machine learning, Generative Artificial Intelligence (Generative AI), personalized behavioral interventions, and gamification. Based on the findings presented throughout this study, all three project objectives were successfully achieved. The literature review established a strong theoretical foundation by identifying the behavioral and motivational factors associated with academic procrastination and suitable gamification strategies. These findings subsequently informed both the machine learning model development and the design of the mobile application.

The machine learning component successfully identified an appropriate predictive model capable of estimating students' procrastination risk based on psychological and behavioral

characteristics. The selected Random Forest classifier demonstrated the most suitable performance among the evaluated algorithms and provided the basis for delivering personalized interventions within the application. Building upon these predictive capabilities, the Nudgr mobile application successfully integrated machine learning prediction with AI-assisted task planning, cloud-based data management, personalized reminders, behavioral nudges, productivity analytics, and gamification into a unified platform designed to support students throughout their academic journey.

The evaluation conducted through functional testing, integration testing, and pilot testing demonstrated that the proposed application was both technically feasible and positively received by its intended users. The successful operation of the integrated system, together with favourable feature evaluation results and an above-average System Usability Scale (SUS) score, indicates that the proposed approach provides an engaging and practical solution for supporting academic task management and encouraging positive behavioral change.

Overall, this study demonstrates that combining predictive analytics with intelligent behavioral interventions within a mobile application has considerable potential to support university students in reducing academic procrastination. Beyond delivering a functional application, this research contributes a practical framework for integrating machine learning, Generative Artificial Intelligence, cloud computing, and gamification into a unified educational support system. It is anticipated that the findings of this study will provide a useful foundation for future research and further development of intelligent technologies that promote self-regulated learning, improve academic productivity, and support students' long-term academic success.

5.2 Limitations of the Study

Although the proposed Nudgr mobile application successfully demonstrated the feasibility of integrating machine learning, Generative Artificial Intelligence (Generative AI), cloud computing, personalised behavioural interventions, and gamification into a unified academic procrastination management platform, several limitations were identified throughout the development and evaluation of the system. These limitations should be considered when

interpreting the findings of this study and provide valuable opportunities for future research and system enhancement.

5.2.1 Dataset Imbalance and Prediction Performance

One of the primary limitations of this study relates to the dataset used to develop the machine learning prediction model. Although the selected dataset contained validated psychological and behavioral indicators of academic procrastination, further analysis conducted during the later stages of the project revealed that the target classes became noticeably imbalanced after the procrastination scores were converted into risk categories. The dataset contained substantially more students with higher procrastination tendencies than students demonstrating consistently low procrastination behavior.

This imbalance influenced the performance of the Random Forest classifier during prediction. While the model successfully achieved a high recall in identifying students who were likely to procrastinate, it also produced a relatively large number of false positive predictions by classifying some low-risk students as high-risk users. Consequently, the model demonstrated stronger sensitivity towards identifying procrastinating students than distinguishing users who consistently exhibited effective self-regulated learning behaviors.

Although the Synthetic Minority Oversampling Technique (SMOTE) was applied during model training to reduce the effects of class imbalance, synthetic oversampling alone was insufficient to overcome the limited representation of genuine low-risk students within the original dataset. Since SMOTE generates new samples based on existing observations rather than collecting new behavioral data, the diversity of the minority class remained limited. As a result, the prediction model continued to learn considerably more behavioral patterns associated with higher procrastination than with consistently low procrastination behavior.

From the perspective of an intervention-based application, prioritising recall was considered appropriate because failing to identify a genuinely procrastinating student may prevent timely behavioral intervention, whereas incorrectly classifying a low-risk student as high-risk primarily results in additional motivational support. Nevertheless, reducing false positive

predictions remains important for improving the reliability and credibility of future versions of the prediction model. Overall, this limitation highlights that machine learning performance depends not only on algorithm selection but also on the quality, diversity, and representativeness of the training dataset.

5.2.2 Rules-Based Fallback Mechanisms

Another limitation of this study concerns the temporary rule-based fallback mechanism incorporated into the mobile application during the system integration phase. After integrating the trained Random Forest classifier into the application, it was observed that some users who consistently selected questionnaire responses representing very low procrastination tendencies were occasionally classified as high-risk. This behavior reduced the practical reliability of the prediction module because users whose responses clearly reflected low procrastination characteristics sometimes received unnecessarily frequent behavioral interventions.

To improve the consistency of prediction results during application deployment, an additional rule-based decision layer was introduced before invoking the machine learning model. Instead of relying entirely on the prediction output, users whose assessment responses remained consistently below a predefined threshold were directly classified as low-risk. This fallback mechanism reduced unnecessary high-risk predictions and provided behavior that more closely matched users' assessment responses during pilot testing.

Although this engineering solution improved the practical behavior of the application, several limitations remain. First, the threshold value was determined empirically through repeated application testing and observation rather than systematic experimental optimization or statistical validation. Consequently, the selected threshold may not represent the optimal decision boundary and cannot be assumed to generalize across different datasets or user populations. Different threshold values may produce different prediction outcomes depending on the characteristics of the assessment responses.

Furthermore, the fallback mechanism operates independently of the trained Random Forest classifier and therefore should not be regarded as an improvement to the machine learning

model itself. Instead, it serves as an application-level control mechanism that compensates for limitations arising from the training dataset. Since the underlying prediction model remains unchanged, the fallback mechanism does not eliminate the original cause of the false positive predictions but merely reduces their impact during application usage.

Therefore, while the fallback mechanism successfully improved the consistency of the deployed application during pilot testing, it should be regarded as a temporary engineering solution. Future work should focus on improving the prediction model through better dataset quality and systematic threshold optimization rather than relying on manually defined decision rules.

5.2.3 Pilot Testing Scope

Another limitation of this study relates to the scope of the pilot testing conducted to evaluate the developed mobile application. The evaluation involved a relatively small number of university students over a limited testing period. Although the participants provided valuable feedback regarding the functionality, usability, and overall user experience of the application, the sample size may not fully represent the diversity of the broader university student population. Students from different academic disciplines, educational backgrounds, institutions, and learning environments may demonstrate different study behaviors, technology acceptance, and procrastination patterns that were not fully represented during this evaluation.

In addition, the pilot testing primarily focused on evaluating whether the application functioned correctly and whether users could successfully interact with its main features. The evaluation therefore measured users' initial perceptions of usability, feature usefulness, and overall satisfaction rather than the long-term effectiveness of the proposed behavioral interventions. Since academic procrastination develops over extended periods, behavioral change cannot be reliably assessed through a short-term pilot study.

Furthermore, several technical improvements were identified during pilot testing through participants' qualitative feedback. Issues such as duplicate task creation, navigation inconsistencies, delayed experience point updates, temporary authentication failures, and AI

service response delays were observed under specific testing conditions. Although these issues were subsequently investigated and many were addressed during development, they demonstrate that additional large-scale testing is necessary before deployment to a wider user population.

Finally, the pilot testing was conducted under controlled conditions using the development version of the application. Consequently, the evaluation did not fully assess system performance under conditions involving a large number of concurrent users, prolonged application usage, or varying network environments. A larger-scale evaluation conducted over several academic semesters would therefore provide stronger evidence regarding both the usability and long-term effectiveness of the proposed system.

5.2.4 Development Infrastructure and Deployment Limitations

The final limitation concerns the development infrastructure used throughout the implementation and evaluation of the proposed application. The system was developed using the free-tier versions of Firebase Authentication, Cloud Firestore, and Google's Gemini API. While these cloud services provided an effective platform for rapid application development and prototyping, they also imposed resource limitations that affected system behavior during pilot testing.

Several participants experienced temporary issues related to user authentication, database synchronization, and cloud service availability. Examples included login failures after account registration, delayed updates to experience points and completed tasks, and occasional inconsistencies in synchronising user data across different application modules. Investigation showed that these issues were primarily associated with temporary service restrictions and quota limitations imposed by the cloud service providers rather than flaws in the application's architecture or business logic.

Similarly, the AI-assisted task generation module occasionally required multiple attempts before successfully generating subtasks during periods of high request volume. These delays were mainly caused by API processing constraints and request quota limitations within the

Gemini API rather than weaknesses in the prompt design or application implementation. Furthermore, certain API-generated error messages exposed technical implementation details, including quota information and model-specific responses, which would not be appropriate in a production environment. More user-friendly exception handling should therefore be implemented to provide informative feedback without exposing internal system details.

These observations also highlight the importance of scalability when deploying intelligent mobile applications. Although the current cloud infrastructure performed adequately for pilot testing involving a small number of users, additional computing resources, larger database capacity, and higher AI service quotas would be required to support large-scale deployment involving many concurrent users. Nevertheless, the development environment successfully demonstrated the feasibility of integrating machine learning, Generative AI, cloud computing, and mobile application development within a single architecture, providing a solid foundation for future production deployment.

5.2.5 Generalizability of the Machine Learning Model

Another limitation of this study concerns the generalizability of the developed machine learning model. The prediction model was trained using a publicly available academic procrastination dataset rather than behavioral data collected directly from users of the Nudgr mobile application. Although the dataset provided validated psychological and behavioral indicators of academic procrastination, the characteristics of its participants may differ from those of the intended users of the proposed application.

Consequently, the prediction model may not fully represent procrastination behaviors across different universities, academic disciplines, cultural backgrounds, or educational environments. Behavioral factors influencing procrastination may also evolve over time as students' learning habits and educational practices change. Therefore, future research should consider collecting longitudinal behavioral data directly from application users to retrain and validate the prediction model. Such an approach would improve the model's ability to generalize to the target population while further enhancing the reliability of personalized interventions delivered by the application.

5.3 Future Work

5.3.1 Enhancement of the Machine Learning Prediction Model

Although the Random Forest classifier achieved the best overall performance among the evaluated machine learning algorithms, several opportunities remain for improving the prediction model. The findings of this study showed that the quality and representativeness of the training dataset had a significant influence on prediction performance, particularly in reducing false positive predictions for students with consistently low procrastination tendencies. Therefore, future research should prioritize collecting a larger and more balanced dataset comprising students from different universities, academic disciplines, educational levels, and learning environments. A more representative dataset would allow the model to learn more reliable decision boundaries while improving its ability to distinguish between different levels of procrastination risk.

Future studies should also investigate alternative approaches for handling class imbalance beyond traditional oversampling methods such as SMOTE. Techniques including cost-sensitive learning, balanced ensemble methods, probability calibration, threshold optimization, and advanced sampling strategies may further improve classification performance while maintaining the model's ability to identify students requiring behavioral intervention. As larger datasets become available, additional machine learning algorithms such as XGBoost, LightGBM, CatBoost, or suitable deep learning architectures may also be evaluated to determine whether they provide improvements in prediction accuracy, robustness, and generalization compared with the Random Forest classifier.

Finally, future research should place greater emphasis on exploratory data analysis during the early stages of model development. Evaluating class distribution, sampling bias, feature representativeness, and data quality before model training would allow potential prediction biases to be identified earlier, resulting in more reliable machine learning models and reducing the need for application-level correction mechanisms during deployment.

5.3.2 Enhancement of Intelligent Personalized Support

The current implementation successfully integrates Generative Artificial Intelligence (Generative AI) and personalized behavioral interventions to support academic task management. However, future versions of the application could provide more adaptive and context-aware assistance by incorporating additional information into the AI planning process. Rather than generating subtasks based solely on the task title and description, future implementations could also consider factors such as assignment type, subject area, workload, available study time, previous task completion behavior, and users' preferred study schedules. Such contextual information would enable the AI model to generate study plans that are more personalized and better aligned with individual learning needs.

Another important area for improvement is the personalization of behavioral interventions. The current reminder notifications and motivational nudges are delivered using predefined rules based on users' predicted procrastination risk levels. Future research may investigate adaptive recommendation systems or reinforcement learning techniques that continuously learn from users' historical behavior to determine the most effective notification timing, message content, and intervention frequency. By analyzing users' response patterns, study consistency, and task completion history, future systems could provide behavioral interventions that become increasingly personalized over time.

The AI component itself may also be extended beyond task decomposition. Future versions of the application could incorporate conversational AI capabilities that allow students to seek study advice, clarify assignment requirements, modify generated subtasks, or receive personalized productivity recommendations through natural language interactions. Furthermore, implementing asynchronous request processing, retry mechanisms, and improved exception handling would enhance the reliability of AI-assisted features while providing more user-friendly feedback when external AI services experience temporary failures.

5.3.3 Expansion of Gamification and Productivity Analytics

The pilot testing demonstrated positive user acceptance of both the gamification system and the analytics dashboard. Nevertheless, future development could further expand these features to encourage long-term engagement and strengthen self-regulated learning behaviors. Additional gamification elements such as achievement badges, daily and weekly challenges, milestone rewards, seasonal events, unlockable virtual garden content, and personalized progression systems may provide users with greater motivation to maintain consistent study habits over extended periods.

Future versions of the application could also introduce social gamification features, including collaborative study groups, accountability partners, team-based challenges, and optional leaderboards. These features have the potential to increase user motivation through positive social interaction while promoting collaborative learning. Appropriate privacy settings and user controls should be incorporated to ensure that participation remains voluntary and respects users' individual preferences.

Similarly, the analytics dashboard may be enhanced by providing richer visualizations of users' academic behavior. In addition to displaying task completion statistics and productivity trends, future implementations could incorporate weekly and monthly progress reports, procrastination behavior analysis, study consistency indicators, heat maps, predictive productivity forecasting, and personalized recommendations for improving study habits. Such enhancements would transform the analytics dashboard from a descriptive reporting tool into a proactive decision-support system capable of guiding users towards more effective academic behaviors.

5.3.4 Large-Scale Deployment and Longitudinal Evaluation

The evaluation conducted in this study primarily focused on pilot testing involving a relatively small number of university students over a limited period. Although the findings demonstrated positive user acceptance and satisfactory usability, future studies should evaluate the application using a substantially larger and more diverse participant population. Including students from different universities, academic disciplines, educational levels, and cultural

backgrounds would improve the generalizability of the evaluation findings while providing a more comprehensive understanding of users' behavioral responses to the proposed interventions.

Future research should also conduct longitudinal evaluations over an academic semester or an entire academic year. Since behavioral change associated with academic procrastination develops gradually, longer-term studies would provide stronger evidence regarding whether sustained use of the application leads to measurable improvements in time management, task completion, self-regulated learning, and academic performance. Such evaluations would also enable researchers to investigate how users' engagement with AI-assisted planning, personalized interventions, and gamification evolves over time.

From a technical perspective, future deployment should utilise production-level cloud infrastructure capable of supporting a larger number of concurrent users. Increasing Firebase database capacity, authentication quotas, AI service request limits, and backend resource monitoring would improve system scalability and reliability. Additional improvements such as stricter validation mechanisms, enhanced exception handling, asynchronous processing, and automatic retry strategies would further increase the robustness of the application under real-world operating conditions.

Overall, large-scale deployment combined with long-term evaluation would provide stronger empirical evidence regarding both the technical performance and behavioral effectiveness of the proposed system while supporting the continued development of intelligent educational technologies for academic procrastination management.

5.4 Conclusion

This study successfully demonstrated the feasibility of integrating machine learning, Generative Artificial Intelligence, cloud computing, personalized behavioral interventions, and gamification into a unified mobile application for managing academic procrastination among university students. The developed Nudgr application combines predictive analytics with

intelligent behavioral support to provide personalized academic assistance beyond the capabilities of conventional task management applications.

The findings indicate that the proposed system successfully achieved its project objectives through the development of an effective Random Forest prediction model, the implementation of AI-assisted task planning, personalized reminders, behavioral nudges, productivity analytics, and an engaging gamification framework. Functional testing, integration testing, and pilot testing further demonstrated that the application operated successfully and was positively received by university students, while the System Usability Scale evaluation indicated good overall usability.

Although several limitations relating to dataset quality, prediction performance, infrastructure constraints, and pilot testing scope were identified, these also provide valuable opportunities for future enhancement. Overall, this research contributes a practical framework for combining predictive analytics, artificial intelligence, and behavioral intervention strategies within an educational mobile application. It is anticipated that the findings of this study will support future research on intelligent educational technologies while providing a foundation for the continued development of personalized digital solutions that promote self-regulated learning and reduce academic procrastination.

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Appendix A

Project Resources

Resource	Link
GitHub Repository (Application – Nudgr)	https://github.com/keets-commits/nudgr_app
GitHub Repository (Machine Learning)	https://github.com/Keertana0307/nudgr_ml
Mobile App Install Link (Android)	https://appdistribution.firebase.dev/i/a15e3f57b1281e1e
App Web version (Recommended for iOS Users)	https://nudgr-95673.web.app/
Pilot Testing Google Form	https://forms.gle/DuKX6gkmDLhuX6p2A
Pilot Testing Responses	https://docs.google.com/spreadsheets/d/1FZMVBcqPX93wNd3ifwnwBnrS472Ke2L5CPQWLSEg5zk/edit?usp=sharing
Machine Learning Dataset	https://docs.google.com/spreadsheets/d/1RsSx-LXfKByj85s6XF68-_ze_mzoozZf/edit?usp=sharing&oid=111400897276427181883&rtpof=true&sd=true